



Chefaa

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Summary

Healthcare services often suffer from fragmented systems, making it difficult for patients to access doctors, laboratories, and pharmacies through a unified platform. Patients frequently face challenges in booking appointments, obtaining medical consultations, accessing laboratory services, and purchasing medications efficiently. On the other hand, healthcare providers often lack integrated tools to manage appointments, services, payments, and communication with patients.

To address these challenges, we developed **Chefaa**, an AI-powered online healthcare platform that connects four main stakeholders: **Patients, Doctors, Pharmacies, and Medical Laboratories**, under the supervision of a secure administration system. To ensure trust and reliability, doctors, pharmacists, and laboratory owners must submit professional licenses and verification documents before gaining access to the platform.

The platform enables doctors to create and manage multiple clinics, organize schedules without conflicts, monitor appointments, manage patient attendance and payments, issue digital prescriptions, and track their financial performance through intelligent analytics dashboards. Pharmacies can manage medicine inventories, receive and track orders, provide delivery services, and monitor profits through AI-generated insights. Laboratories can publish their available services and securely deliver medical test results to patients through the application.

For patients, the platform offers a seamless healthcare experience by allowing them to search for healthcare providers, book, cancel, or reschedule appointments, order medications online, and make secure electronic payments. In addition, patients can benefit from AI-powered services, including a medical chatbot for healthcare guidance and a diagnostic assistant capable of analyzing uploaded medical reports, providing simplified explanations and recommending suitable medical specialists when necessary.

The platform also integrates advanced Artificial Intelligence features such as daily briefings, business analytics, financial insights, and intelligent healthcare assistance for all stakeholders. By combining healthcare services, digital payments, appointment

management, pharmacy operations, laboratory services, and AI technologies into a single ecosystem, Chefaa aims to improve healthcare accessibility, efficiency, and user experience while reducing administrative complexity for both patients and healthcare providers.

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Abbreviations list

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CBC	Complete Blood Count
DFD	Data Flow Diagram
DS	Data Store
ERD	Entity Relationship Diagram
GPT	Generative Pre-trained Transformer
HTTP	Hypertext Transfer Protocol
ID	Identifier
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model View Controller
NoSQL	Not Only Structured Query Language
ODM	Object Document Mapping
ORM	Object Relational Mapping
PDF	Portable Document Format
REST	Representational State Transfer

RTK	Redux Toolkit
SQL	Structured Query Language
UC	Use Case
UI	User Interface
UX	User Experience

Chapter 1: Introduction

1.1 Introduction

The rapid growth of digital technologies has transformed many sectors, including healthcare. Despite these advancements, many healthcare services still suffer from fragmented systems that make it difficult for patients to access medical care efficiently. Patients often face challenges in finding suitable doctors, booking appointments, obtaining laboratory services, purchasing medications, and managing medical records. Healthcare providers also struggle with appointment scheduling, patient management, communication, and business analytics.

To address these issues, we developed **Chefaa**, an AI-powered healthcare platform that integrates patients, doctors, pharmacies, laboratories, and administrators into a unified digital ecosystem. The platform streamlines healthcare processes, improves accessibility, enhances communication, and provides intelligent services through Artificial Intelligence technologies.

1.2 Problem Definition

The healthcare sector faces several challenges that affect both patients and healthcare providers:

1.2.1 Difficulty Accessing Healthcare Services

Patients often spend significant time searching for suitable doctors, pharmacies, and laboratories. The lack of centralized healthcare platforms makes access to services inefficient and inconvenient.

1.2.2 Inefficient Appointment Management

Traditional appointment systems frequently lead to scheduling conflicts, missed appointments, and poor clinic organization, affecting both patients and doctors.

1.2.3 Limited Communication Between Stakeholders

Doctors, patients, pharmacies, and laboratories usually operate independently, creating communication gaps and delays in delivering healthcare services.

1.2.4 Lack of Intelligent Healthcare Assistance

Many patients require quick medical guidance, interpretation of medical reports, and recommendations for appropriate specialists, which are not always readily available.

1.2.5 Limited Business and Performance Analytics

Healthcare providers often lack tools that help them analyze appointments, revenues, service performance, and operational efficiency.

1.3 Objectives

The main objectives of the proposed system are:

- Develop an integrated healthcare platform connecting patients, doctors, pharmacies, and laboratories.
- Facilitate online appointment booking, cancellation, and rescheduling.
- Enable doctors to manage clinics, schedules, prescriptions, and patient records efficiently.
- Provide pharmacies with inventory management and order-tracking capabilities.
- Allow laboratories to manage services and securely deliver test results.
- Implement secure online payment services.
- Integrate Artificial Intelligence features for healthcare assistance, analytics, and daily summaries.
- Improve healthcare accessibility, efficiency, and user experience.
- Ensure security and trust through verification and authentication mechanisms.

1.4 Tools Used to Build and Develop the System

1.4.1 Frontend Tools & Technologies

- Visual Studio Code
- React.js
- Redux Toolkit
- React Router DOM
- Axios
- HTML5
- CSS3
- JavaScript (ES6+)
- Bootstrap / Custom CSS
- Chart.js
- SweetAlert2
- Socket.io Client

1.4.2 Backend Tools & Technologies

- Visual Studio Code
- Node.js
- Express.js
- MongoDB
- Mongoose
- JSON Web Token (JWT)
- Multer
- Socket.io
- Node-Cron
- Cloudinary
- OpenAI API
- google-auth-library
- helmet

- moment
- morgan
- nodemailer
- passport
- passport-google-oauth20
- uuid
- validator
- winston

1.4.3 Artificial Intelligence Technologies

- OpenAI GPT Models
- Medical Report Analysis
- AI Medical Chatbot
- Daily Brief Generation
- Revenue and Performance Analytics

1.4.4 UI/UX Design Tools

- Figma

1.4.5 Mobile Application Development Tools

- AndroidStudio
- Git & GitHub
- Dart Programming Language
- Flutter Framework
- StateManagement(BLoC/Cubit)
- Dependency Injection (GetIt & Injectable)
- RESTful APIs (Dio)
- LocalDatabase(Hive)
- LocalPreferencesStorage(SharedPreferences)
- Secure Data Storage (Flutter Secure Storage)

- Google Authentication (Google Sign-In)
- MapsIntegration(GoogleMaps)
- Geolocation Services (Geolocator & Geocoding)
- Responsive User Interface Design (Flutter ScreenUtil)
- Data Visualization (FL Chart)
- EasyLocalization
- Internationalization & Localization (Intl)
- File Management (File Picker)
- Code Generation (Build Runner)

Chapter 2: Previous Work

2.1 Introduction

Many digital healthcare platforms have been developed to improve access to medical services, appointment booking, online consultations, pharmacy services, and laboratory management. These systems aim to reduce waiting times, enhance communication between patients and healthcare providers, and simplify healthcare management. However, most existing solutions focus on a limited set of services and rarely provide a fully integrated healthcare ecosystem supported by Artificial Intelligence.

2.2 Comparative Study: Similar Applications

The following applications were studied as references during the development of Chefaa.

1. Vezeeta

Vezeeta is a leading healthcare booking platform across Egypt, Saudi Arabia, and other MENA countries. It enables patients to discover doctors, book clinic appointments, and access pharmacy services in select regions.

Technical Gaps Addressed by Chefaa:

- No integrated clinic management system: Vezeeta handles booking only. Doctors have no tools for managing patient attendance, multiple clinic schedules, or conflict-free appointment organization.
- Digital prescriptions are limited to online consultations only and are not linked to a pharmacy ordering or delivery system.
- No AI-powered analytics or daily briefings for healthcare providers — doctors and pharmacies have no revenue insights or business intelligence tools.
- No real-time chat between patient and doctor or pharmacy; communication is limited to SMS appointment reminders.
- No support for uploading/receiving lab results within the platform.

2. Altibbi

Altibbi is the largest Arabic digital health platform in the MENA region, focusing on online consultations, medical content, and AI-based symptom checking. It operates across Jordan, Saudi Arabia, Egypt, and the UAE.

Technical Gaps Addressed by Chefaa:

- Appointment booking is limited to online (telemedicine) consultations only — no in-clinic appointment booking, scheduling, or physical attendance management.
- No physical clinic management tools: doctors cannot manage multiple clinics, organize physical schedules, or track patient attendance.
- Digital prescriptions exist only as a post-consultation output and are not connected to a pharmacy system for direct ordering or delivery.
- No pharmacy integration, inventory management, or medicine ordering system.
- No laboratory services — patients cannot book lab tests or receive results through the platform.
- AI functionality is limited to a symptom checker for specialist selection. No AI analysis of medical reports, or business intelligence for providers.

3. Sehaty (Sehhaty)

Sehaty is the official national health platform of Saudi Arabia's Ministry of Health, serving over 24 million users. It provides access to government healthcare services, medical records, telemedicine, and vaccination management.

Technical Gaps Addressed by Chefaa:

- Exclusively tied to Saudi government health infrastructure — not accessible or applicable for private clinics, independent pharmacies, or private medical laboratories.
- No private clinic management tools: no multiple clinic support, no conflict-free scheduling, no patient attendance or payment tracking for private providers.
- Pharmacy services are limited to a medication search tool (Rassd) — no ordering, delivery, inventory management, or integration with private pharmacies.
- Laboratory support covers only government lab result viewing — no booking with private laboratories or result delivery to patients from private labs.
- No AI-powered features for medical report analysis, or business intelligence for healthcare providers.
- No revenue analytics, daily briefings, or platform commission tracking — the platform has no tools for private healthcare business management.
- Real-time chat is not available — communication is handled through the Ministry of Health's call center (937), not through the app.

Comparison Between Existing Systems and Chefaa

Feature	Vezeeta	Altibbi	Sehaty	Chefaa
Appointment Management				
Appointment Cancellation & Rescheduling	✓	✗	✓	✓
Multiple Clinic Management	✓	✗	✗	✓
Clinic Schedule Organization	✓	✗	✗	✓
Patient Attendance Tracking	✗	✗	✗	✓
Prescriptions & Pharmacy				
Digital Prescriptions	Limited	Limited	✗	✓
Medicine Ordering System	✓	✗	✗	✓
Order Tracking	✓	✗	✗	✓
Delivery & Pickup Options	✓	✗	✗	✓
Laboratory Services				
Laboratory Services	✓	✗	Limited	✓
Upload & Receive Lab Results	✗	✗	✓	✓
Payments & Communication				

Feature	Vezeeta	Altibbi	Sehaty	Chefaa
Secure Online Payment	✓	✓	✓	✓
Real-Time Notifications	Limited	✓	✓	✓
Real-Time Chat System	X	✓	X	✓
Provider Verification System	✓	Limited	✓	✓
AI-Powered Features				
AI Medical Chatbot	X	Limited	X	✓
AI Healthcare Guidance	Limited	✓	X	✓
AI Analysis of Medical Reports	X	X	X	✓
Analytics & Administration				
Doctor Daily Briefing	X	X	X	✓
Pharmacy Daily Briefing	X	X	X	✓
Revenue Analytics	X	X	X	✓
AI Business Insights	X	X	X	✓
Platform Commission Tracking	X	X	X	✓
Overall Integration				
Integrated Healthcare Ecosystem	Partial	Partial	Partial	✓

Discussion

The comparative study reveals that existing healthcare platforms each address a narrow segment of the healthcare delivery chain. Vezeeta focuses on doctor discovery and appointment booking but lacks integrated clinic management, AI capabilities, and a unified pharmacy-laboratory ecosystem. Altibbi excels in telemedicine consultations and Arabic medical content, but provides no tools for physical clinic operations, pharmacy management, or laboratory services. Sehaty, while comprehensive for Saudi government healthcare, is architecturally restricted to public sector infrastructure and offers no support for private clinics, independent pharmacies, or private laboratories.

Chefaa addresses these technical gaps by delivering a fully integrated healthcare ecosystem covering four stakeholder groups: Patients, Doctors, Pharmacies, and Medical Laboratories. Key differentiators include multi-clinic management with conflict-free scheduling, a complete pharmacy workflow from ordering to delivery, private laboratory integration with secure result delivery, and a suite of AI-powered features including medical report and, daily operational briefings, revenue analytics, and AI-driven business insights — none of which are available in the studied platforms.

Chapter 3: System Analysis

3.1 Introduction

System analysis is a crucial phase in software development that focuses on understanding the requirements, functionalities, and interactions within the proposed system. This chapter presents the analysis of the Chefaa platform, including stakeholders, functional requirements, non-functional requirements, and system use cases.

3.2 System Overview

Chefaa is an AI-powered healthcare platform designed to connect patients, doctors, pharmacies, laboratories, and administrators within a unified ecosystem. The platform aims to simplify healthcare access, appointment management, medicine ordering, laboratory services, online payments, and healthcare analytics.

The system consists of five main actors:

- ❖ Administrator
- ❖ Doctor
- ❖ Pharmacy
- ❖ Laboratory
- ❖ Patient

Each actor has specific responsibilities and permissions that contribute to the overall operation of the platform.

3.3 Stakeholders Analysis

3.3.1 Administrator

The administrator is responsible for managing the entire platform and ensuring the authenticity of healthcare providers.

Responsibilities:

- Verify doctors, pharmacies, and laboratories.

- Approve or reject registration requests.
- Manage platform users.
- Monitor system activities.
- Manage platform commissions and revenues.

3.3.2 Doctor

Responsibilities:

- Create and manage clinics.
- Organize schedules and appointments.
- Manage patient visits.
- Issue prescriptions.
- View analytics and revenues.
- Access AI-powered daily briefings.

3.3.3 Pharmacy

Responsibilities:

- Manage medicine inventory.
- Receive and process orders.
- Track deliveries.
- View financial reports and analytics.

3.3.4 Laboratory

Responsibilities:

- Manage laboratory services.
- Receive test requests.
- Upload and send test results.
- Manage laboratory information.

3.3.5 Patient

Responsibilities:

- Search for doctors and healthcare services.
- Book, cancel, and reschedule appointments.

- Order medicines online.
- View prescriptions and laboratory results.
- Use AI healthcare services.

3.4 Functional Requirements

Functional requirements describe the services, operations, and behaviors that the Chefaa platform must provide to its users. The system supports five main actors: Patients, Doctors, Pharmacies, Laboratories, and Administrators.

3.4.1 Authentication and Authorization

The system shall provide a secure authentication mechanism that allows users to register and log into the platform according to their roles. Patients can register directly, while doctors, pharmacies, and laboratories must submit professional verification documents during registration. These documents are reviewed by the administrator before account activation.

The system shall support:

- User registration and login.
- Role-based access control.
- Password recovery and reset functionality.
- Secure session management using JWT authentication.
- Account verification for healthcare providers.

3.4.2 Profile Management

The platform shall allow all users to create and maintain their profiles. Each user type has a customized profile according to their role within the system.

The system shall allow users to:

- View profile information.
- Update personal details.
- Change passwords.
- Upload profile images and supporting documents.

- Manage account settings.

Doctors can additionally provide information such as specialization, qualifications, clinic information, consultation fees, and available schedules. Pharmacies and laboratories can maintain their business information, services, and contact details.

3.4.3 Notification Management

The platform shall provide a notification system to keep users informed about important events and activities.

The system shall allow users to:

- Receive real-time notifications.
- View notification history.
- Mark notifications as read.
- Configure notification preferences.
- Medicine Alarm

Notifications may include appointment confirmations, prescription updates, laboratory results, payment confirmations, order status updates, and administrative announcements.

3.4.4 Appointment Management

The appointment management module enables patients and doctors to efficiently organize healthcare visits.

Patients shall be able to:

- Search for doctors.
- View doctor schedules and available time slots.
- Book appointments online.
- Cancel appointments.
- Reschedule appointments.
- View appointment history.

Doctors shall be able to:

- View all appointments.
- Manage clinic schedules.
- Mark appointments as completed.
- Track attendance status.
- Identify missed appointments.
- Block patients who repeatedly fail to attend appointments.

The system shall automatically prevent scheduling conflicts and overlapping appointments.

3.4.5 Prescription Management

The prescription module enables doctors to generate and manage digital prescriptions while allowing patients to access their treatment information.

Doctors shall be able to:

- Create prescriptions.
- Update existing prescriptions.
- Review prescription history.
- Add medications, dosage instructions, and medical notes.
- Request laboratory tests.

Patients shall be able to:

- View prescriptions.
- Access previous prescriptions.
- Share prescriptions with pharmacies when ordering medications.

3.4.6 Clinic Management

The clinic management module enables doctors to organize and maintain multiple clinics from a centralized dashboard.

Doctors shall be able to:

- Create clinics.
- Update clinic information.
- Delete clinics.
- Manage clinic addresses.
- Configure working days and operating hours.
- Define appointment durations.
- View clinic schedules and statistics.

Patients shall be able to:

- View clinic information.
- View available appointment slots.
- Select preferred clinic locations.

3.4.7 Pharmacy Management

The pharmacy module provides a complete medicine ordering and inventory management system.

Patients shall be able to:

- Search for pharmacies.
- Search for medicines.
- View pharmacy profiles.
- View medicine information.
- Create medicine orders.
- Track order status.
- Confirm order receipt.
- Submit reviews and ratings.
- Pay online.

Pharmacies shall be able to:

- Receive customer orders.
- Process and update order status.
- Manage deliveries.
- Monitor inventory levels.
- Track revenue and sales performance.

3.4.8 Laboratory Management

The laboratory module facilitates communication between laboratories and patients.

Laboratories shall be able to:

- Manage laboratory profiles.
- Add and update services.
- Activate or deactivate services.
- Request laboratory services.
- Upload test results.
- Manage patient records related to laboratory services.

Patients shall be able to:

- View laboratory information.
- Receive and download laboratory reports.

3.4.9 Catalog and Inventory Management

The platform shall provide inventory and catalog management functionality for pharmacies and laboratories.

Pharmacies shall be able to:

- Add medicines.
- Edit medicine information.
- Update stock quantities.
- Restock inventory.

- Remove unavailable products.

Laboratories shall be able to:

- Add services.
- Modify service details.
- Activate or deactivate services.

3.4.10 Billing and Financial Management

The platform shall provide financial management capabilities for users and administrators.

Users shall be able to:

- View billing summaries.
- Access payment history.
- Make secure online payments.

Administrators shall be able to:

- Generate billing reports.
- Manage financial transactions.
- Track platform commissions.
- View revenue analytics.
- Monitor payment activities.

3.4.11 Artificial Intelligence Services

The system shall integrate Artificial Intelligence technologies to improve healthcare accessibility and decision support.

The AI module shall provide:

- Medical chatbot assistance for patients.
- Healthcare guidance and recommendations.
- AI-generated daily briefings for doctors and pharmacies.

- Financial analysis and performance insights.
- Pharmacy stock analysis and recommendations.
- Medical report interpretation.
- Specialist recommendations based on uploaded medical documents.

The AI services are designed to support healthcare decision-making and should not replace professional medical consultation.

3.4.12 Administration Module

The administration module provides complete control over the platform.

Administrators shall be able to:

- Verify doctors, pharmacies, and laboratories.
- Approve or reject registration requests.
- Block or delete accounts.
- Manage roles and permissions.
- Monitor system logs.
- Configure platform settings.
- View system-wide analytics.
- Manage backup and restoration operations.
- Monitor platform health and performance.

The administrator is responsible for maintaining system security, reliability, and compliance with platform policies.

3.5 Non-Functional Requirements

Security

- The system shall use JWT-based authentication.
- Passwords shall be securely encrypted.
- Role-based authorization shall be enforced.
- Payment transactions shall be processed securely.
- Only verified healthcare providers shall access professional features.

Performance

- System response time should be minimal during normal operation.
- The platform shall support concurrent users efficiently.
- Real-time updates shall be delivered using Socket.io.

Reliability

- The system shall maintain high availability.
- Data loss shall be minimized through backup mechanisms.
- Critical transactions shall be logged and tracked.

Scalability

- The system shall support future growth in users, clinics, pharmacies, and laboratories.
- Additional AI services and healthcare modules can be integrated without major redesign.

Usability

- The user interface shall be intuitive and easy to navigate.
- The platform shall support responsive web and mobile access.
- Healthcare services shall be accessible with minimal user effort.

Maintainability

- The system architecture shall support future updates and enhancements.
- Modules shall be designed independently to simplify maintenance.

Availability

- The platform shall be available 24/7 except during scheduled maintenance periods.

Compatibility

- The system shall function correctly on modern web browsers.
- The platform shall support both desktop and mobile devices.

Data Integrity

- Medical, financial, and appointment records shall remain accurate and consistent.
- Unauthorized modifications to critical data shall be prevented.

3 . 6 System Modeling

We will represent our system using the following diagrams:

- Use case diagram.
- Class Diagram.
- Sequence diagram.
- Data flow diagram (DFD).
- Entity relationship diagram (ERD).

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3 . 6 . 1 Use Case Specifications

1. UC-01: User Login

Item	Description
Use Case ID	UC-01
Use Case Name	User Login
Primary Actor	Patient, Doctor, Pharmacy, Laboratory, Admin
Description	Allows users to access the platform securely.
Preconditions	User account exists and is active.
Main Flow	1. Enter email and password.2. System validates credentials.3. System identifies user role.4. User is redirected to dashboard.
Alternative Flow	Invalid credentials or inactive account.
Postconditions	User gains access to authorized functions.

2. UC-02: Book Appointment

Item	Description
Use Case ID	UC-02
Use Case Name	Book Appointment
Primary Actor	Patient
Description	Allows patients to reserve appointments with doctors.
Preconditions	Patient is logged in and slot is available.
Main Flow	1. Search doctor.2. Select clinic.3. Choose available slot.4. Confirm booking.5. System creates appointment.
Alternative Flow	Selected slot becomes unavailable.
Postconditions	Appointment is successfully scheduled.

3. UC-03:Cancel / Reschedule Appointment

Item	Description
Use Case ID	UC-03
Use Case Name	Cancel or Reschedule Appointment
Primary Actor	Patient
Description	Allows patients to modify existing appointments.
Preconditions	Appointment exists.
Main Flow	1. Open appointment list.2. Select appointment.3. Choose cancel or reschedule.4. Confirm action.
Alternative Flow	Appointment already completed.
Postconditions	Appointment status is updated.

4. UC-04: Create Clinic

Item	Description
Use Case ID	UC-04
Use Case Name	Create Clinic
Primary Actor	Doctor
Description	Allows doctors to create and manage clinics.
Preconditions	Doctor account is approved.
Main Flow	1. Enter clinic information.2. Define working days.3. Define available slots.4. Save clinic.
Alternative Flow	Invalid clinic data.
Postconditions	Clinic becomes visible to patients.

5. UC-05: Manage Appointments

Item	Description
Use Case ID	UC-05
Use Case Name	Manage Appointments
Primary Actor	Doctor
Description	Allows doctors to manage patient appointments.
Preconditions	Doctor is logged in.
Main Flow	1. View appointments.2. Mark patient arrived.3. Mark appointment completed.4. Mark payment status.
Alternative Flow	Patient does not attend appointment.
Postconditions	Appointment status is updated.

6. UC-06: Create Prescription

Item	Description
Use Case ID	UC-06
Use Case Name	Create Prescription
Primary Actor	Doctor
Description	Allows doctors to issue electronic prescriptions.
Preconditions	Patient appointment exists.
Main Flow	1. Open patient record.2. Add medicines.3. Add dosage instructions.4. Save prescription.
Alternative Flow	Missing prescription information.
Postconditions	Prescription becomes available to patient.

7. UC-07: Create Medicine Order

Item	Description
Use Case ID	UC-07
Use Case Name	Create Medicine Order
Primary Actor	Patient
Description	Allows patients to order medicines from pharmacies.
Preconditions	Medicine is available.
Main Flow	1. Search medicine.2. Add to cart.3. Select delivery method.4. Confirm order.
Alternative Flow	Medicine out of stock.
Postconditions	Order is created successfully.

8. UC-08: Online Payment

Item	Description
Use Case ID	UC-08
Use Case Name	Online Payment
Primary Actor	Patient
Description	Allows users to pay for appointments and medicine orders online.
Preconditions	Payment request exists.
Main Flow	1. Select payment method.2. Enter payment details.3. Complete transaction.4. Receive confirmation.
Alternative Flow	Payment fails.
Postconditions	Payment status becomes Paid.

UC-09: Manage Inventory

Item	Description
Use Case ID	UC-09
Use Case Name	Manage Inventory
Primary Actor	Pharmacy
Description	Allows pharmacies to manage medicine stock.
Preconditions	Pharmacy account is approved.
Main Flow	1. Add medicine.2. Update medicine information.3. Restock inventory.4. Delete medicine if necessary.
Alternative Flow	Invalid medicine information.
Postconditions	Inventory database is updated.

9. UC-10: Process Pharmacy Order

Item	Description
Use Case ID	UC-10
Use Case Name	Process Pharmacy Order
Primary Actor	Pharmacy
Description	Allows pharmacies to process medicine orders.
Preconditions	Order exists.
Main Flow	1. View order.2. Accept order.3. Prepare medicines.4. Mark order ready.5. Complete delivery.
Alternative Flow	Order cancelled.
Postconditions	Order status becomes completed.

10. UC-11: Upload Lab Result

Item	Description
Use Case ID	UC-11
Use Case Name	Upload Lab Result
Primary Actor	Laboratory
Description	Allows laboratories to upload test results.
Preconditions	Lab request exists.
Main Flow	1. Open request.2. Upload result file.3. Save result.4. Notify patient.
Alternative Flow	Invalid file upload.
Postconditions	Result becomes available to patient.

11. UC-12: Analyze Medical Report using AI

Item	Description
Use Case ID	UC-12
Use Case Name	Analyze Medical Report
Primary Actor	Patient
Description	Allows patients to upload reports and receive AI analysis.
Preconditions	Patient is logged in.
Main Flow	1. Upload report.2. AI processes file.3. Generate explanation.4. Recommend specialist if needed.
Alternative Flow	Unsupported file format.
Postconditions	Analysis results are displayed.

12. UC-13: Verify Healthcare Provider

Item	Description
Use Case ID	UC-13
Use Case Name	Verify Healthcare Provider
Primary Actor	Admin
Description	Allows administrators to verify doctors, pharmacies, and laboratories.
Preconditions	Registration request exists.
Main Flow	1. Review submitted documents.2. Validate credentials.3. Approve or reject request.4. Notify applicant.
Alternative Flow	Missing or invalid documents.
Postconditions	Account status is updated.

13. UC-14: Manage Users

Item	Description
Use Case ID	UC-14
Use Case Name	Manage Users
Primary Actor	Admin
Description	Allows administrators to manage platform users.
Preconditions	Administrator logged in.
Main Flow	1. View users.2. Search users.3. Activate, deactivate, or block accounts.4. Save changes.
Alternative Flow	User not found.
Postconditions	User status is updated.

14. UC-15: Generate Billing & Reports

Item	Description
Use Case ID	UC-15
Use Case Name	Generate Billing Reports
Primary Actor	Admin
Description	Allows administrators to monitor revenues and transactions.
Preconditions	Financial data exists.
Main Flow	1. Access billing module.2. Generate report.3. View revenue statistics.4. Export report if needed.
Alternative Flow	No available data.
Postconditions	Financial reports are generated successfully.

3.6.2 Use Case Diagram

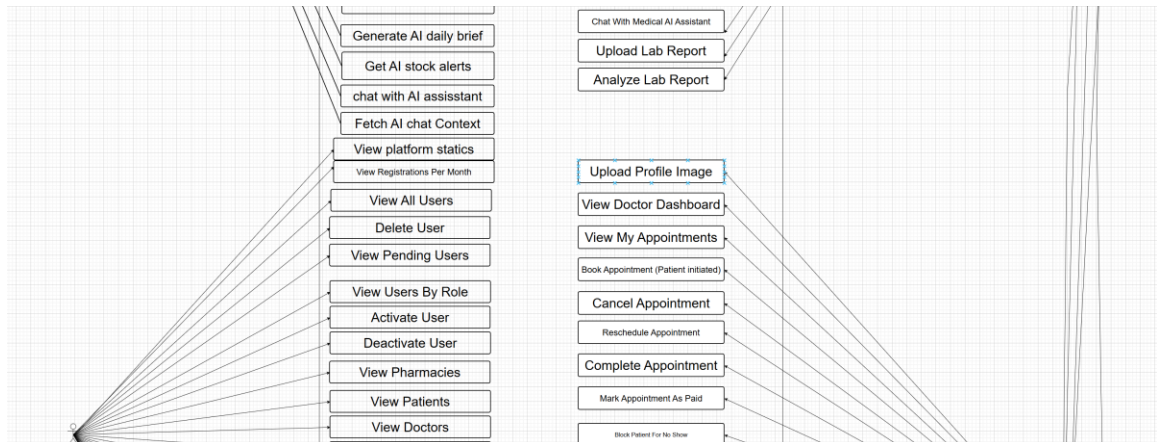
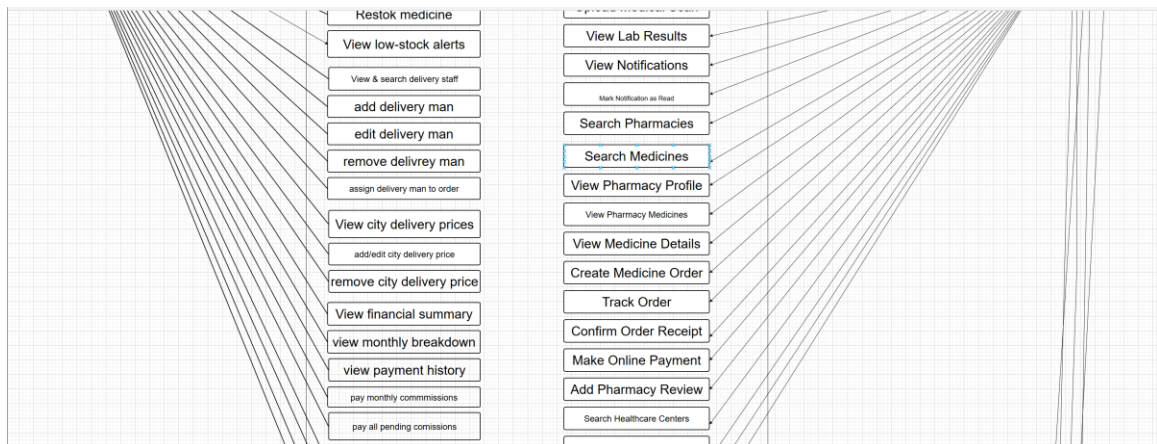
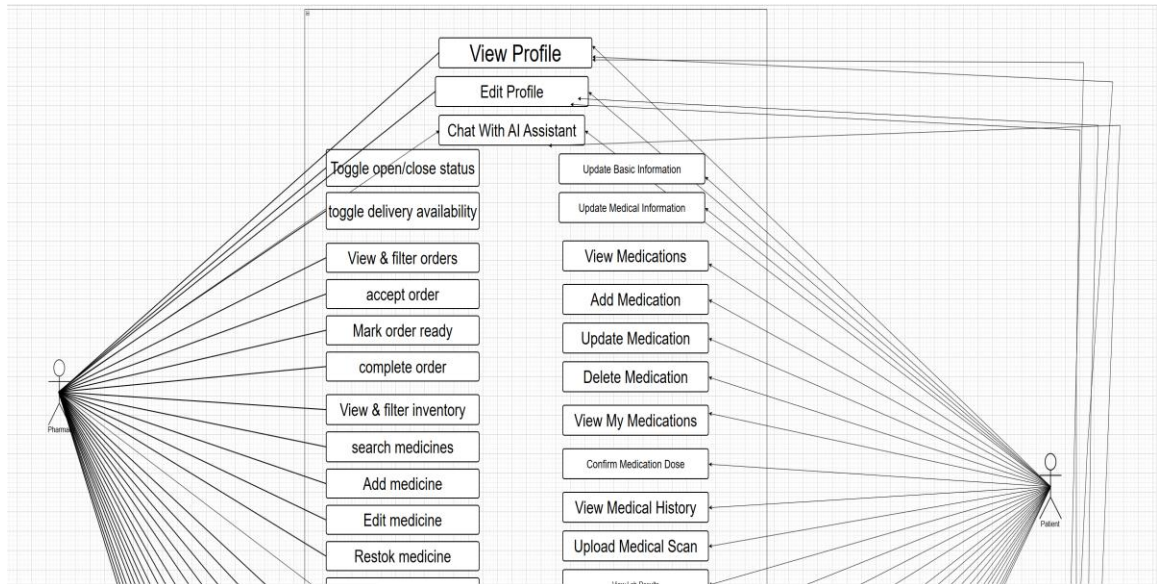




Figure 3.1: Use Case Diagram of the Chefaa Healthcare Platform

3.6.2 Class Diagram

The following class diagram illustrates the main classes of the Chefaa Healthcare Platform and the relationships between them. The diagram was derived directly from the MongoDB collections and Mongoose schemas implemented in the system.

Classes

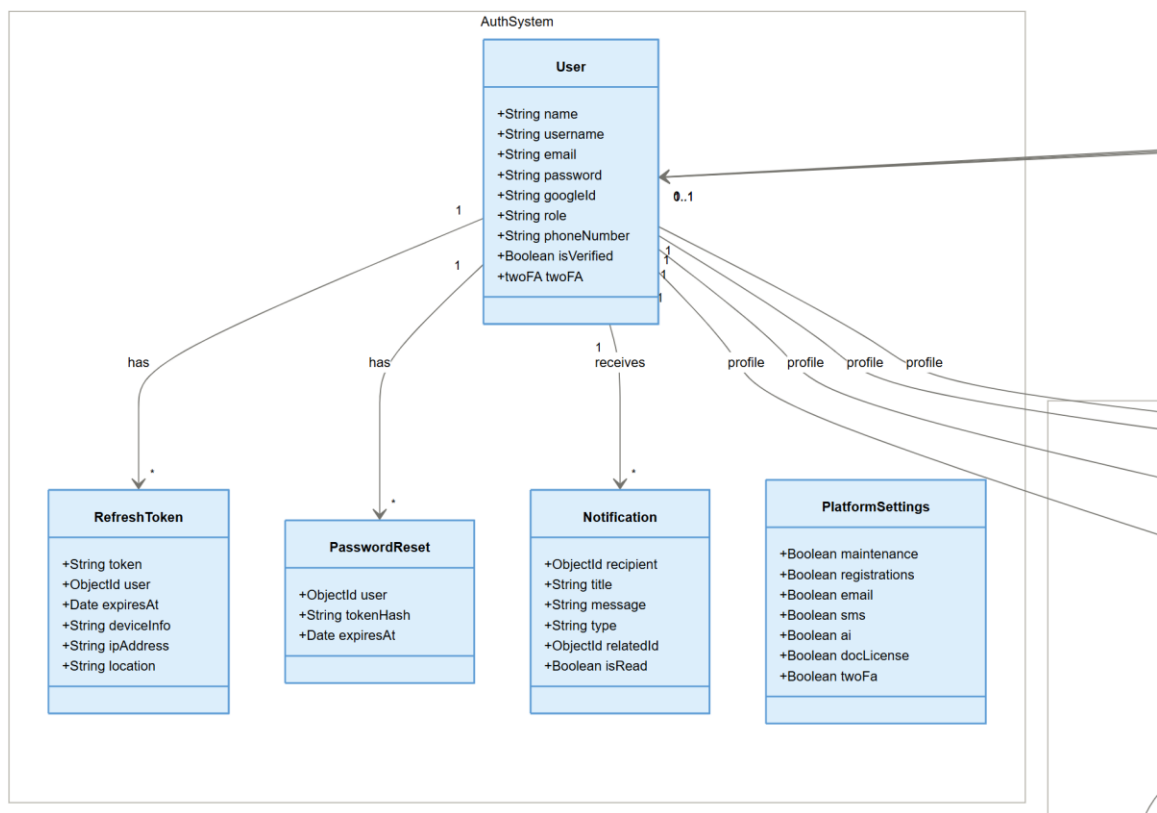
- **User Class** represents all registered users in the platform, including patients, doctors, pharmacies, laboratories, and administrators. It contains authentication, authorization, and account management information.
- **Patient Class** represents patients who use the platform to book appointments, order medications, upload medical reports, and access healthcare services.
- **Doctor Class** represents healthcare providers who manage clinics, appointments, prescriptions, and patient medical records.
- **Clinic Class** represents medical clinics associated with doctors and contains scheduling, pricing, and location information.
- **Appointment Class** manages appointment reservations between patients and doctors, including scheduling and payment information.
- **Prescription Class** represents electronic prescriptions created by doctors and may include medications, laboratory tests, imaging requests, and follow-up recommendations.
- **MedicalRecord Class** stores patient medical history, diagnoses, treatments, prescriptions, and attachments.
- **Lab Class** represents laboratories and radiology centers registered in the system.
- **Service Class** represents laboratory and radiology services offered by laboratories.
- **LabRequest Class** manages laboratory requests submitted by Lab staff and stores uploaded test results.

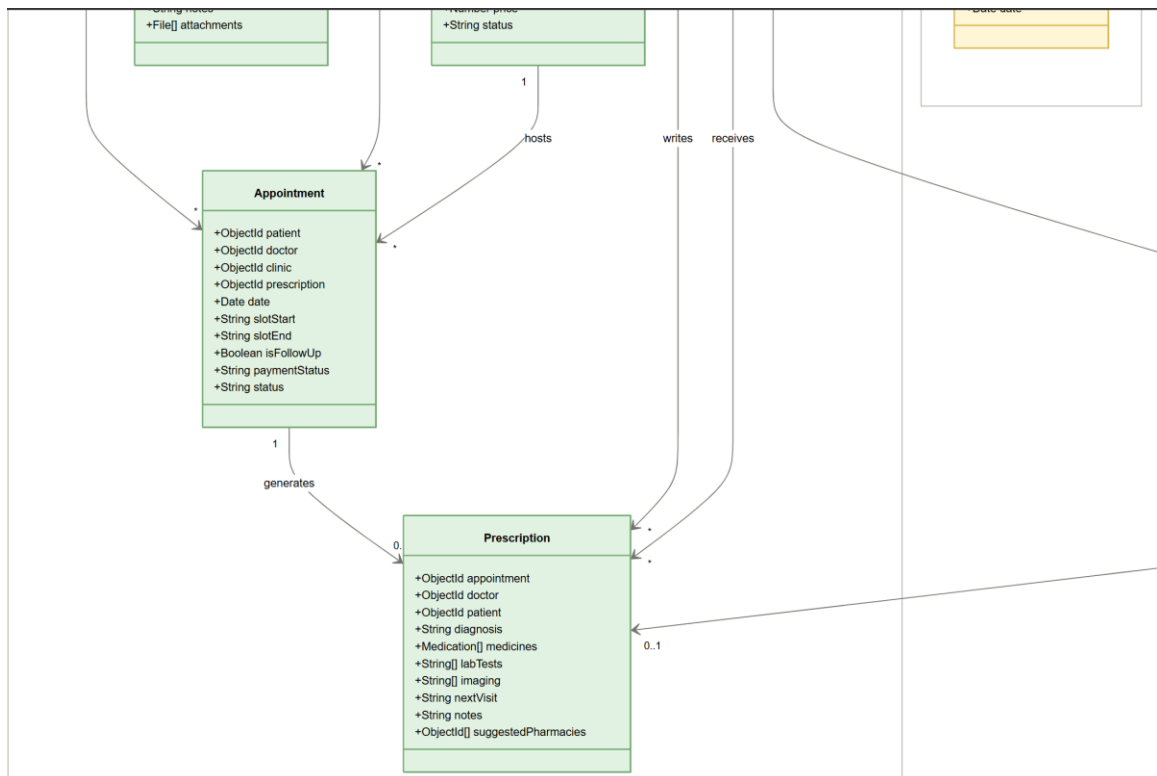
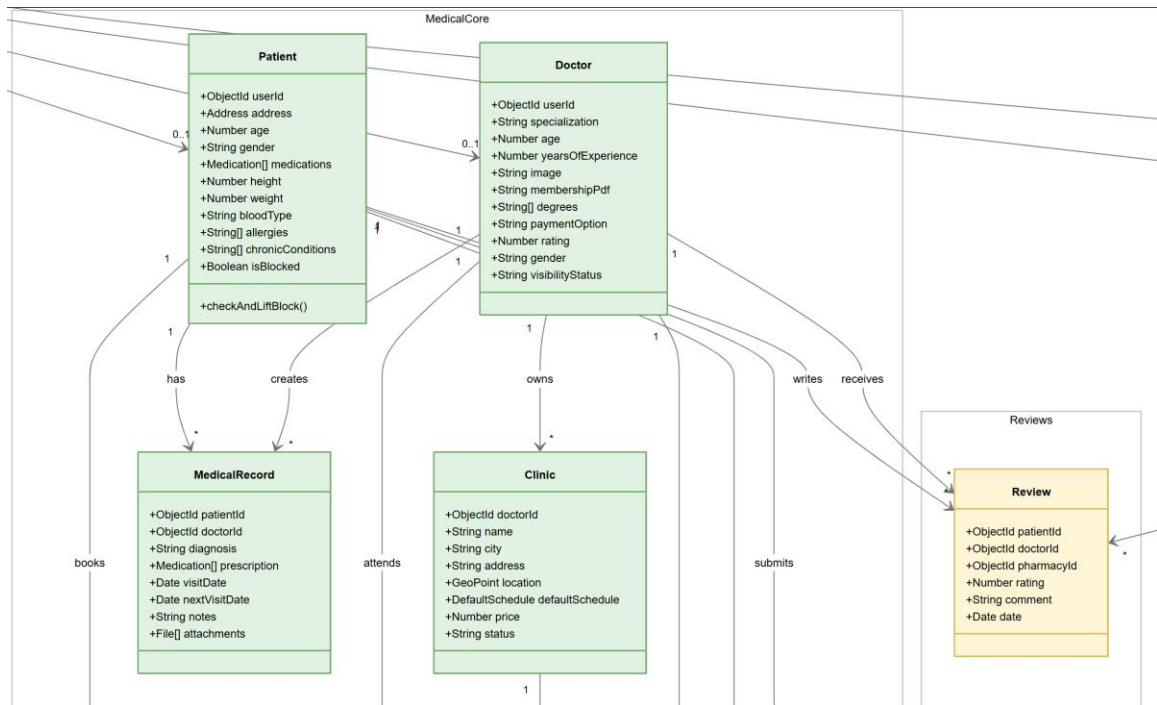
- Pharmacy Class represents pharmacies that provide medication ordering and delivery services.
- MedicineStock Class manages pharmacy inventory, medication availability, pricing, and stock quantities.
- Order Class represents medication orders created by patients and processed by pharmacies.
- DeliveryMan Class represents delivery personnel responsible for delivering pharmacy orders.
- Review Class stores ratings and feedback submitted by patients regarding doctors and pharmacies.
- Transaction Class manages financial transactions, online payments, commissions, and platform fees.
- MonthlyPayment Class stores pharmacy commission settlements and monthly payment records.
- BillingRecord Class tracks financial obligations, revenues, and billing information for platform entities.
- Notification Class manages notifications and alerts sent to users across the platform.
- RefreshToken Class handles user session management and authentication tokens.
- PasswordReset Class manages password recovery requests and reset tokens.
- PlatformSettings Class stores global platform configurations such as AI services, registration controls, notifications, and maintenance settings.

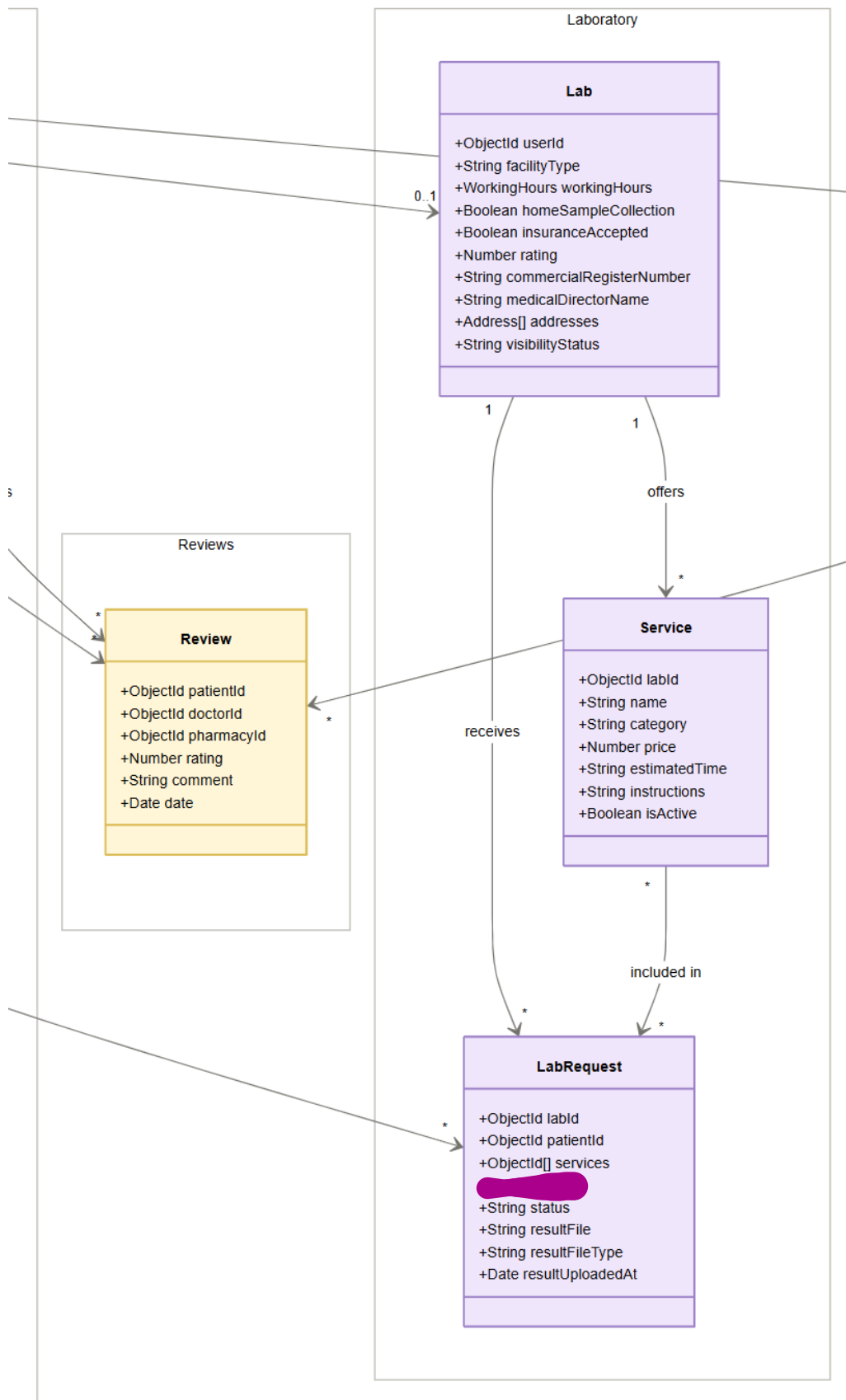
Relationships

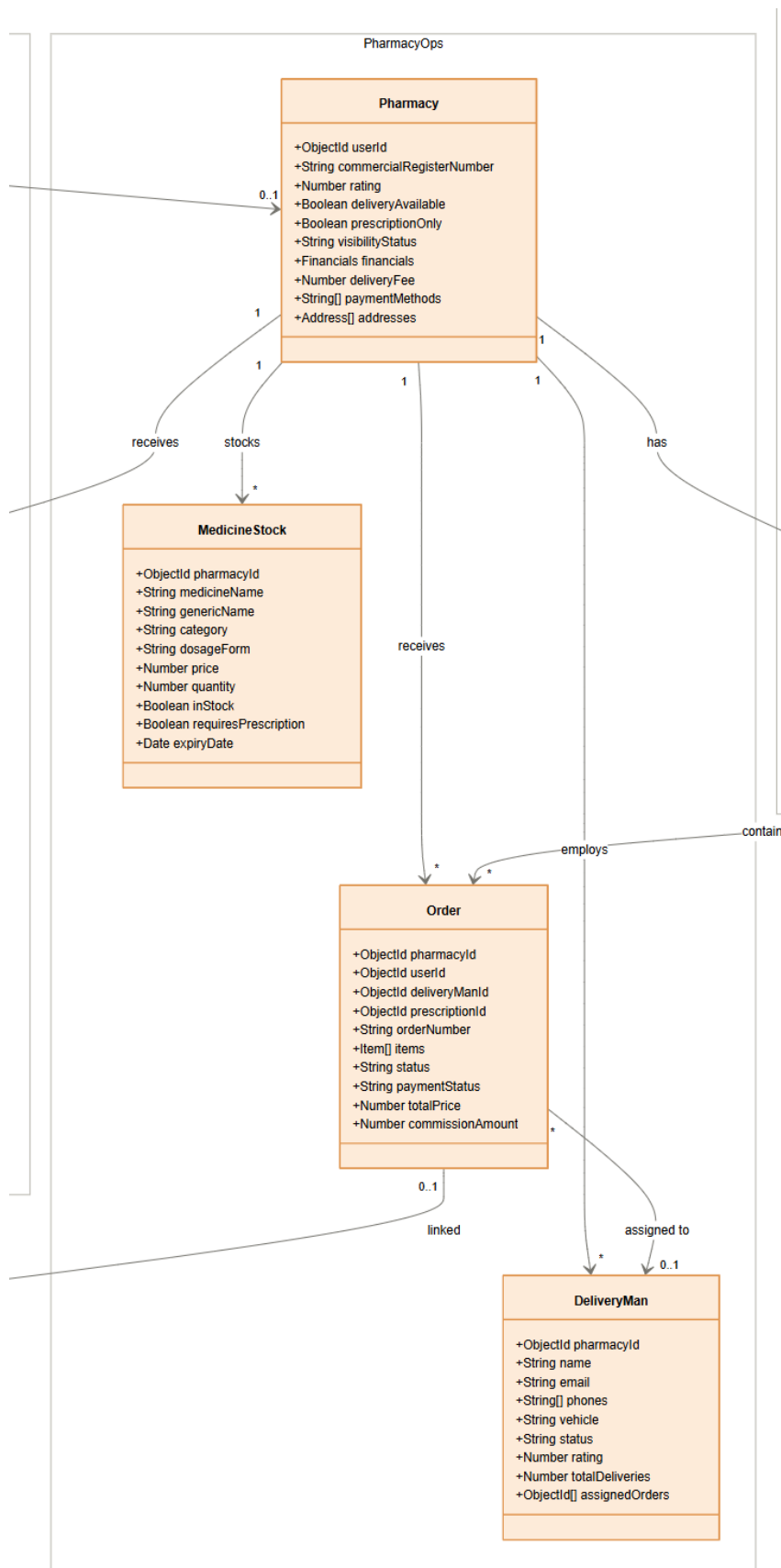
The relationships between classes reflect the healthcare workflow implemented within the platform. A user may have a patient, doctor, pharmacy, or laboratory profile. Doctors manage clinics and appointments, while patients can book appointments and receive prescriptions. Prescriptions may generate medication orders through pharmacies, while laboratories process service requests and upload test results. Financial transactions, billing records, notifications, and authentication services support the overall operation of the healthcare ecosystem.

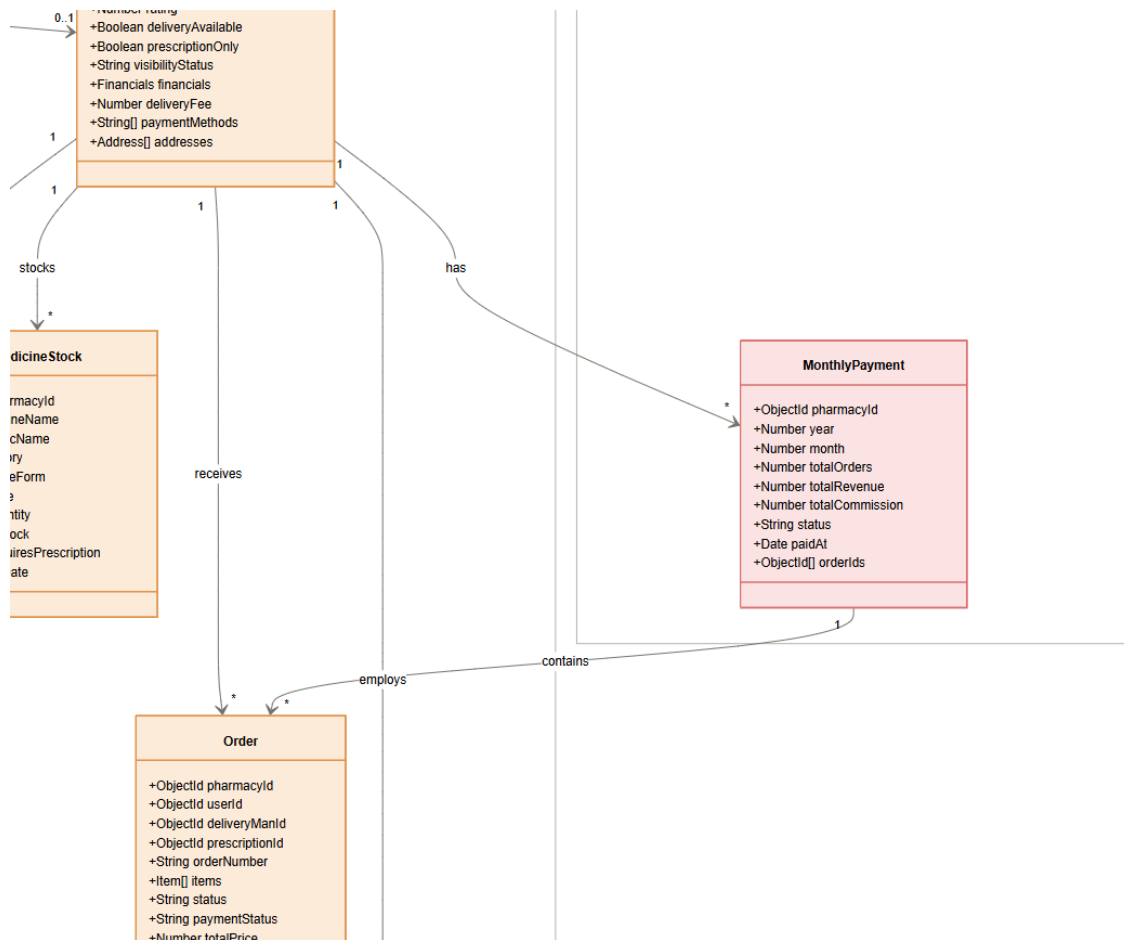
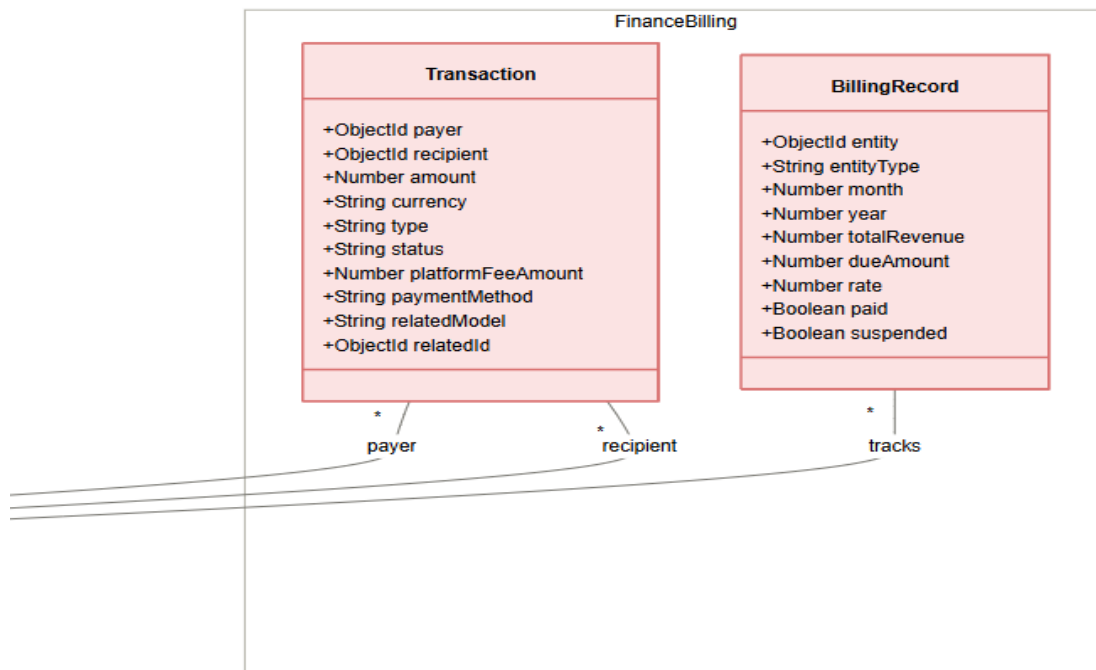
The following figure presents the complete Class Diagram of the Chefaa Healthcare Platform.











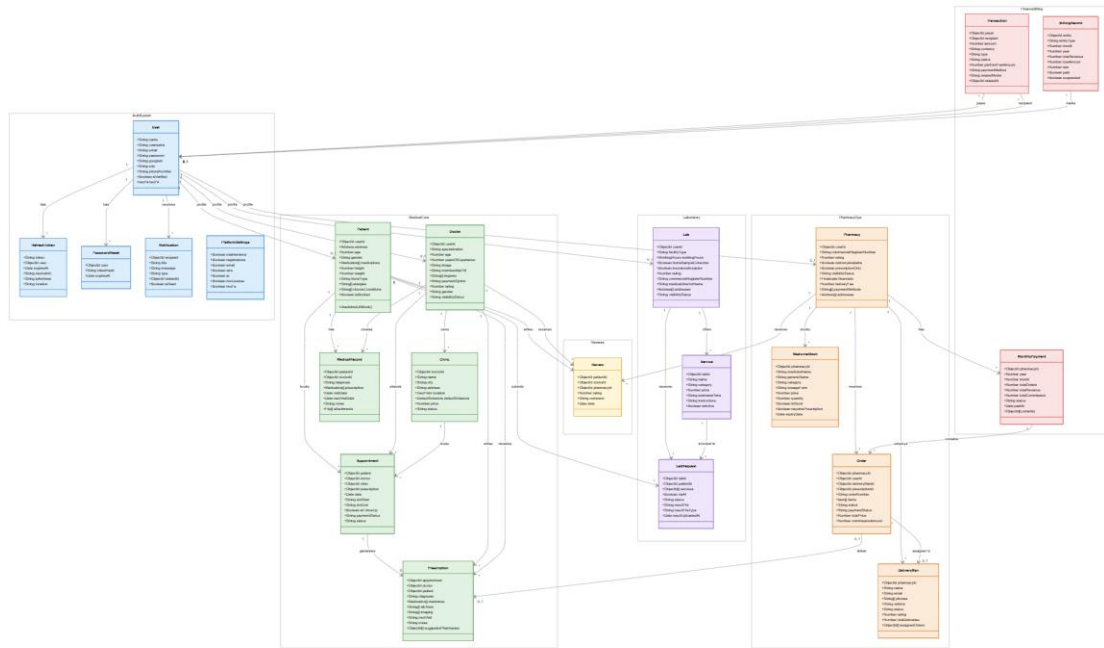


Figure 3.2: Class Diagram of the Chefaa Healthcare Platform

3.6.3 Sequence Diagram

Doctor Registration and Admin Approval Sequence Diagram

The following sequence diagram illustrates the interactions between a doctor, the system, and the administrator during the doctor registration and verification process in the Chefaa Healthcare Platform. The diagram demonstrates how doctors create accounts, submit professional verification documents, undergo administrative review, and receive approval or rejection notifications.

Participants:

- **Doctor:** The healthcare professional who registers and submits verification documents.
 - **System:** Handles account creation, document storage, verification requests, and notifications.
 - **Admin:** Reviews submitted documents and decides whether to approve or reject the doctor's application.
 - **Notification:** Sends approval or rejection messages to doctors.
-

Flow of Interactions:

1. Doctor Registration

- Doctor → System: Submits registration information including personal details, specialization, and account credentials.
- System → Doctor: Validates registration data and creates a doctor account with an "Unverified" status.

2. Upload Verification Documents

- Doctor → System: Uploads professional verification documents (medical license, membership certificate, or professional credentials).

- System → System: Stores uploaded documents securely in the database.
- System → Admin: Generates a verification request and notifies the administrator.

3. Review Verification Request

- Admin → System: Requests the list of pending doctor verification applications.
- System → Admin: Returns submitted doctor information and uploaded documents.
- Admin → System: Reviews the provided credentials and supporting documents.

4. Verification Decision

Alt [Documents Approved]

- Admin → System: Approves the doctor's verification request.
- System → System: Updates the doctor's account status from "Unverified" to "Verified".
- System → Notification: Generates an approval notification.
- Notification → Doctor: Sends account approval confirmation.

5. Rejection Flow

Alt [Documents Rejected]

- Admin → System: Rejects the verification request and optionally provides a rejection reason.
- System → System: Records the rejection decision.
- System → Notification: Generates a rejection notification.
- Notification → Doctor: Sends rejection message with the reason for rejection.

6. Access Granted

- Doctor → System: Attempts to access doctor services.

- System → Doctor: Grants access only if the account status is verified.

7. Send Notifications

- System → Notification: Creates status update notifications.
- Notification → Doctor: Sends verification status updates and account activation information.

This sequence diagram ensures that all doctors undergo a secure verification process before being allowed to provide healthcare services through the platform. The workflow improves platform credibility, protects patients from unverified practitioners, and maintains compliance with healthcare service requirements.

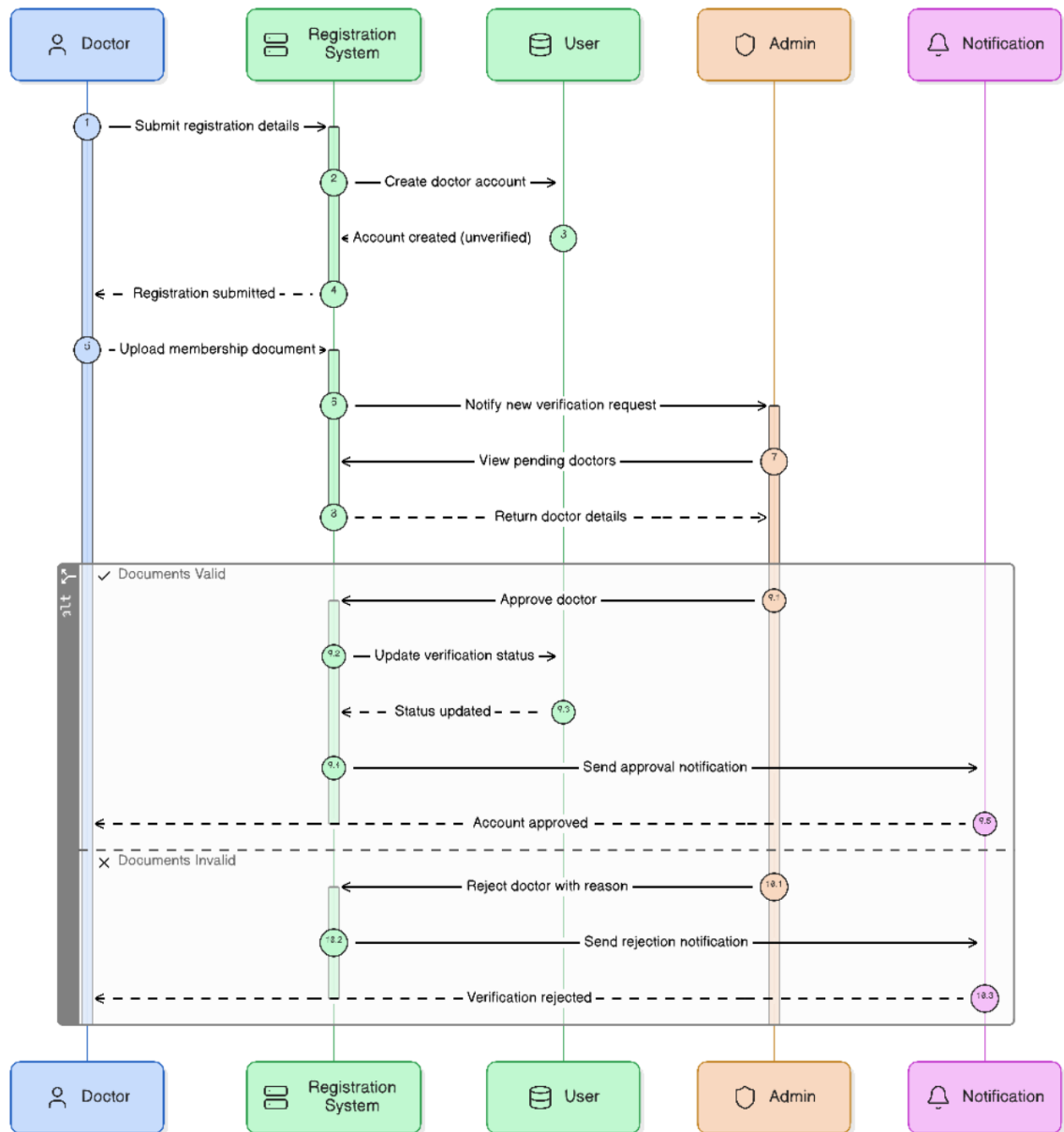


Figure 3.6.1: Doctor Registration and Admin Approval Sequence Diagram

Create Prescription Sequence Diagram

The following sequence diagram illustrates the workflow of a doctor creating and issuing an electronic prescription within the Chefaa Healthcare Platform during a patient appointment. It demonstrates how medical data is retrieved, how prescriptions are generated, and how patients are notified once a prescription is issued.

Participants:

- **Doctor:** The healthcare provider responsible for diagnosing the patient and issuing the prescription.
 - **System:** Manages appointment data, patient medical records, prescription creation, and data storage.
 - **Appointment System:** Provides access to appointment details and links prescriptions to specific visits.
 - **Patient Record:** Stores and retrieves patient medical history and clinical information.
 - **Notification Service:** Sends alerts to patients when a new prescription is created.
-

Flow of Interactions:

1. Access Appointment

- Doctor → Appointment System: Opens a specific patient appointment.
- Appointment System → Doctor: Returns appointment details including patient information and visit context.

2. Retrieve Patient Medical Data

- Doctor → System: Requests patient medical history.
- System → Patient Record: Fetches stored medical data including previous diagnoses, allergies, and treatments.
- Patient Record → System: Returns complete patient medical information.

- System → Doctor: Displays patient medical data for clinical evaluation.

3. Create Prescription

- Doctor → System: Initiates prescription creation.
- Doctor → System: Adds diagnosis, medications, dosage instructions, laboratory tests, imaging requests, and medical notes.
- System → System: Validates and stores prescription data.

4. Link Prescription to Appointment

- System → Appointment System: Attaches prescription to the corresponding appointment record.
- Appointment System → System: Confirms successful linking of prescription and visit.
- System → Doctor: Confirms that the prescription has been saved successfully.

5. Send Patient Notification

- System → Notification Service: Triggers prescription notification event.
- Notification Service → Patient: Sends alert informing the patient that a new prescription has been issued and is available for viewing.

6. Finalization

- Doctor → System: Completes the consultation session.
- System → Appointment System: Updates appointment status to "Completed" with prescription attached.

This sequence ensures that prescriptions are accurately generated, securely stored, and properly associated with patient appointments. It also guarantees immediate communication with patients, improving continuity of care and overall healthcare efficiency.

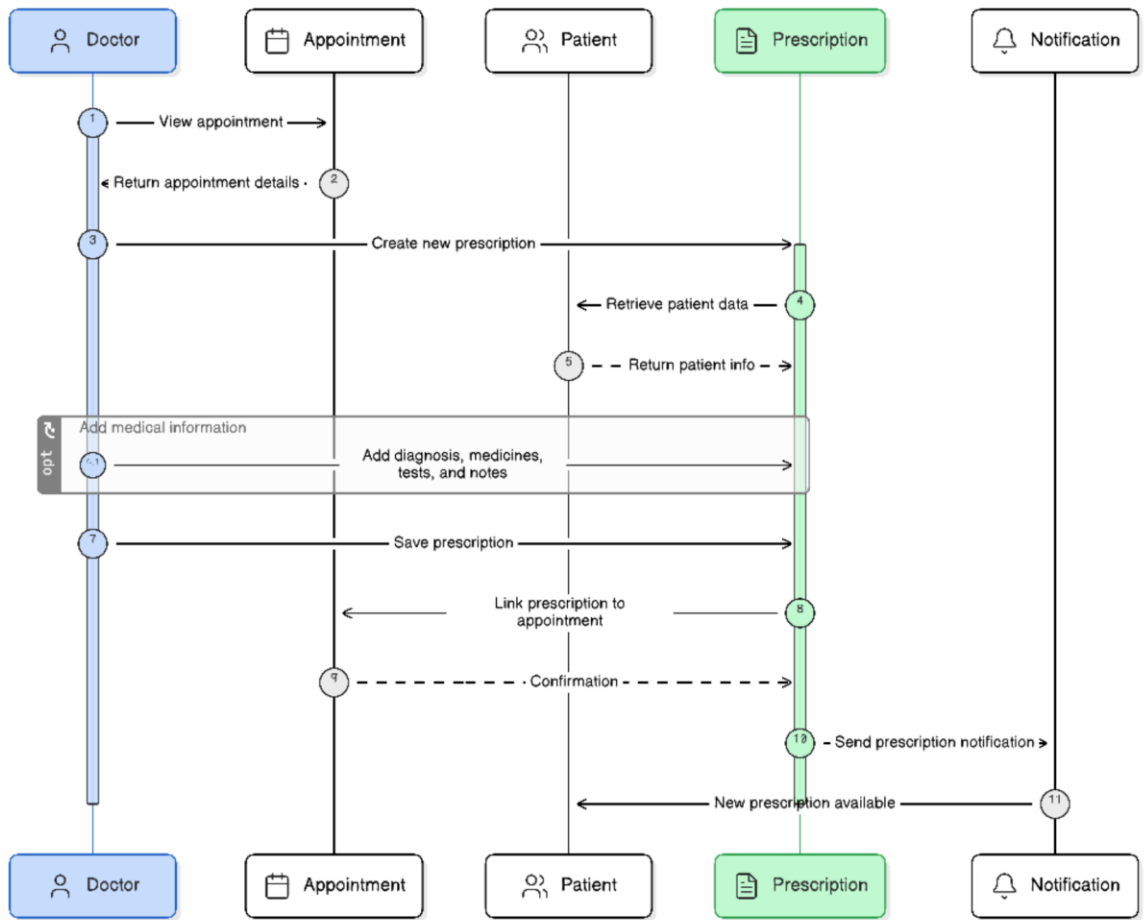


Figure 3.6.2: Create Prescription Sequence Diagram

Pharmacy Order Process Sequence Diagram

The following sequence diagram illustrates the complete lifecycle of a medicine ordering process within the Pharmacy module of the Healthcare Platform. It covers all stages starting from product search, order creation, payment processing, pharmacy handling, and final delivery to the patient.

Participants:

- Patient: The user who searches for medicines and places an order.
 - Pharmacy System: Handles medicine search, stock validation, order processing, and status management.
 - Stock Database: Stores real-time information about medicine availability and inventory levels.
 - Payment Gateway: Processes online payments when prepayment is selected.
 - Pharmacy: Responsible for accepting, preparing, and packaging the order.
 - Delivery Personnel: Handles delivery of the order to the patient.
 - Notification Service: Sends updates regarding order status changes.
-

Flow of Interactions:

1. Search for Medicines

- Patient → Pharmacy System: Searches for required medicines.
- Pharmacy System → Stock Database: Queries medicine availability and stock status.
- Stock Database → Pharmacy System: Returns available medicines and quantities.
- Pharmacy System → Patient: Displays available products.

2. Create Order

- Patient → Pharmacy System: Selects medicines and creates an order request.
- Pharmacy System → Stock Database: Validates stock availability for selected items.
- Stock Database → Pharmacy System: Confirms or rejects availability.
- Pharmacy System → Patient: Displays order summary and confirmation details.

3. Payment Process (Alternative Flow)

Alt [Online Payment]

- Patient → Payment Gateway: Initiates online payment.
- Payment Gateway → Pharmacy System: Confirms successful transaction.
- Pharmacy System → Patient: Updates order status as "Paid" and confirmed.

Alt [Cash on Delivery]

- Pharmacy System → Patient: Marks order as "Pending Payment" to be collected upon delivery.

4. Order Confirmation

- Pharmacy System → Pharmacy: Sends confirmed order details for preparation.
- Pharmacy → Pharmacy System: Accepts the order and updates status to "Preparing."

5. Delivery Assignment

- Pharmacy → Delivery Personnel: Assigns order for delivery.
- Delivery Personnel → Pharmacy System: Updates status to "On the Way."
- Delivery Personnel → Patient: Delivers the order.

6. Order Completion

- Patient → Pharmacy System: Confirms receipt of order.
- Pharmacy System → Stock Database: Updates inventory after order fulfillment.
- Pharmacy System → Notification Service: Sends order completion notification.

- Notification Service → Patient: Confirms successful delivery.

This sequence ensures a complete and structured workflow for pharmacy operations, including inventory validation, payment handling, order tracking, and final delivery, improving efficiency and reliability of the healthcare platform.

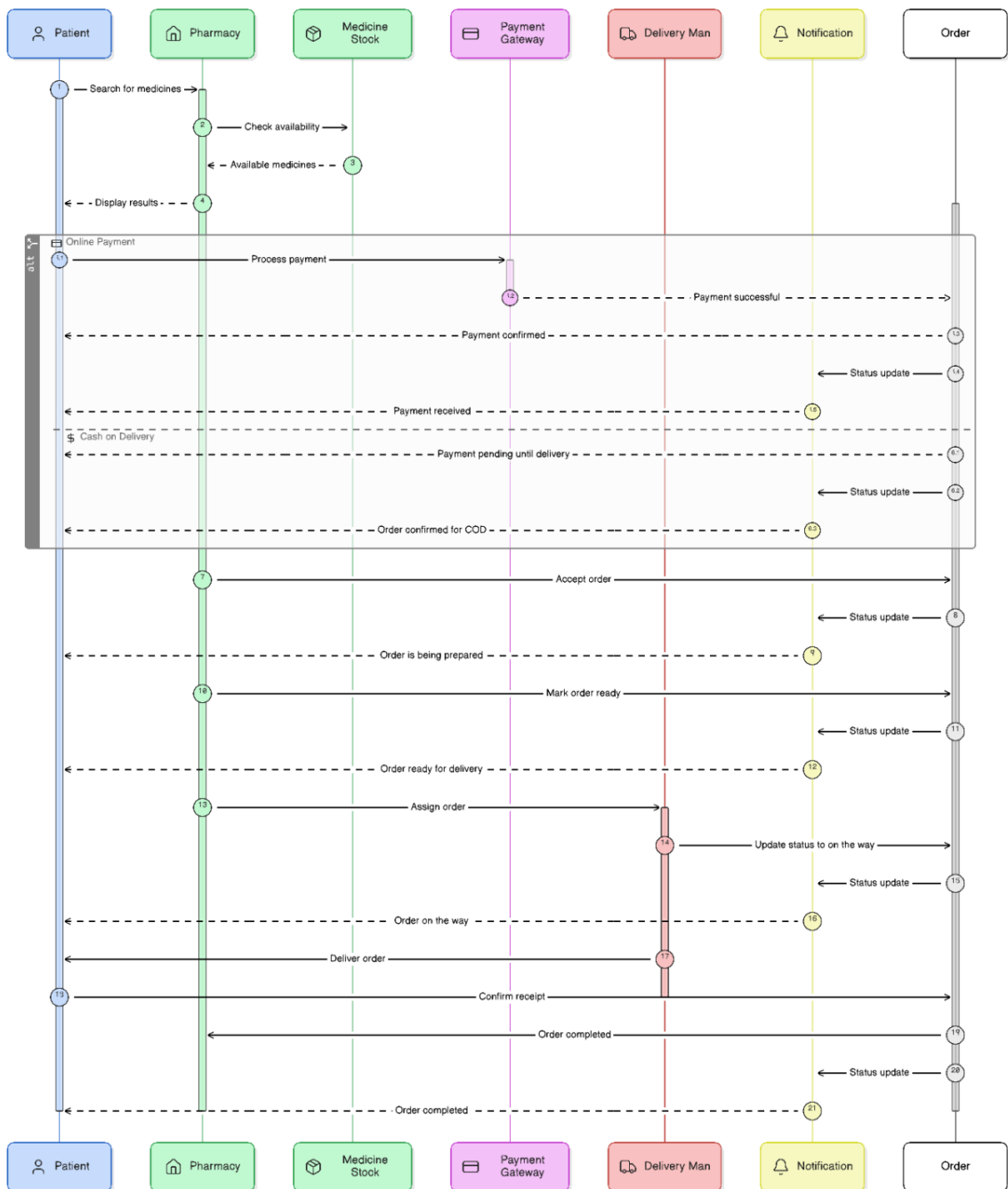


Figure 3.6.3: Pharmacy Order Process Sequence Diagram

Lab Request and Upload Result Sequence Diagram

The following sequence diagram describes the laboratory workflow within the Healthcare Platform. In this process, the laboratory initiates the laboratory test request, notifies the patient through the application, performs the required test, uploads the results, and informs the patient once the results are available.

Participants:

- Patient: Receives notifications about laboratory tests and views the uploaded results.
- Lab System: Manages laboratory requests, test status, result storage, and communication.
- Laboratory: Creates laboratory test requests, performs tests, and uploads results.
- Lab Service Database: Stores laboratory requests, test information, and results.
- Notification Service: Sends notifications to patients through the application.

Flow of Interactions

1. Create Laboratory Test Request

- Laboratory → Lab System: Creates a new laboratory test request for a patient.
- Lab System → Lab Service Database: Stores the laboratory request and its details.
- Lab Service Database → Lab System: Confirms successful storage of the request.

2. Notify Patient

- Lab System → Notification Service: Triggers a notification informing the patient that a laboratory test has been requested.
- Notification Service → Patient: Sends an in-app notification indicating that a laboratory test has been scheduled.

3. Test Processing

- Laboratory → Laboratory: Performs the required laboratory test.
- Laboratory → Lab System: Updates the test status to "In Progress" during processing.

4. Upload Lab Results

- Laboratory → Lab System: Uploads the completed laboratory results and any related attachments.
- Lab System → Lab Service Database: Stores the results securely and links them to the patient's medical record.

5. Notify Patient of Result Availability

- Lab System → Notification Service: Triggers a notification indicating that the laboratory results are available.
- Notification Service → Patient: Sends an in-app notification informing the patient that the test results are ready for viewing.

6. View Results

- Patient → Lab System: Requests to view the laboratory results.
- Lab System → Patient: Displays the complete laboratory report and uploaded test results.

This sequence ensures an efficient laboratory workflow where laboratories initiate test requests, patients receive timely notifications through the application, and laboratory results are securely stored and delivered while maintaining accurate tracking and communication throughout the process.

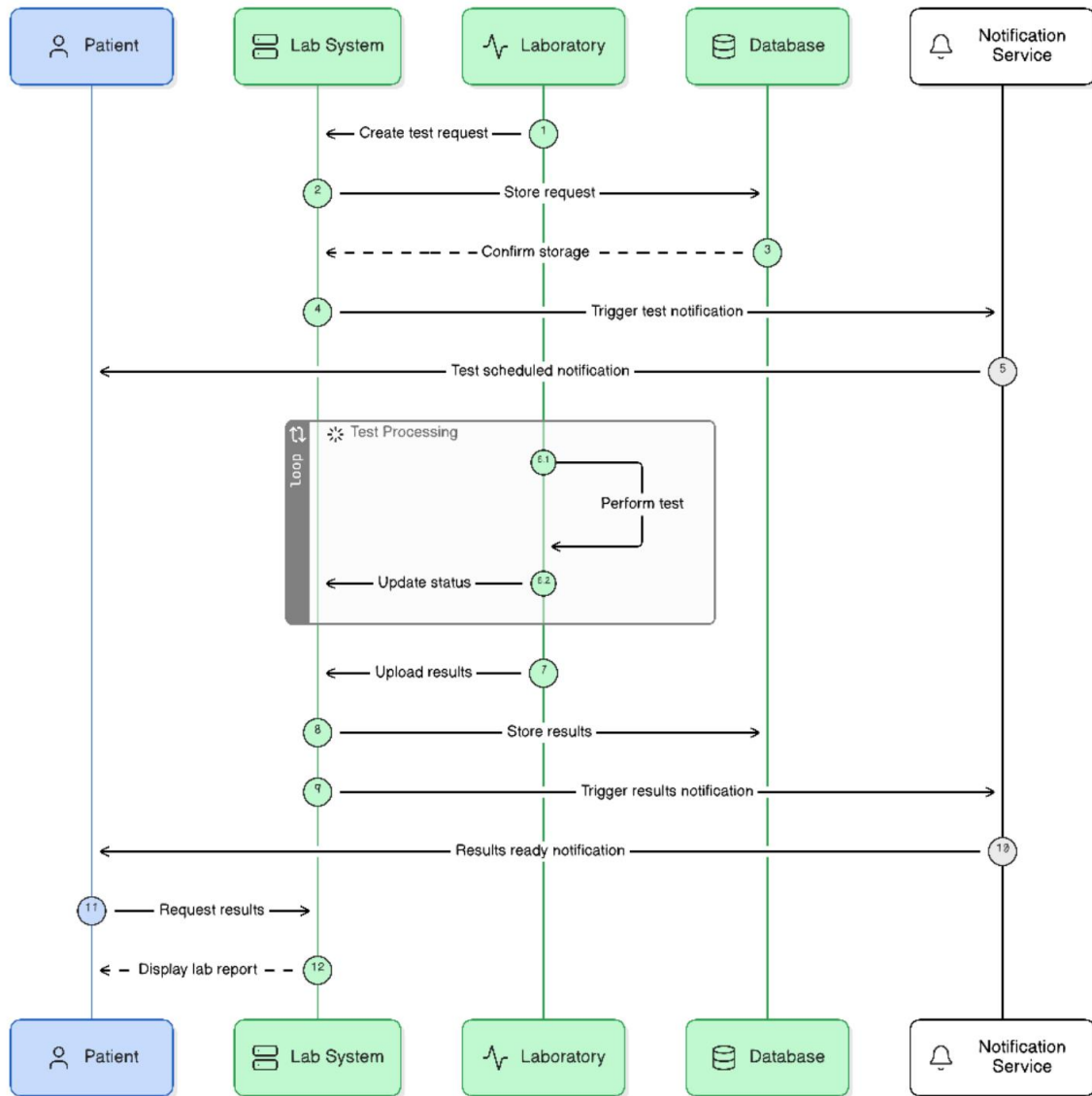


Figure 3.6.4: Lab Request and Upload Result Sequence Diagram

AI Medical Report Analysis Sequence Diagram

The following sequence diagram describes the workflow of the AI-powered medical report analysis feature within the Healthcare Platform. It illustrates how patient-uploaded medical reports are validated, processed using AI models, analyzed, and finally presented to the user along with recommendations.

Participants:

- **Patient:** The user who uploads medical reports for AI-based analysis and receives results.
- **AI Analysis System:** Handles file validation, data extraction, AI processing, and result generation.
- **File Validator:** Ensures uploaded medical files meet format and quality requirements before processing.
- **AI Engine:** Performs intelligent analysis of extracted medical data using trained AI models.
- **Recommendation Engine:** Generates interpretations, insights, and suggestions for relevant medical specialists.
- **Database:** Stores uploaded reports and generated analysis results.
- **Notification Service:** Notifies the patient when analysis is completed.

Flow of Interactions:

1. Upload Medical Report

- Patient → AI Analysis System: Uploads medical report or diagnostic file.
- AI Analysis System → File Validator: Sends file for validation.
- File Validator → AI Analysis System: Confirms whether the file is valid or rejected.

2. Data Extraction and Processing

- AI Analysis System → AI Engine: Sends validated report for processing.

- AI Engine → AI Engine: Extracts relevant medical data and performs AI-based interpretation.
- AI Engine → AI Analysis System: Returns analyzed medical insights.

3. Generate Recommendations

- AI Analysis System → Recommendation Engine: Sends AI analysis results.
- Recommendation Engine → AI Analysis System: Generates possible interpretations and recommended specialist consultations.
- AI Analysis System → Patient: Displays analysis results and suggested medical guidance.

4. Store Results

- AI Analysis System → Database: Stores uploaded reports and generated analysis outputs.
- Database → AI Analysis System: Confirms successful storage.

5. Send Notification

- AI Analysis System → Notification Service: Triggers completion notification.
- Notification Service → Patient: Sends alert that medical report analysis is completed and available.

6. Access Results

- Patient → AI Analysis System: Requests to view previous analysis reports.
- AI Analysis System → Patient: Retrieves and displays stored analysis history.

This sequence demonstrates the integration of artificial intelligence within the healthcare platform to enhance patient support by providing automated medical report interpretation, improving accessibility to insights, and assisting in early medical guidance.

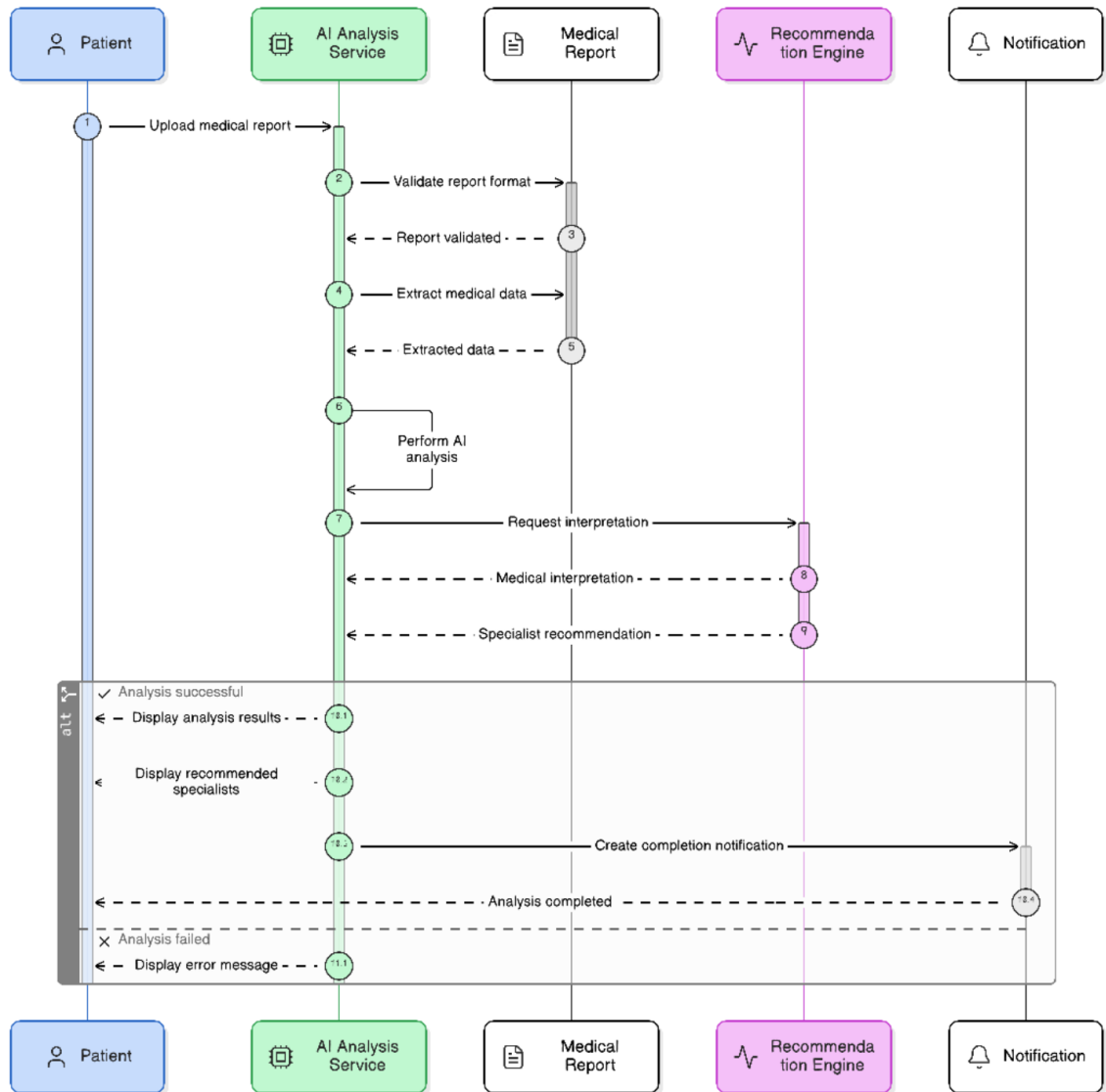


Figure 3.6.5:AI Medical Report Analysis Sequence Diagram

3.6.4 Data Flow Diagram

Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation used to illustrate how data moves throughout a system. It shows the flow of information between external entities, system processes, and data stores. The DFD helps in understanding how the Chefaa Healthcare Platform processes, stores, and exchanges data among different stakeholders, including patients, doctors, pharmacies, laboratories, administrators, and external services.

The Chefaa platform integrates multiple healthcare services within a single ecosystem, enabling appointment management, prescription handling, laboratory services, pharmacy orders, online payments, and AI-powered healthcare assistance. The DFD provides a clear visualization of these interactions and demonstrates how information flows across the system components.

1. Level 0 Data Flow Diagram (Context Diagram)

The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram, presents a high-level view of the Chefaa Healthcare Platform. At this level, the entire platform is represented as a single process that interacts with external entities through data flows. The purpose of this diagram is to define the system boundaries and illustrate how information enters and leaves the platform without showing internal processing details.

Main Process

0.0 Chefaa Healthcare Platform

The Chefaa Healthcare Platform serves as an integrated healthcare ecosystem that connects patients, doctors, pharmacies, laboratories, administrators, and external services. The platform facilitates healthcare service delivery by managing appointments, prescriptions, laboratory requests, medication orders, online payments, notifications, and AI-powered healthcare assistance.

External Entities and Their Roles

1. Patient

The Patient is the primary user of the platform who utilizes healthcare services provided through the system.

Data Flow from Patient to System:

- Appointment booking requests.
- Medication orders.
- Medical reports and laboratory documents.
- Payment information.
- Search and consultation requests.

Data Flow from System to Patient:

- Appointment confirmations.
- Prescription information.
- Laboratory results.
- Order status updates.
- Notifications and reminders.

2. Doctor

Doctors provide healthcare consultations and medical services through the platform.

Data Flow from Doctor to System:

- Profile information.
- Clinic schedules.
- Prescriptions.
- Appointment updates.

Data Flow from System to Doctor:

- Appointment requests.
- Patient information.
- Revenue analytics.
- Notifications and reminders.

3. Pharmacy

Pharmacies manage medication inventories and process patient orders.

Data Flow from Pharmacy to System:

- Inventory information.
- Medicine availability.
- Order processing updates.

Data Flow from System to Pharmacy:

- Medication orders.
- Prescription details.
- Delivery requests.
- Financial reports.

4. Laboratory

Laboratories provide diagnostic and radiology services through the platform.

Data Flow from Laboratory to System:

- Laboratory reports.
- Diagnostic results.
- Service information.

Data Flow from System to Laboratory:

- Laboratory requests.
- Patient information.
- Test requirements.

5. Administrator

The Administrator is responsible for monitoring and managing the entire platform.

Data Flow from Administrator to System:

- User management operations.
- Configuration settings.
- Approval and verification decisions.

Data Flow from System to Administrator:

- Analytics reports.
- System monitoring data.
- Financial reports.

6. Payment Gateway

The Payment Gateway is an external service responsible for processing online payments.

Data Flow from System to Payment Gateway:

- Payment requests.
- Transaction details.

Data Flow from Payment Gateway to System:

- Payment confirmations.
- Transaction status updates.

7. AI Services

AI Services provide intelligent healthcare support and data analysis.

Data Flow from System to AI Services:

- Medical reports.
- Laboratory results.
- User questions and requests.

Data Flow from AI Services to System:

- Analysis results.
- Medical recommendations.
- AI chatbot responses.

8. SMS / Push Notification Service

The Notification Service is responsible for delivering alerts, reminders, and updates to users.

Data Flow from System to Notification Service:

- Appointment reminders.
- Prescription notifications.
- Order updates.

Data Flow from Notification Service to System:

- Delivery status confirmations.

9. Email / Cloud Storage Service

The Email and Cloud Storage Service handles document storage, backups, and email communications.

Data Flow from System to Email / Cloud Storage:

- Medical reports.
- Prescriptions.

- User documents.
- Backup files.

Data Flow from Email / Cloud Storage to System:

- Stored documents.
- Backup data.
- Email delivery confirmations.

The following figure illustrates the Context Diagram (Level 0 DFD) of the Chefaa Healthcare Platform and highlights the interactions between the platform and its external entities.

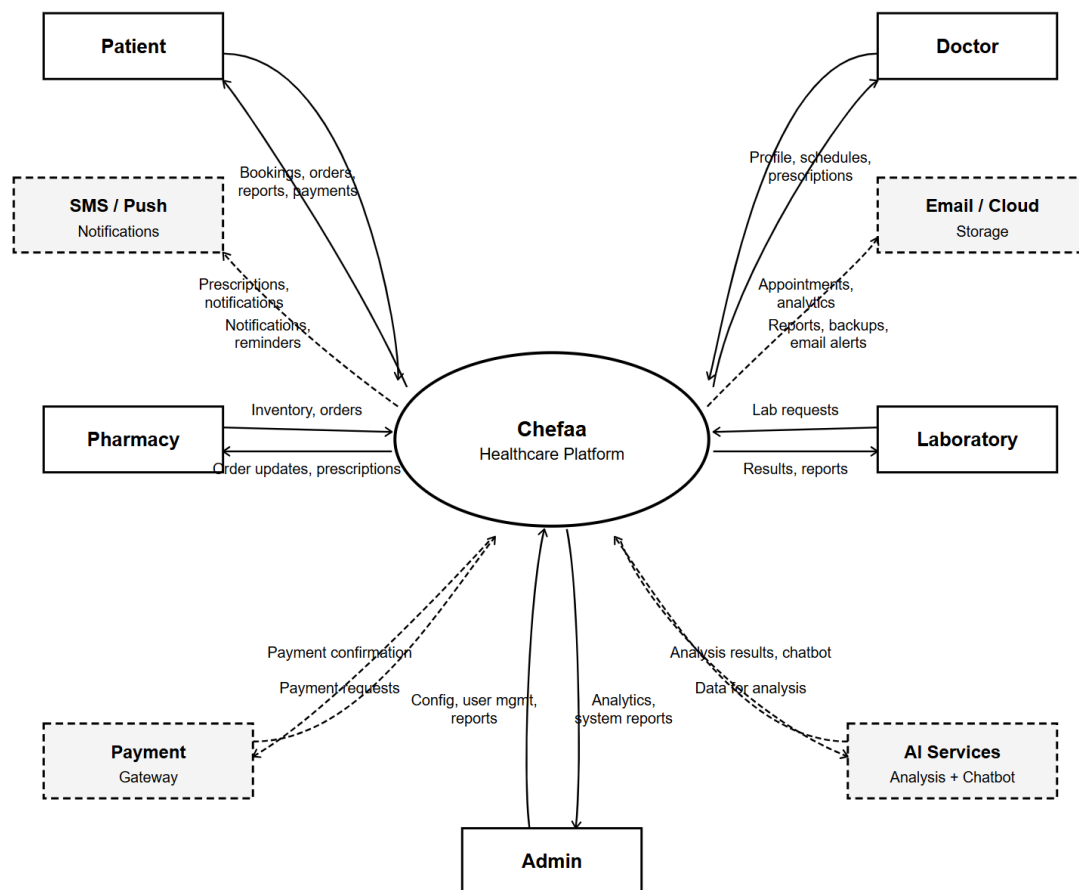


Figure 3.3: Class Diagram of the Chefaa Healthcare Platform

2. Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram (DFD) provides a detailed decomposition of the Healthcare Management System into its major functional processes. Unlike the Context Diagram (Level 0), which represents the entire system as a single process, the Level 1 DFD illustrates the main subsystems, their interactions with external entities, and the data stores used to support system operations.

The system is divided into eight major processes:

P1 – Registration & Verification

This process handles user registration and login activities. Patients can create accounts and authenticate their identities. User information is stored and retrieved from the Users Database (DS1) to ensure secure access to the system.

P2 – Appointment Management

This process enables patients to search for doctors and book appointments. Doctors can manage their available schedules and appointment slots. Appointment information is maintained in the Appointments Database (DS2) to support scheduling and appointment tracking.

P3 – Prescription Management

Doctors create and manage electronic prescriptions for patients through this process. Prescription details are stored in the Prescriptions Database (DS3), while relevant medical information is recorded in the Medical Records Database (DS4). Patients can access their prescriptions through the system.

P4 – Pharmacy Orders

This process allows patients to order prescribed medications from participating pharmacies. Pharmacies receive and process medicine orders, update order statuses, and manage inventory information. Order records are stored in the Orders Database (DS5), while medicine availability is maintained in the Medicine Inventory Database (DS6).

P5 – Lab Result Management

Doctors can submit laboratory test requests, and laboratories can upload test results once analyses are completed. Laboratory requests are stored in the Lab Requests Database (DS7), while notifications related to test results are maintained in the Notifications Database (DS8).

P6 – AI Medical Analysis

This process provides AI-powered medical insights and recommendations. Patients can submit queries or request analysis of medical information. The system communicates with the external AI Analysis Service, which processes the request and returns analytical results to support healthcare decision-making.

P7 – Payment Processing

The Payment Processing module manages financial transactions related to appointments, laboratory services, and pharmacy orders. Payment requests are sent to the Payment Gateway, and transaction confirmations are recorded in the Payments Database (DS9).

P8 – Admin Management

This process supports administrative operations, including reviewing pending requests, approving actions, monitoring system activities, and maintaining audit records. Administrative data is stored and retrieved from the Admin/Audit Log Database (DS10).

The Level 1 DFD demonstrates how the Healthcare Management System integrates multiple healthcare services into a single platform. It illustrates the flow of information between patients, doctors, pharmacies, laboratories, administrators, external services, and internal databases, ensuring efficient management of healthcare operations and patient care.



Figure 3.4: Level 1 Data Flow Diagram of the Chefaa Healthcare Platform

Level 2 Data Flow Diagram (P2 – Appointment Management)

The Level 2 Data Flow Diagram (DFD) provides a detailed breakdown of the Appointment Management process. It illustrates how patients search for doctors, book appointments, how doctors manage appointment schedules, how notifications are generated, and how payment information is recorded. The diagram also shows the interaction between external entities, system processes, and data stores involved in the appointment lifecycle.

The Appointment Management process consists of five main sub-processes:

P2.1 Search Doctors

In this process, the patient submits a search query to find suitable doctors based on criteria such as specialization, location, or availability. The system retrieves the relevant appointment and availability information from the Appointments Database (DS2) and presents the available options. Once the patient selects a suitable time slot, the information is passed to the appointment booking process.

P2.2 Book Appointment

After selecting a doctor and an available slot, the patient submits a booking request. The system creates the appointment record and stores the associated clinic information in the Clinics Database (DS-C). A confirmation message is then sent to the patient, and the booked appointment is forwarded for further management.

P2.3 Manage Appointment

This process allows doctors to manage their appointments by updating schedules, canceling appointments, or rescheduling them when necessary. The system maintains and updates doctor-related information in the Doctors Database (DS-D). Any modifications to the appointment status are recorded and forwarded to the notification process.

P2.4 Track & Notify

Once an appointment status changes, the system generates and manages notifications for the relevant users. Notification records are stored in the Notifications Database (DS8). This

process ensures that patients and doctors are informed about confirmations, cancellations, rescheduling requests, and other appointment updates. If payment is required, the process triggers the payment recording stage.

P2.5 Record Payment

In this process, payment information is exchanged with the Payment Gateway to complete the transaction. The system stores payment records in the Payments Database (DS9) and receives transaction confirmation from the payment service. This ensures that appointment-related payments are securely recorded and verified.

Overall, this Level 2 DFD demonstrates the complete appointment management workflow, beginning with doctor search and appointment booking, followed by appointment management, notification handling, and payment recording. The process ensures efficient coordination between patients, doctors, clinics, and payment services while maintaining accurate and up-to-date records throughout the system.

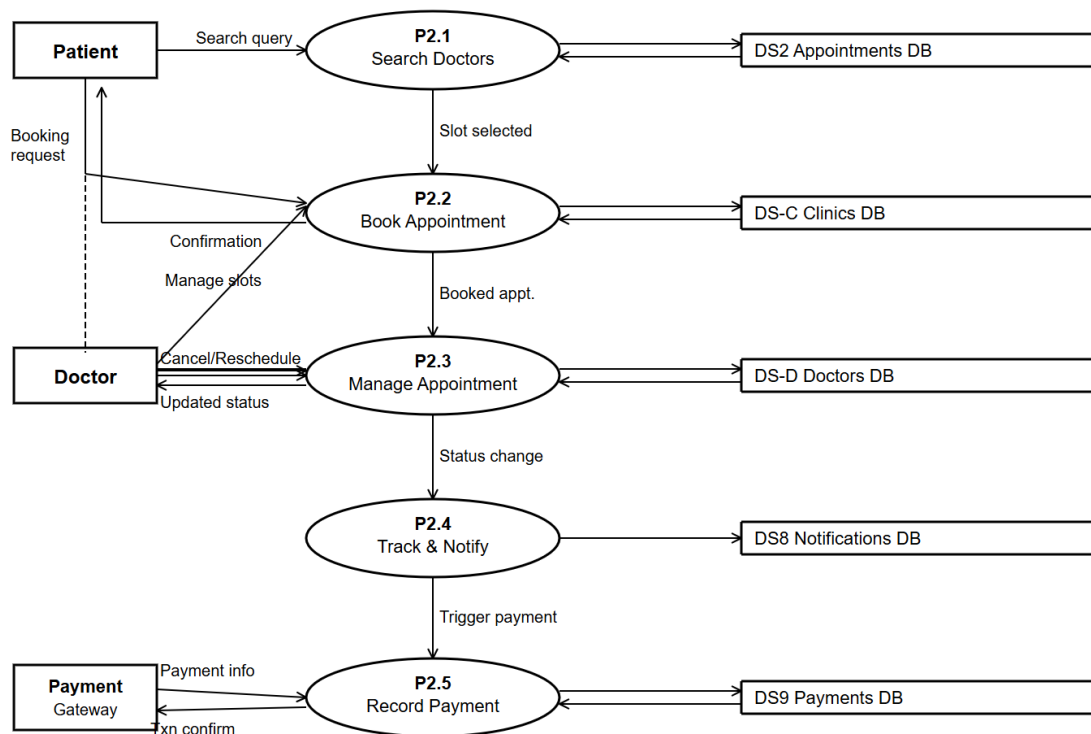


Figure 3.5: Level 2 Data Flow Diagram Appointment Management of the Chefaa Healthcare Platform

Level 2 Data Flow Diagram (P4 – Pharmacy Order Management)

The Level 2 Data Flow Diagram (DFD) provides a more detailed view of the Pharmacy Order Management process. It breaks down the main process into several sub-processes that describe how a patient searches for medicines, places an order, how the pharmacy processes and prepares the order, how delivery is managed, and finally how the payment is completed. The diagram also illustrates the interaction between external entities, internal processes, and data stores used throughout the order lifecycle.

The Pharmacy Order Management process consists of five main sub-processes:

P4.1 Browse Medicines

In this process, the patient searches for available medicines by submitting a search query. The system retrieves medicine information from the Medicine Inventory database (DS6) and displays the available products. Once the patient selects the desired medicine, the information is forwarded to the order creation process.

P4.2 Create Order

After selecting the required medicines, the patient places an order. The system creates a new order record and stores the order details in the Orders Database (DS5). An order confirmation is then provided to the patient, and the order is forwarded for processing.

P4.3 Process Order

The pharmacy receives the order details and verifies medicine availability and order information. The pharmacy prepares and fulfills the order, while the system updates the notification records in the Notifications Database (DS8). Once the order is ready, it is assigned for delivery.

P4.4 Manage Delivery

This process manages the delivery operation. Delivery information is stored and retrieved from the Delivery Database. The pharmacy can track the delivery status, and the system monitors the order until successful delivery is completed.

P4.5 Process Payment

After the delivery process is completed, payment information is exchanged with the Payment Gateway. The system records payment details in the Payments Database (DS9) and receives a transaction confirmation, ensuring that the payment has been successfully processed.

Overall, this Level 2 DFD demonstrates the complete workflow of pharmacy order management, from medicine browsing and order placement to delivery tracking and payment processing, ensuring efficient coordination between patients, pharmacy staff, delivery services, and payment systems.

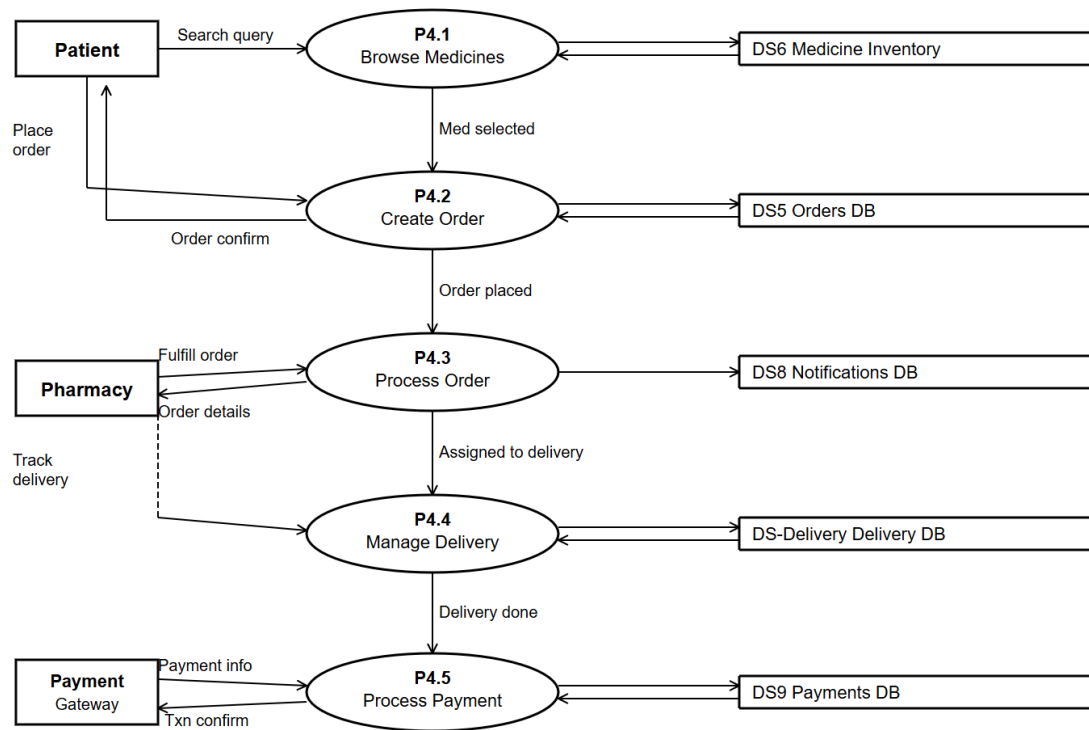


Figure 3.4: Level 2 Data Flow Diagram Pharmacy Order Management of the Chefaa Healthcare Platform.

Level 2 Data Flow Diagram (P5 – Laboratory Services Management)

The Level 2 Data Flow Diagram (DFD) provides a detailed representation of the Laboratory Services Management process. It illustrates how laboratory test requests are received from doctors, how analyses are conducted by laboratory staff, how test results are uploaded and stored, how patients are notified, and how medical records are updated. The diagram also shows the interaction between external entities, internal processes, and data stores involved throughout the laboratory workflow.

The Laboratory Services Management process consists of five main sub-processes:

P5.1 Receive Lab Request

In this process, the staff submits a laboratory test request for a patient. The system records the request details and stores them in the Lab Requests Database (DS7). Once the request is successfully received, it is forwarded to the laboratory analysis process for further action.

P5.2 Conduct Analysis

After receiving the laboratory request, the laboratory performs the required medical tests and analyses. Sample results are processed and associated with the corresponding laboratory services stored in the Lab Services Database (DS-Lab Services). Once the analysis is completed, the generated results are prepared for uploading.

P5.3 Upload Results

In this process, laboratory staff upload the completed test results and supporting documents. The files are stored in cloud storage, and the system receives the generated file URLs for future access. Notification records are also created and stored in the Notifications Database (DS8) to inform relevant users that results are available.

P5.4 Notify & Deliver

Once the laboratory results have been uploaded, the system notifies the patient that the results are ready. Patients can access and view their laboratory reports through the system. At the same time, the results are linked to the patient's information stored in the Medical Records Database (DS4), ensuring secure and organized access to medical data.

P5.5 Update Medical Record

After successful delivery and notification, the patient's medical record is updated with the newly generated laboratory results. This ensures that the patient's health history remains complete and up-to-date, allowing healthcare providers to access accurate information for future consultations and treatments.

Overall, this Level 2 DFD demonstrates the complete laboratory services workflow, starting from the doctor's laboratory request and ending with the integration of laboratory results into the patient's medical record. The process ensures efficient communication between doctors, laboratory staff, patients, cloud storage services, and healthcare databases while maintaining accurate and accessible medical information.

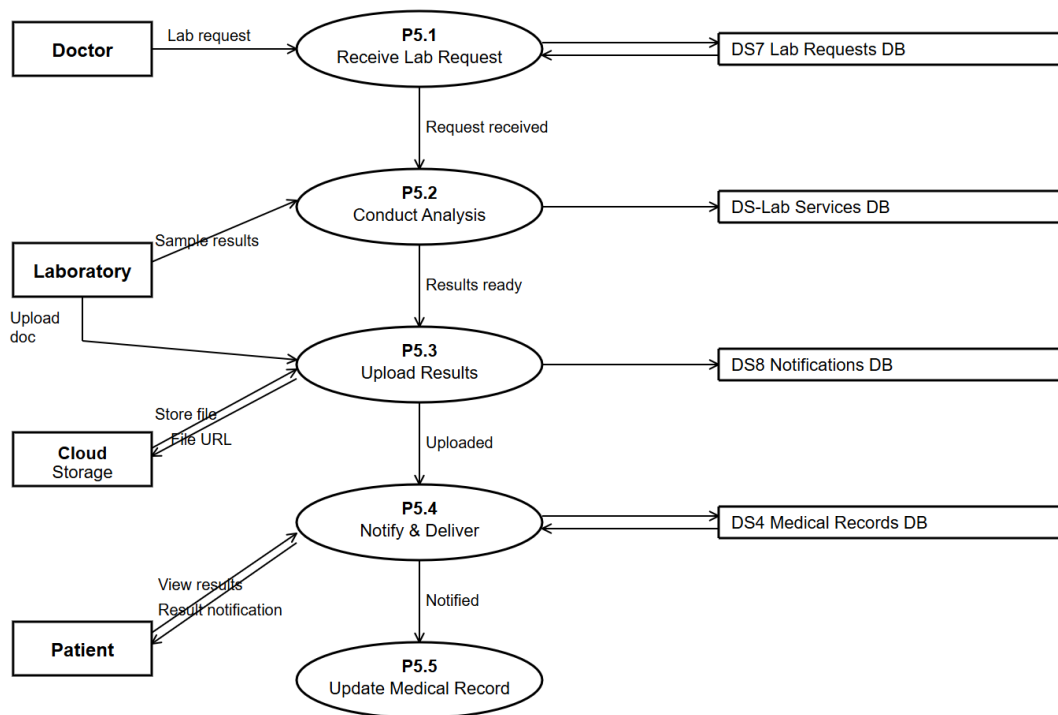


Figure 3.4: Level 2 Data Flow Diagram Laboratory Services Management of the Chefaa Healthcare Platform.

Level 2 Data Flow Diagram (P6 – AI-Based Medical Report Analysis & Specialist Recommendation)

The Level 2 Data Flow Diagram (DFD) provides a detailed representation of the AI-Based Medical Report Analysis and Specialist Recommendation process. It illustrates how patients upload medical reports, how reports are validated and processed by the AI engine, how analytical insights are generated and stored, how suitable specialists are recommended, and how patients are notified of the final results. The diagram also shows the interaction between external entities, internal processes, cloud storage services, AI services, and healthcare databases involved throughout the workflow.

The AI-Based Medical Report Analysis and Specialist Recommendation process consists of six main sub-processes:

P6.1 Upload Report

In this process, the patient uploads a medical report, scan, or laboratory document through the system. The uploaded file is stored and linked to the patient's information in the Medical Records Database (DS4). Once the file is successfully received, the system confirms the upload and forwards the report to the validation process for further processing.

P6.2 Validate & Store

After receiving the uploaded report, the system validates the file format, completeness, and required metadata. Valid reports are stored and prepared for AI processing. Relevant report information is also recorded in the AI Insights Database (DS-AI Insights) to support future analysis and tracking. Once validation is completed, the report is forwarded to the AI analysis process.

P6.3 Send to AI Engine

In this process, the validated report data is sent to the AI Analysis Service for automated examination. Supporting files may be stored in cloud storage, and document URLs are generated for secure access. The AI service receives the report data and performs advanced analysis to identify medical findings, patterns, and potential health concerns. The generated analysis results are then returned to the system.

P6.4ReceiveAIAnalysis

Once the AI engine completes its processing, the system receives the generated analytical insights and recommendations. The process retrieves relevant specialist information from the Doctors Database (DS-Doctors) and matches the patient's medical condition with the most appropriate healthcare professionals. The resulting insights and doctor matches are prepared for recommendation delivery.

P6.5RecommendSpecialist

Based on the AI-generated analysis and matching results, the system recommends the most suitable medical specialist for the patient. The recommendation process may interact with the AI Chatbot Service to provide additional explanations, answer patient inquiries, and present personalized guidance regarding the recommended specialist and medical findings.

P6.6NotifyPatient

After the specialist recommendation is generated, the system notifies the patient that the analysis results and recommendations are available. Notification records are stored in the Notifications Database (DS8), ensuring that patients can securely access their analysis reports, AI insights, and specialist recommendations through the platform.

Overall, this Level 2 DFD demonstrates the complete AI-powered medical report analysis workflow, starting from the patient's report upload and ending with specialist recommendation and patient notification. The process ensures efficient integration between patients, AI services, cloud storage, chatbot services, healthcare databases, and notification systems while providing accurate medical insights and personalized healthcare recommendations.

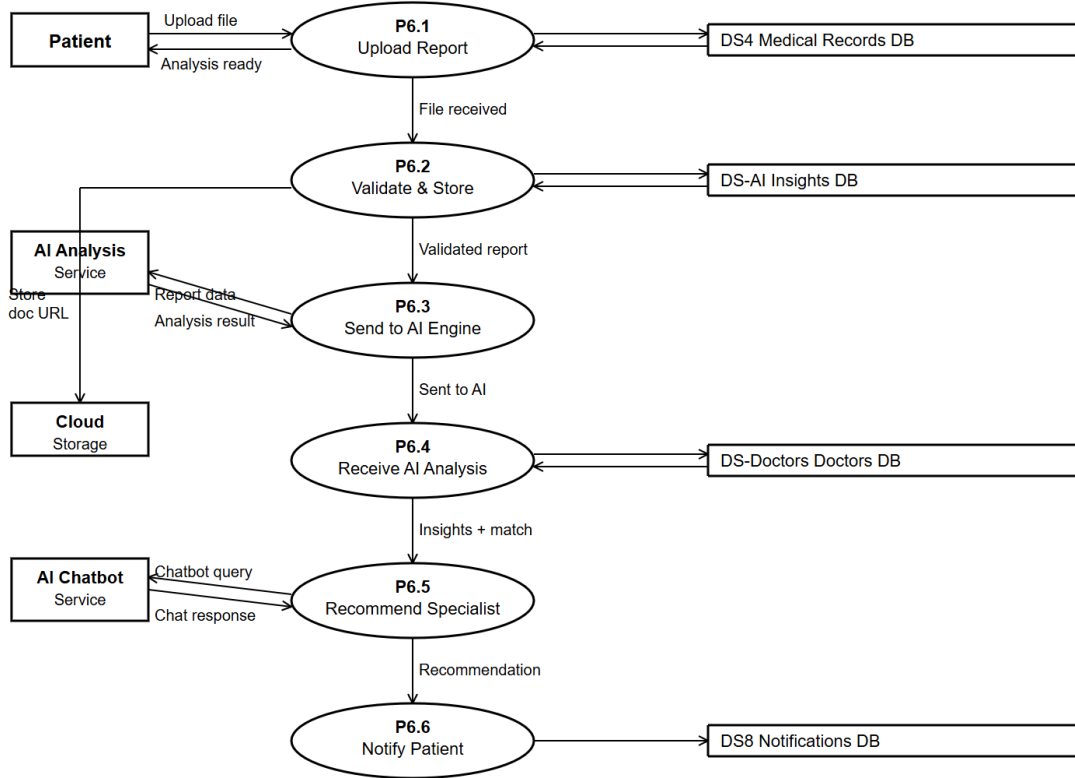


Figure 3.4: Level 2 Data Flow Diagram AI-Based Medical Report Analysis & Specialist Recommendation of the Chefaa Healthcare Platform.

3.6.5 ERD (Entity Relationship diagram)

The Entity-Relationship Diagram (ERD) is a visual representation of the system's data model. It defines the entities used in the Healthcare Management System, their attributes, and the relationships between them. This diagram helps design the underlying database schema and ensures proper data integrity, consistency, and organization across all system modules.

1. Entity Identification

Based on the system's main functionalities, the following entities have been identified:

- User
- Doctor
- Patient
- Clinic
- Appointment
- MedicalRecord
- MedicalRecordAttachment
- MedicalRecordPrescription
- Review
- ClinicScheduledDay
- ClinicScheduleBreak
- PatientMedication
- PatientMedicationAdherence
- Lab
- LabRequest
- LabService
- LabAddress
- Transaction
- Pharmacy
- PharmacyAddress
- PharmacyWorkingHour

- PharmacyCityDeliveryPrice
- DeliveryMan
- Order
- OrderItem
- OrderStatusHistory
- MedicineStock
- Prescription
- PrescriptionFile
- PrescriptionMedicine
- Notification
- RefreshToken
- PasswordReset
- MonthlyPayment
- BillingRecord
- Platform Setting

2. Relationship Identification

Relationships between the entities are identified based on system operations and user interactions:

User – Doctor (One-to-One)

Each doctor account is associated with one user account, and each doctor profile belongs to one user.

User – Patient (One-to-One)

Each patient profile is linked to one user account.

User – Lab (One-to-One)

Each laboratory account is associated with one user account.

User – Pharmacy (One-to-One)

Each pharmacy account is linked to one user account.

User – Notification (One-to-Many)

A user can receive multiple notifications, while each notification belongs to one user.

User – RefreshToken (One-to-Many)

A user may have multiple active authentication tokens for different sessions.

User – PasswordReset (One-to-Many)

A user can request multiple password reset operations over time.

Doctor – Clinic (One-to-Many)

A doctor can manage multiple clinics, while each clinic belongs to one doctor.

Doctor – Appointment (One-to-Many)

A doctor can have many appointments with patients.

Patient – Appointment (One-to-Many)

A patient can book multiple appointments.

Clinic – Appointment (One-to-Many)

A clinic can host multiple appointments.

Appointment – Medical Record (One-to-One)

Each appointment may generate a medical record.

Patient – Medical Record (One-to-Many)

A patient can have multiple medical records throughout their medical history.

Medical Record – Medical Record Attachment (One-to-Many)

A medical record may contain multiple attached files and reports.

Medical Record – Medical Record Prescription (One-to-Many)

A medical record can include multiple prescribed medications.

Patient – Review (One-to-Many)

A patient can submit multiple reviews.

Doctor – Review (One-to-Many)

A doctor can receive multiple reviews from patients.

Clinic – Clinic Scheduled Day (One-to-Many)

A clinic can have multiple scheduled working days.

Clinic Scheduled Day – Clinic Schedule Break (One-to-Many)

Each scheduled day may contain multiple break periods.

Patient – Patient Medication (One-to-Many)

A patient can have multiple medications assigned.

Patient Medication – Patient Medication Adherence (One-to-Many)

Medication adherence records track a patient's compliance with medication schedules.

Lab – Lab Request (One-to-Many)

A laboratory can process multiple lab requests.

Lab – Service (One-to-Many)

A laboratory can offer multiple medical services.

Lab – Lab Address (One-to-Many)

A laboratory may have one or more registered addresses.

Lab Request – Service (Many-to-Many)

A lab request may include multiple services, and each service may be included in multiple requests.

Pharmacy – Pharmacy Address (One-to-Many)

A pharmacy can have multiple registered addresses.

Pharmacy – Pharmacy Working Hour (One-to-Many)

A pharmacy can define multiple working-hour records.

Pharmacy – Delivery Man (One-to-Many)

A pharmacy can manage multiple delivery personnel.

Pharmacy – Medicine Stock (One-to-Many)

A pharmacy maintains multiple medicine inventory records.

Pharmacy – Order (One-to-Many)

A pharmacy can receive and process multiple orders.

Delivery Man – Order (One-to-Many)

A delivery man can deliver multiple orders.

Order – Order Item (One-to-Many)

An order contains multiple order items.

Order – Order Status History (One-to-Many)

An order can have multiple status updates throughout its lifecycle.

Medicine Stock – Order Item (One-to-Many)

Each order item references a medicine from the pharmacy inventory.

Appointment – Prescription (One-to-One)

An appointment may generate a prescription.

Prescription – Prescription File (One-to-Many)

A prescription can contain multiple attached files or images.

Prescription – Prescription Medicine (One-to-Many)

A prescription can include multiple prescribed medicines.

Prescription – Order (One-to-Many)

A prescription may be used to create one or more pharmacy orders.

Monthly Payment – Billing Record (One-to-Many)

Each monthly payment can generate one or more billing records.

3. Entities and Their Attributes

User

- user_id(PrimaryKey)
- name
- email
- password
- role
- status

Doctor

- doctor_id(PrimaryKey)
- user_id(ForeignKeyreferencingUser)
- name
- status

Patient

- patient_id(PrimaryKey)
- user_id(ForeignKeyreferencingUser)
- name
- status

Clinic

- clinic_id(PrimaryKey)
- doctor_id(ForeignKeyreferencingDoctor)

- name
- status

Appointment

- appointment_id(PrimaryKey)
- patient_id(ForeignKeyreferencingPatient)
- doctor_id(ForeignKeyreferencingDoctor)
- clinic_id(ForeignKeyreferencingClinic)
- status

Medical Record

- medical_record_id(PrimaryKey)
- patient_id(ForeignKeyreferencingPatient)
- doctor_id(ForeignKeyreferencingDoctor)
- status

Review

- review_id(PrimaryKey)
- status

Lab

- lab_id(PrimaryKey)
- user_id(ForeignKeyreferencingUser)
- name
- status

Lab Request

- request_id(PrimaryKey)
- patient_id(ForeignKeyreferencingPatient)
- lab_id(ForeignKeyreferencingLab)
- status

Service

- service_id(PrimaryKey)
- lab_id (Foreign Key referencing Lab)

Pharmacy

- pharmacy_id(PrimaryKey)
- user_id(ForeignKeyreferencingUser)
- name
- status

Order

- order_id(PrimaryKey)
- pharmacy_id(ForeignKeyreferencingPharmacy)
- user_id(ForeignKeyreferencingUser)
- delivery_man_id(ForeignKeyreferencingDeliveryMan)
- prescription_id(ForeignKeyreferencingPrescription)
- status

Prescription

- prescription_id(PrimaryKey)
- appointment_id(ForeignKeyreferencingAppointment)
- status

Notification

- notification_id(PrimaryKey)
- user_id (Foreign Key referencing User)

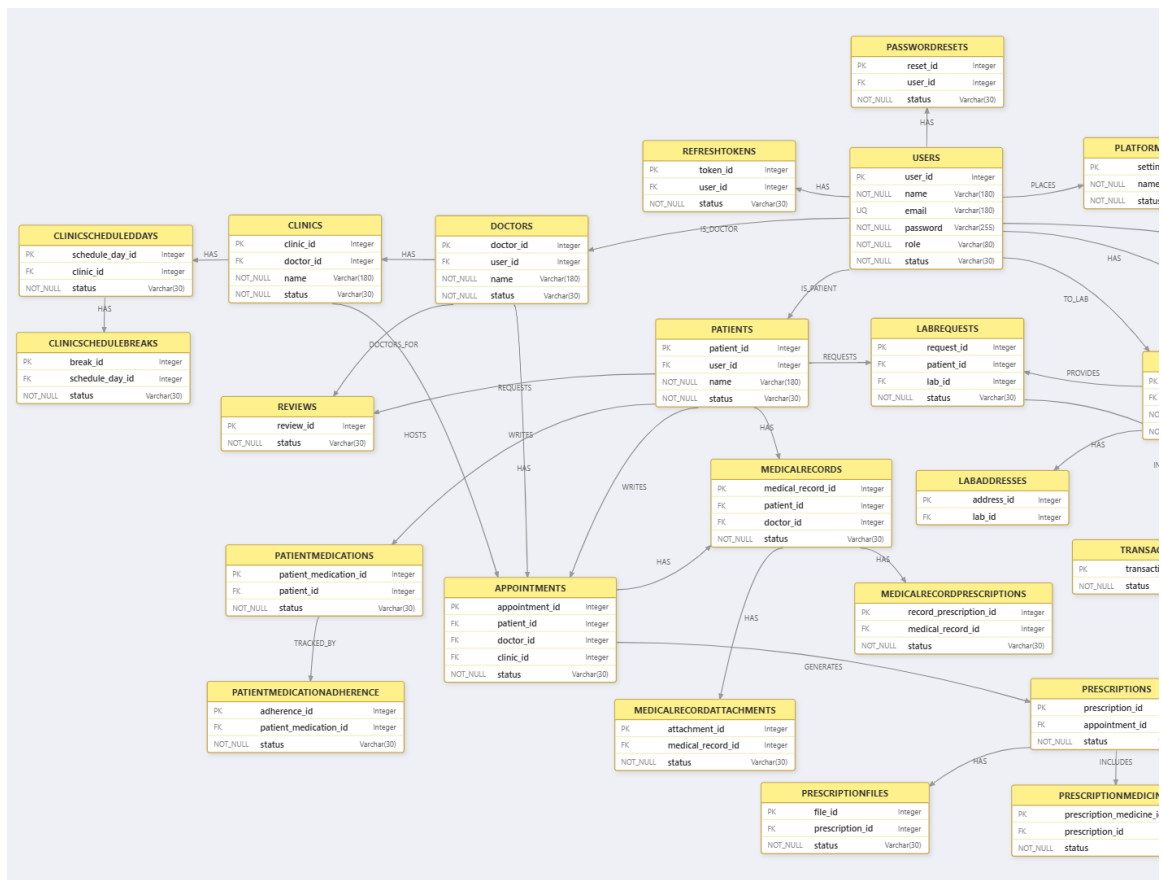
Refresh Token

- token_id(PrimaryKey)
- user_id(ForeignKeyreferencingUser)
- status

Password Reset

- reset_id(PrimaryKey)
- user_id(ForeignKeyreferencingUser)
- status

Additional entities such as MedicalRecordAttachment, PatientMedication, PharmacyAddress, MedicineStock, OrderItem, BillingRecord, and others follow the same structure using primary keys and foreign keys to maintain referential integrity throughout the system.



Chapter 4: Business Model

Customer Segments

- **Patients**

Individuals seeking accessible healthcare services through a unified platform. They can book appointments, receive prescriptions, order medications, access laboratory services, and benefit from AI-powered medical assistance. The platform targets patients of all age groups who value convenience and remote healthcare access.

- **Doctors and Healthcare Providers**

General practitioners and specialists looking for efficient tools to manage clinics, schedules, appointments, electronic prescriptions, and financial analytics. The platform helps healthcare providers digitize their daily operations.

- **Pharmacies**

Community and retail pharmacies aiming to expand their reach through online medication ordering, inventory management, delivery services, and digital payment integration.

- **Laboratories and Radiology Centers**

Medical laboratories and diagnostic centers that provide testing and imaging services and require a secure channel to communicate results and interact with patients and physicians.

- **Healthcare Organizations**

Clinics and healthcare institutions interested in improving patient engagement, reducing administrative work, and integrating digital healthcare services.

Value Propositions

- **Integrated Healthcare Platform**

Chefaa provides a complete healthcare ecosystem that connects patients, doctors, pharmacies, laboratories, and administrators within a single platform, simplifying communication and healthcare management.

- **Intelligent Appointment Management**

Patients can search for doctors, view profiles, schedule appointments, and receive notifications, while doctors can organize clinics and manage appointment slots efficiently.

- **Electronic Prescription System**

Doctors can create digital prescriptions containing medications, diagnoses, laboratory tests, imaging requests, and follow-up recommendations, reducing paperwork and improving accessibility.

- **AI Medical Analysis**

Artificial Intelligence assists patients by analyzing laboratory reports and providing explanations and recommendations for suitable specialists.

- **AI Chatbot Assistant**

An intelligent chatbot offers medical guidance, answers healthcare-related questions, and provides daily summaries and insights for doctors and pharmacies.

- **Pharmacy Order Management**

Patients can order medications online, choose delivery or pickup options, track orders, and complete payments electronically.

- **Laboratory Result Management**

Laboratories can upload test results directly to patients through the platform, enabling secure and efficient communication.

- **Financial Analytics and Reporting**

Healthcare providers receive AI-generated statistics, revenue reports, and performance analyses to support decision-making.

Channels

- **Web Application**

A responsive web platform developed using Next.js and React, allowing users to access healthcare services from desktop devices.

- **Mobile Application**

A cross-platform mobile application built using Flutter, providing users with healthcare services anywhere and anytime.

- **Notifications and Emails**

Push notifications and email services are used to inform users about appointments, prescriptions, laboratory results, and order status updates.

- **Social Media Platforms**

Marketing and awareness campaigns are conducted through social media channels to increase user engagement and promote healthcare services.

Customer Relationships

- **AI-Powered Assistance**

Patients and healthcare providers receive continuous support through intelligent chatbots and personalized recommendations.

- **Customer Support Services**

Users can contact the support team through chat, email, or phone for technical assistance and inquiries.

- **Notification System**

The system sends reminders for appointments, order updates, laboratory results, and payment confirmations.

- **Feedback and Rating System**

Patients can evaluate doctors, pharmacies, and laboratories, helping maintain service quality and improving user experience.

Key Resources

- **Software Infrastructure**

The platform consists of web and mobile applications supported by scalable backend servers and cloud databases.

- **Artificial Intelligence Models**

AI services are used for medical report analysis, chatbot functionality, and business analytics.

- **Databases**

MongoDB databases store users, appointments, prescriptions, orders, laboratory results, and financial records.

- **Payment Gateway Integration**

Secure payment services support online transactions for appointments and pharmacy orders.

- **Development Team**

Frontend, backend, mobile, AI engineers, and designers contribute to building and maintaining the platform.

Key Activities

- **Software Development and Maintenance**

Continuous development, testing, and updating of web and mobile applications.

- **AI Model Development**

Improving AI-based healthcare assistance, report analysis, and business intelligence systems.

- **Database and Security Management**

Maintaining secure storage and protecting medical records and user information.

- **Customer Support and Monitoring**

Providing user support and ensuring smooth platform operation.

- **Healthcare Provider Verification**

Validating doctors, pharmacies, and laboratories before granting access to the platform.

Key Partnerships

- **Hospitals and Clinics**

Collaboration with healthcare institutions to expand service coverage.

- **Pharmacies and Laboratories**

Partnerships with pharmacies and diagnostic centers to provide integrated healthcare services.

- **Payment Service Providers**

Integration with secure payment gateways to facilitate online transactions.

- **Cloud Service Providers**

Cloud infrastructure providers support storage, scalability, and system reliability.

- **AI Service Providers**

External AI technologies enhance medical analysis and intelligent assistance capabilities.

Cost Structure

- **Development Costs**

Expenses related to web application, mobile application, backend services, and AI model development.

- **Cloud Hosting Costs**

Database hosting, file storage, and server infrastructure expenses.

- **Operational Costs**

Customer support, maintenance, monitoring, and administrative operations.

- **Marketing Costs**

Digital marketing campaigns and promotional activities.

- **Third-Party Service Costs**

Payment gateways, email services, cloud storage, and AI APIs.

Revenue Streams

- **Commission on Appointments**

A percentage of consultation fees is collected from completed appointments.

- **Pharmacy Order Commissions**

The platform receives a commission from completed medication orders.

- **Premium Subscription Services**

Doctors, pharmacies, and laboratories can subscribe to advanced analytics and premium features.

- **Advertising and Promotion Services**

Healthcare providers can promote their services through featured listings and advertisements.

- **Healthcare Analytics Services**

Advanced AI-powered reports and business intelligence tools can be offered as premium services.

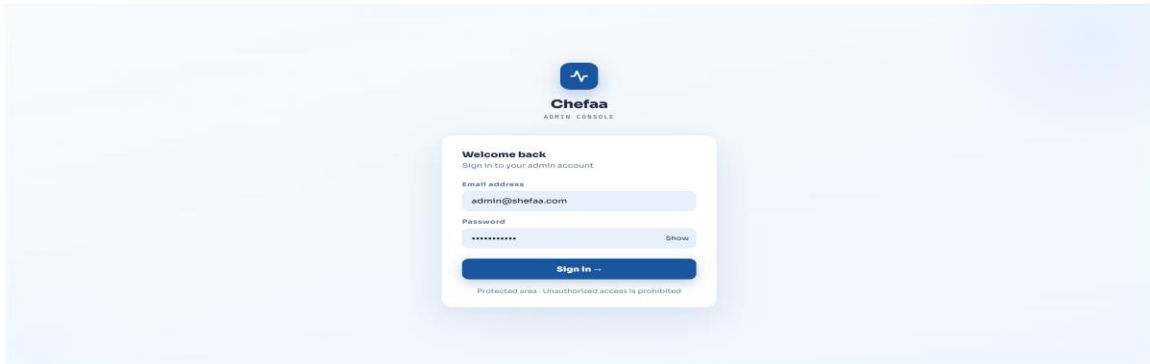
- **Future Partnerships with Healthcare Organizations**

Additional revenue can be generated through collaborations with hospitals, clinics, and insurance providers.

Chapter 5: System Design

<ADMIN>

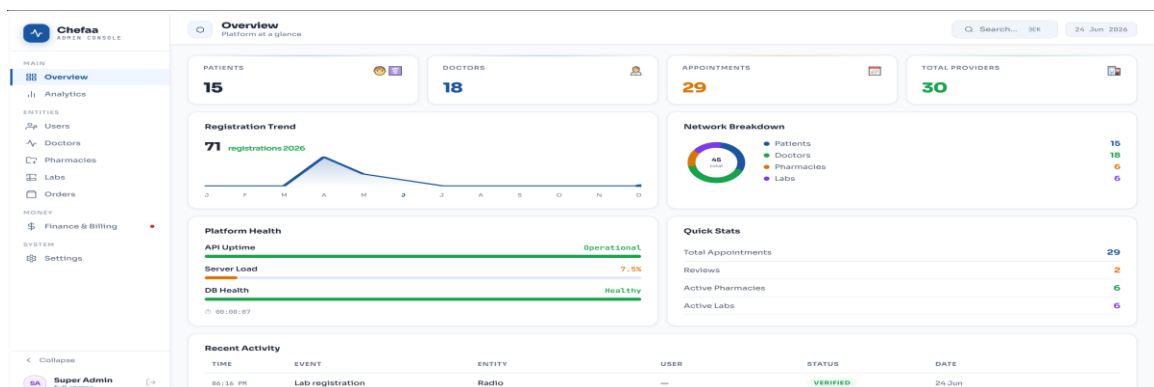
1. Admin Login Page



Description:

This screen represents the administrator login interface of the Chefaa platform. It provides a secure authentication mechanism that allows authorized administrators to access the system dashboard. The login process ensures that only verified users can manage platform operations, maintain security, and monitor healthcare service providers.

2. Admin Dashboard (Overview)

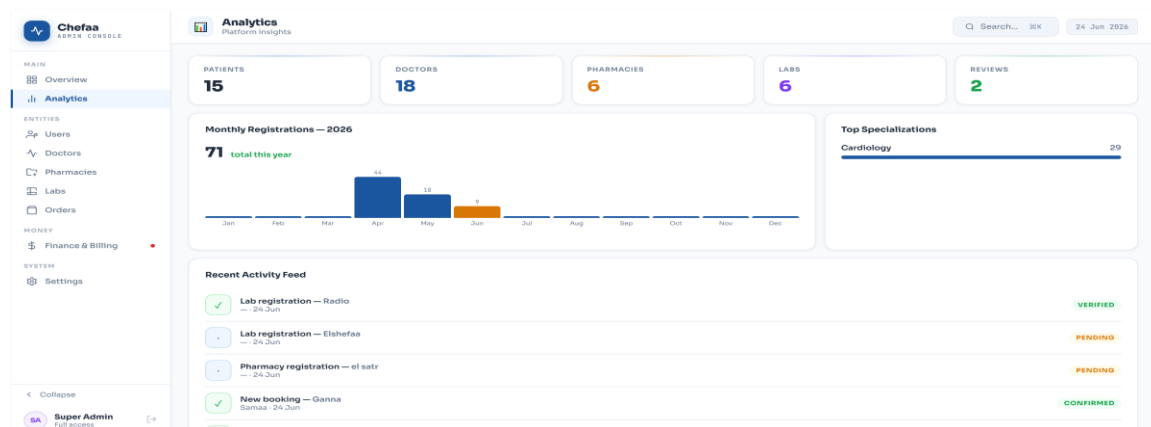


Description:

The Overview Dashboard serves as the central control panel for the administrator. It

presents key platform statistics, including the number of patients, doctors, appointments, pharmacies, and laboratories. Interactive charts and activity logs provide real-time insights into system performance, helping administrators monitor platform growth and operational health efficiently.

3. Analytics Page



Description:

The Analytics module provides comprehensive statistical reports and visual representations of platform activities. Administrators can track registration trends, user growth, provider engagement, and specialty distributions. This feature supports data-driven decision-making and helps evaluate overall system performance.

4. User Management Page

The Users management page displays a list of all registered users. It includes a sidebar with navigation options: MAIN (Overview, Analytics), ENTITIES (Users, Doctors, Pharmacies, Labs, Orders), MONEY (Finance & Billing), and SYSTEM (Settings). The main content area shows a summary of 71 TOTAL USERS, 40 PATIENTS, 18 DOCTORS, 6 PHARMACIES, and 6 LABS. A table lists individual users with columns for NAME, EMAIL, ROLE, STATUS, and ACTIONS. The table is filtered by 'All' and sorted by 'All'. The 'ACTIONS' column includes 'Suspend' and 'Delete' buttons for each user.

NAME	EMAIL	ROLE	STATUS	ACTIONS
Sara Ahmed	sara@example.com	DOCTOR	ACTIVE	Suspend Delete
Ahmed	ahmed2@email.com	PATIENT	ACTIVE	Suspend Delete
Super Admin	admin@shefaa.com	ADMIN	ACTIVE	Suspend Delete
Ali	ali@email.com	PATIENT	ACTIVE	Suspend Delete
grad parm	gradationprot929@gmail.com	PATIENT	ACTIVE	Suspend Delete
abdallah	abdall@e.com	DOCTOR	ACTIVE	Suspend Delete
Ahmed	ahmed@gmail.com	PATIENT	ACTIVE	Suspend Delete
Ali	ali@gmail.com	PATIENT	ACTIVE	Suspend Delete
Amany	amany@gmail.com	PATIENT	ACTIVE	Suspend Delete
Feras	feras@ferrazall.com	PATIENT	ACTIVE	Suspend Delete

Description:

This page enables administrators to manage all registered users within the platform. User information, roles, and account statuses are displayed in an organized table. Administrators can suspend, activate, or review accounts to ensure compliance with platform policies and maintain a secure healthcare environment.

5. Doctor Verification and Management Page

The screenshot displays the 'Doctors' management interface. At the top, a summary bar shows 'TOTAL DOCTORS: 18', 'ACTIVE: 14', and 'PENDING REVIEW: 4'. Below this, a 'Pending doctors verification' section features four cards for doctors whose requests are pending approval. Each card includes the doctor's name, email, specialization, a 'Membership PDF' link, and 'Approve'/'Reject' buttons. The bottom section, 'All Doctors', is a table listing all registered doctors with their contact information, specializations, and current status.

DOCTOR	EMAIL	SPECIALIZATION	STATUS
Sara Ahmed	sara@example.com	Cardiology	ACTIVE
abdallah	abdall@e.com	Ophthalmology	ACTIVE
Ahmed	ahmedd@gmail.com	Gastroenterologist	ACTIVE
esraa ahmed	esraaO@example.com	general	ACTIVE

Description:

The Doctor Management module allows administrators to review healthcare professionals' registration requests. Submitted medical licenses and verification documents are evaluated before approval. This process guarantees that only qualified and licensed doctors can provide medical services through the platform.

6. Pharmacy Verification and Management Page

The screenshot shows the 'Pharmacies' management page. The left sidebar contains navigation links for Overview, Analytics, Users, Doctors, Pharmacies (selected), Labs, Orders, Finance & Billing, and Settings. The main content area has a top summary bar with 'TOTAL PHARMACIES: 6', 'ACTIVE: 5', and 'PENDING REVIEW: 1'. Below this is a 'Pending pharmacies verification' section showing a card for 'el satr' (abduallah@gmail.com) with a 'Medical Licence PDF' and 'Approve'/'Reject' buttons. The bottom section, 'All Pharmacies', is a table with columns: PHARMACY, EMAIL, CITY, STATUS, and ACTIONS. It lists four pharmacies: 'elne3na3a pharmacy', 'chefaa pharmacy', 'elshaimaa pharmacy', and 'El-Ezaby', all with 'ACTIVE' status and a 'Suspend' button.

PHARMACY	EMAIL	CITY	STATUS	ACTIONS
elne3na3a pharmacy	ne3na3a@example.com	—	ACTIVE	Suspend
chefaa pharmacy	chefaa@example.com	—	ACTIVE	Suspend
elshaimaa pharmacy	elshaimaa@example.com	—	ACTIVE	Suspend
El-Ezaby	esaby@gmail.com	—	ACTIVE	Suspend

Description:

This interface is dedicated to pharmacy management and verification. Administrators can review pharmacy licenses, approve registration requests, and monitor active pharmacies. The verification process ensures the authenticity and credibility of pharmaceutical service providers.

7. Laboratory Verification and Management Page

The screenshot shows the 'Labs' management page. The left sidebar is identical to the previous page, with 'Labs' selected. The main content area has a top summary bar with 'TOTAL LABS: 6', 'ACTIVE: 3', and 'PENDING REVIEW: 3'. Below this is a 'Pending labs verification' section showing three cards for 'Al koko', 'el mama', and 'Elshafaa', each with a 'Medical Licence PDF' and 'Approve'/'Reject' buttons. The bottom section, 'All Labs', is a table with columns: LAB, EMAIL, CITY, STATUS, and ACTIONS. It lists four labs: 'Al koko', 'el mama', 'Calro MRI & Lab Center (Updated)', and 'Elite Lab'. The first two are 'PENDING', and the last two are 'ACTIVE'. Each row has an 'Approve' or 'Suspend' button.

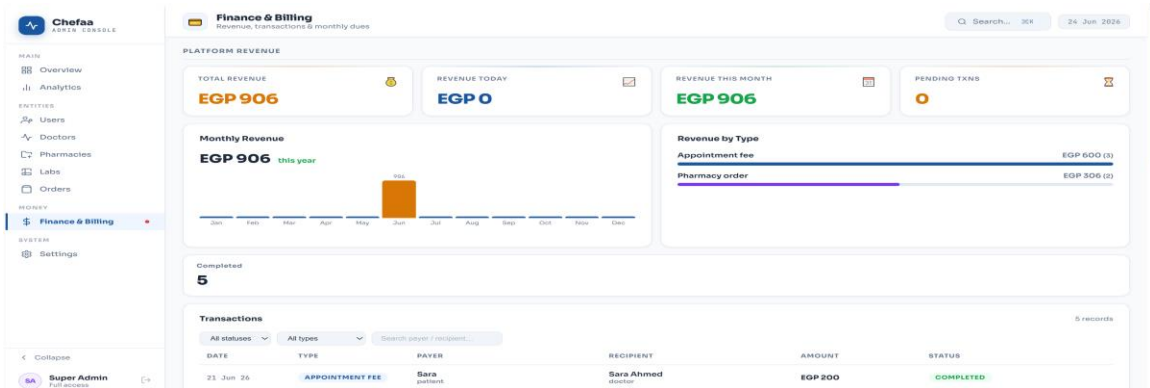
LAB	EMAIL	CITY	STATUS	ACTIONS
Al koko	koko@gmail.com	—	PENDING	Approve
el mama	mama@gmail.com	—	PENDING	Approve
Calro MRI & Lab Center (Updated)	ne3na3a@example.com	—	ACTIVE	Suspend
Elite Lab	abduallah@gmail.com	—	ACTIVE	Suspend

Description:

The Laboratory Management module enables administrators to verify laboratories before granting access to the platform. Uploaded licenses and accreditation documents are

reviewed to ensure compliance with healthcare regulations and maintain service quality standards.

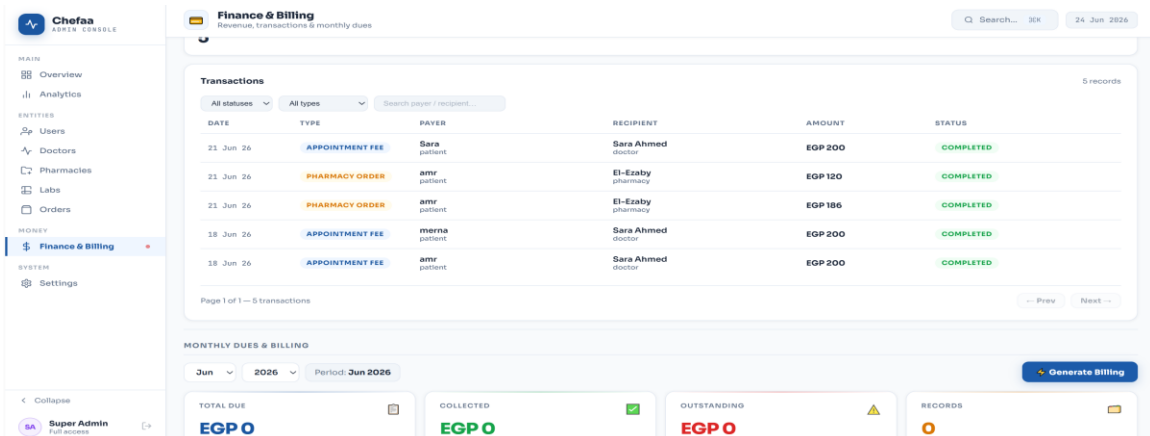
8. Finance and Billing Dashboard



Description:

The Finance and Billing Dashboard provides a detailed overview of platform revenues, completed transactions, and payment distributions. Administrators can monitor financial performance through charts and summaries, allowing effective management of platform earnings and billing activities.

9. Transactions Management

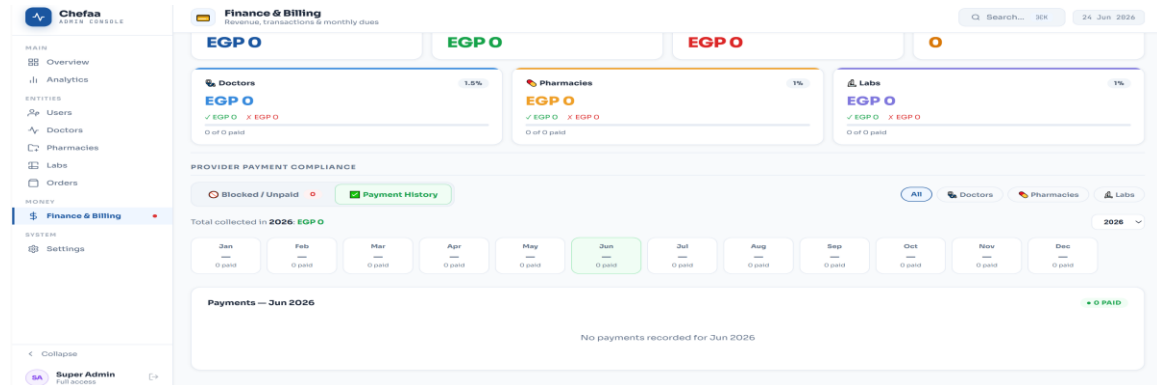


Description:

This screen displays all financial transactions conducted through the platform, including appointment fees and pharmacy orders. The transaction history provides transparency,

allowing administrators to track payments, monitor transaction statuses, and maintain financial accountability.

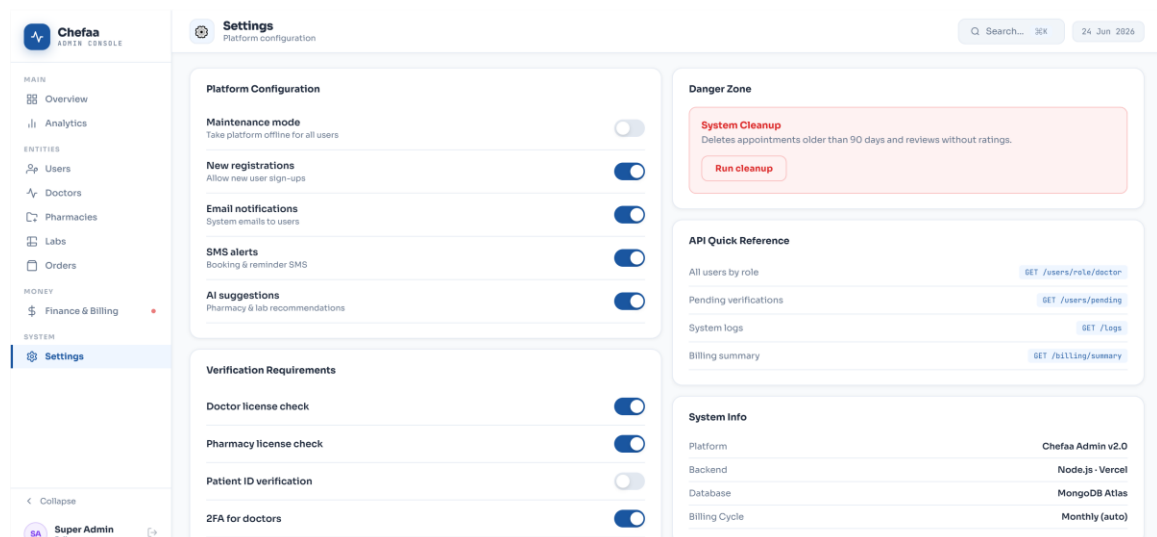
10. Monthly Dues and Provider Compliance& Payment History Monitoring



Description:

This feature provides a historical view of provider payments and billing records. Administrators can review monthly payment statuses, analyze payment trends, and ensure that all healthcare providers fulfill their financial obligations to the platform.

12. System Settings Page



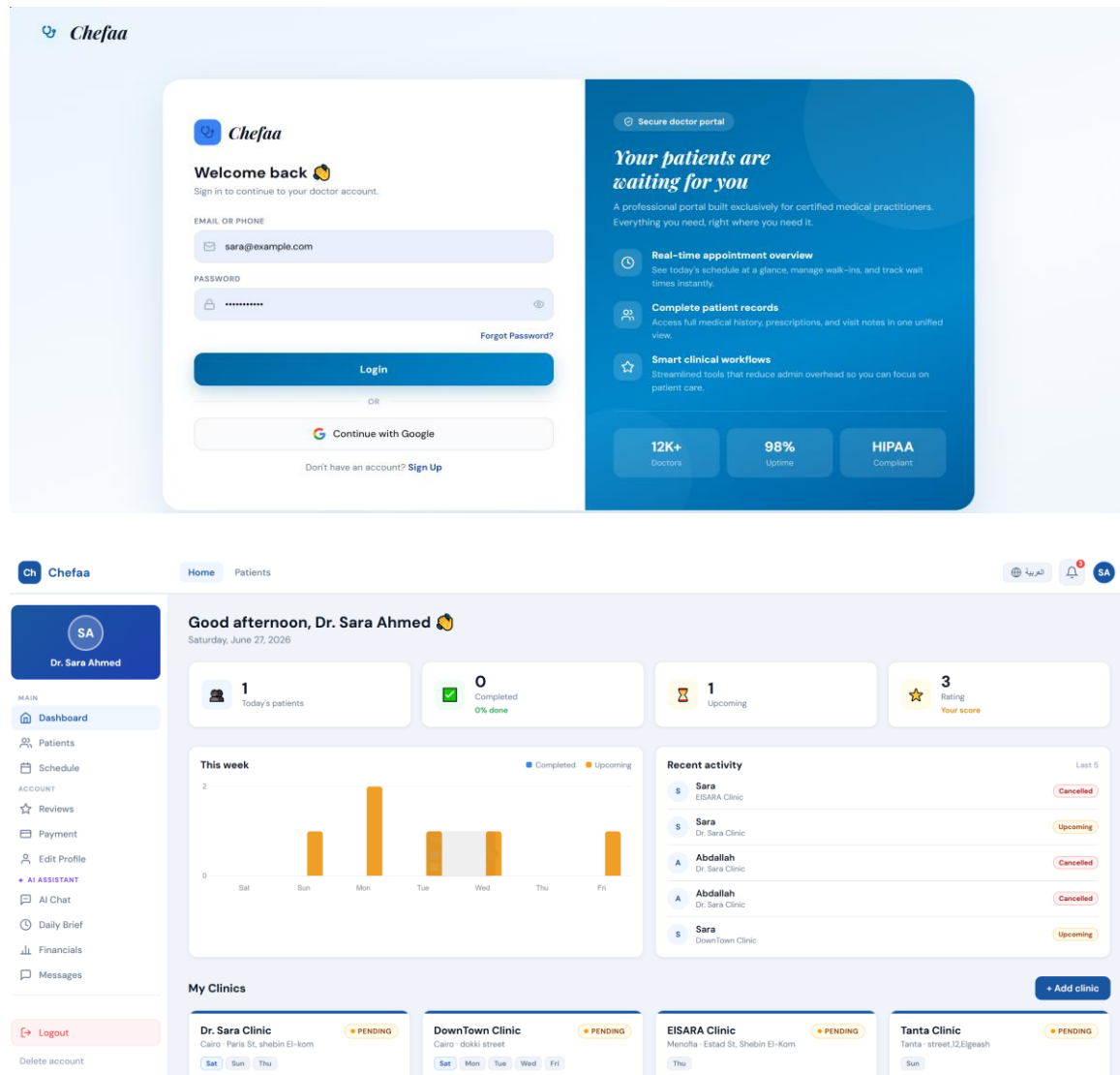
Description:

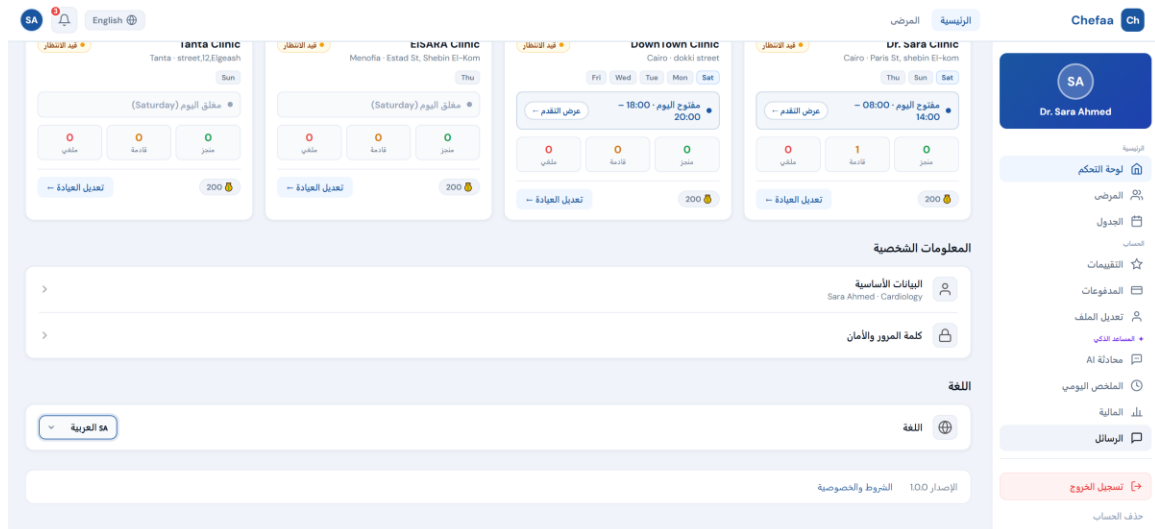
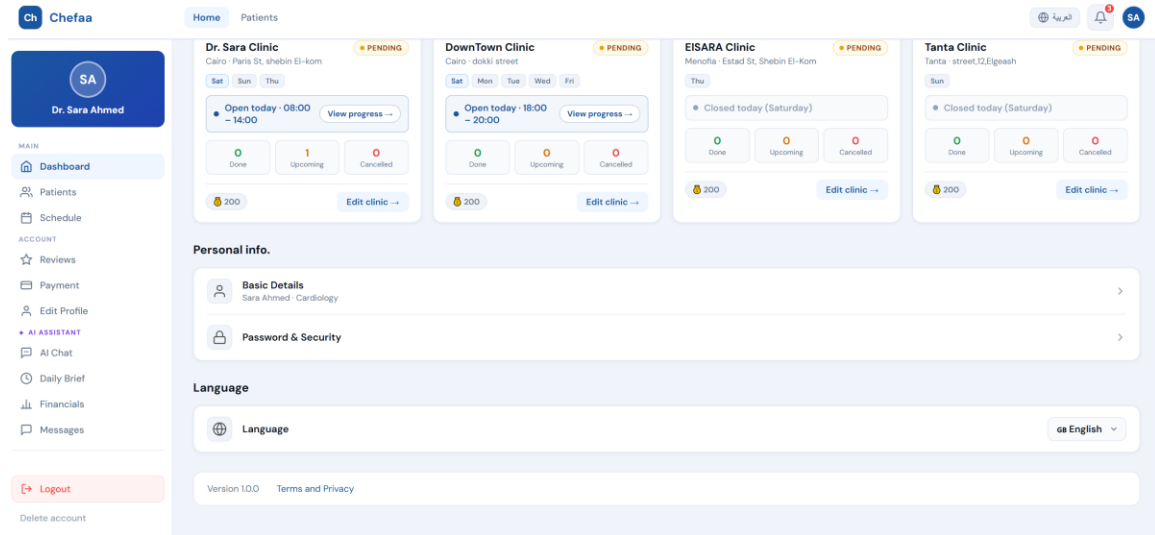
The Settings module allows administrators to configure platform-wide preferences and

operational rules. Features include maintenance mode, registration controls, notification settings, verification requirements, and security configurations. This module ensures flexibility and efficient management of the entire healthcare ecosystem.

<DOCTOR>

Doctor Authentication & Dashboard

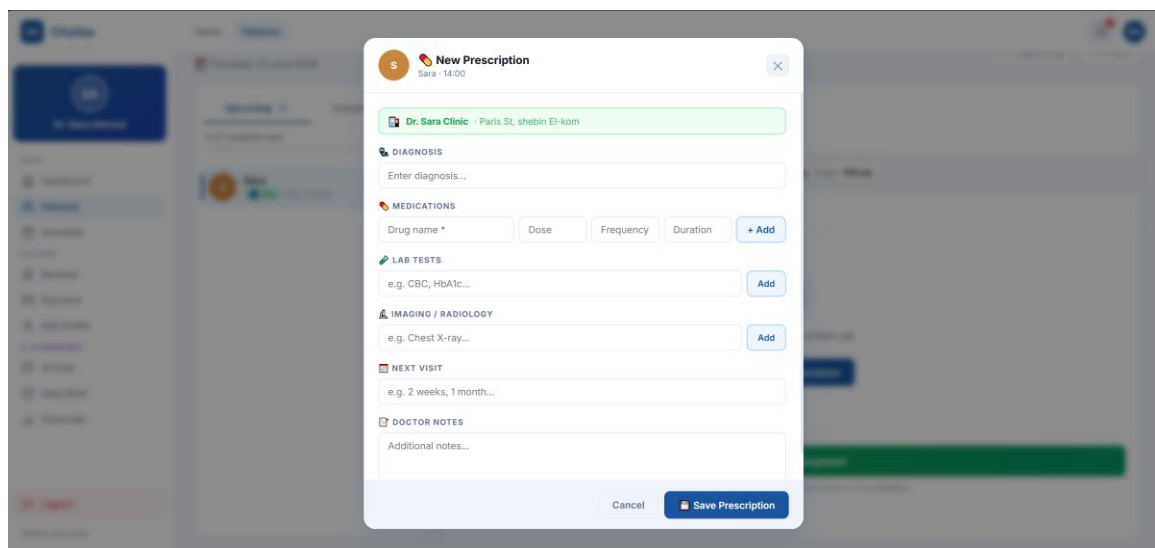
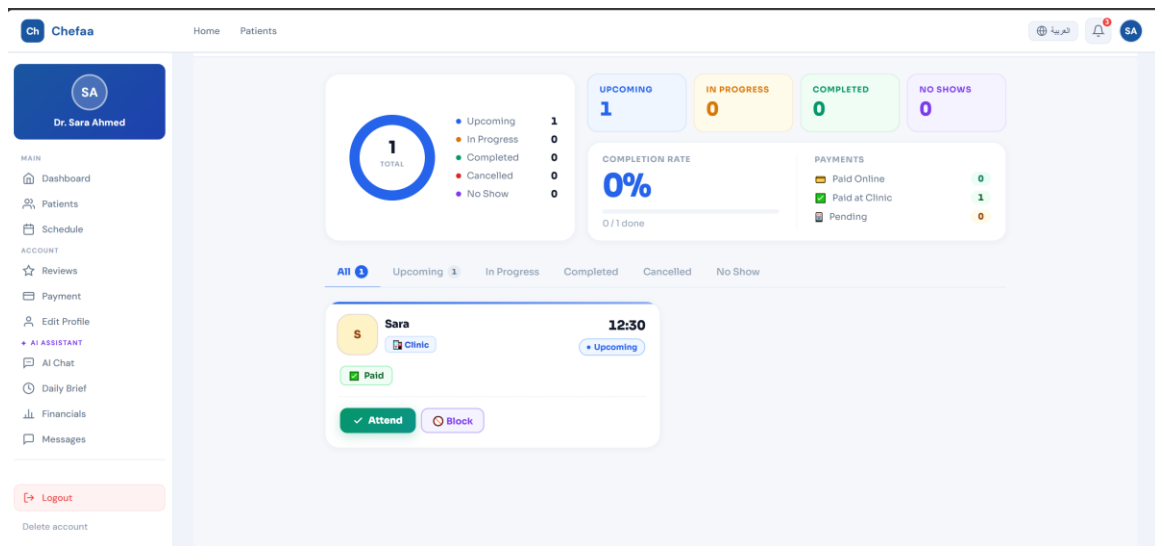




Description:

These screenshots demonstrate the doctor's authentication process and main dashboard. After logging in securely, doctors can access an overview of their clinics, appointments, patient statistics, upcoming schedules, and recent activities. The dashboard serves as a centralized workspace that provides quick access to essential clinical and administrative information. Also supports bilingual display (Arabic / English).

Patient Attendance Management & Consultation Process



Description:

These screenshots illustrate the clinic management and patient consultation workflow. Doctors can monitor appointment statuses, manage patient visits, access medical records, create prescriptions, record diagnoses, request laboratory tests, and document treatment plans. This module streamlines patient care and supports efficient clinical decision-making.

Scheduling & Reviews Management

The image displays three screenshots of the Chefaa app interface, illustrating the Scheduling and Reviews Management modules.

Screenshot 1: My Schedule

The interface shows the user's profile (Dr. Sara Ahmed) and the 'My Schedule' section. The schedule is for Thursday, June 25, 2026. It displays two clinics: Dr. Sara Clinic (OPEN) and DownTown Clinic (CLOSED). Dr. Sara Clinic has 1 booked appointment and 11 open slots. The schedule for Dr. Sara Clinic is as follows:

Time Slot	Status
09:00 - 09:30	Available
09:30 - 10:00	Available
10:00 - 10:30	Available
10:30 - 11:00	Available
11:00 - 11:30	Available
11:30 - 12:00	Available
12:00 - 12:30	Available
12:30 - 13:00	Available

DownTown Clinic is closed today. The interface also shows a 'Today's schedule' section with a 'Closed today' message.

Screenshot 2: Patient Reviews

The interface shows the user's profile (Dr. Sara Ahmed) and the 'Patient Reviews' section. The average rating is 3.0 (Good - 1 review). The rating breakdown is as follows:

Rating	Count
5 stars	0
4 stars	0
3 stars	1
2 stars	0
1 star	0

The review is from Omar Shaban, dated 6 May 2026, with the text: "she is a good doctor." The review is marked as 'OS' (Official) and has a 'Remove' button.

Description:

The Scheduling and Reviews modules enable doctors to manage clinic availability,

appointment slots, and daily schedules across multiple clinics. Additionally, doctors can view patient feedback, ratings, and reviews, helping them evaluate service quality and improve the overall patient experience.

Payments & Financial Transactions

The image displays two screenshots of the Chefaa doctor dashboard, specifically the 'Payments & Financial Transactions' section. The top screenshot shows the 'Revenue Summary' for June 2024, while the bottom screenshot shows the 'Secure Card Payment' form.

Top Screenshot: Revenue Summary

- Header:** Ch Chefaa, Home, Patients, SA (Dr. Sara Ahmed), 31% done.
- Left Sidebar:** MAIN (Dashboard, Patients, Schedule, ACCOUNT (Reviews, Payment, Edit Profile, AI ASSISTANT, AI Chat, Daily Brief, Financials), Logout, Delete account).
- MONTH:** Jan, Feb, Mar, Apr, May, **Jun**, Jul, Aug, Sep, Oct, Nov, Dec.
- JUNE 2024 AT A GLANCE:**
 - Revenue: EGP 800.00
 - Platform Fee: EGP 12.00
 - Net Profit: EGP 788.00
 - Completed: 4
 - Cancelled: 3
 - Upcoming: 6
- COMPLETION RATE:** 31% done, 4 of 13 appointments completed.
- Revenue Summary:**
 - JUNE 2024:** Total Clinic Revenue: EGP 800.00, 4 completed sessions.
 - ONLINE REVENUE:** EGP 0.00
 - AT-CLINIC REVENUE:** EGP 800.00
 - NET PROFIT:** EGP 788.00
- FEE BREAKDOWN - JUNE 2024:**
 - Completed Appointments: 4
 - Total Session Revenue: EGP 800.00
 - Platform Fee (1.5%): - EGP 12.00
 - Net Clinic Earnings: EGP 788.00
- Action:** Pay Platform Fee - EGP 12.00

Bottom Screenshot: Secure Card Payment

- Header:** Ch Chefaa, Home, Patients, SA (Dr. Sara Ahmed), 31% done.
- Left Sidebar:** MAIN (Dashboard, Patients, Schedule, ACCOUNT (Reviews, Payment, Edit Profile, AI ASSISTANT, AI Chat, Daily Brief, Financials), Logout, Delete account).
- COMPLETION RATE:** 31% done, 4 of 13 appointments completed.
- SECURE CARD PAYMENT:**
 - CARDHOLDER NAME:** [Input field]
 - Card Number:** 1234 5678 9012 3456
 - Cardholder Name:** As it appears on your card
 - Expiry Date:** MM/YY
 - CVV:** ***
 - Amount to pay:** EGP 12.00
 - Buttons:** Cancel, Confirm Payment - EGP 12.00

Description:

These screenshots present the payment management functionality available to doctors. The system provides detailed revenue summaries, transaction breakdowns, platform fee

calculations, and secure payment processing. This module ensures financial transparency and allows healthcare providers to monitor earnings effectively.

Doctor Profile Management

Ch

Chefaa

HomePatients

SA

Dr. Sara Ahmed

MAIN

Dashboard

Patients

Schedule

ACCOUNT

Reviews

Payment

Edit Profile

AI ASSISTANT

AI Chat

Daily Brief

Financials

Logout

Delete account

Sara Ahmed

Cardiology

Change photo

BASIC INFORMATION

FULL NAME

Sara Ahmed

SPECIALIZATION

Cardiology

GENDER

MaleFemale

AGE

26

YEARS OF EXPERIENCE

4 yrs

CONTACT NUMBER

0113359506

ABOUT / BIO

Click Click Symphony

DEGREES & QUALIFICATIONS

Menoufia University

e.g. MD – Cairo University

Add

CONSULTATION SETTINGS

CLINIC CONSULTATION PRICE (EGP)

350

PAYMENT OPTION

Pay at clinicPre-payment onlyBoth options

PRE-PAYMENT PHONE NUMBERS

0113359506

+20 1xx xxx xxx

Add

MEMBERSHIP & LICENSE

CLINIC CONSULTATION PRICE (EGP)

350

PAYMENT OPTION

Pay at clinicPre-payment onlyBoth options

PRE-PAYMENT PHONE NUMBERS

0113359506

+20 1xx xxx xxx

Add

MEMBERSHIP & LICENSE

MEMBERSHIP CERTIFICATE

Medical syndicate certificateClick to open in new tab

To update your certificate, please contact support

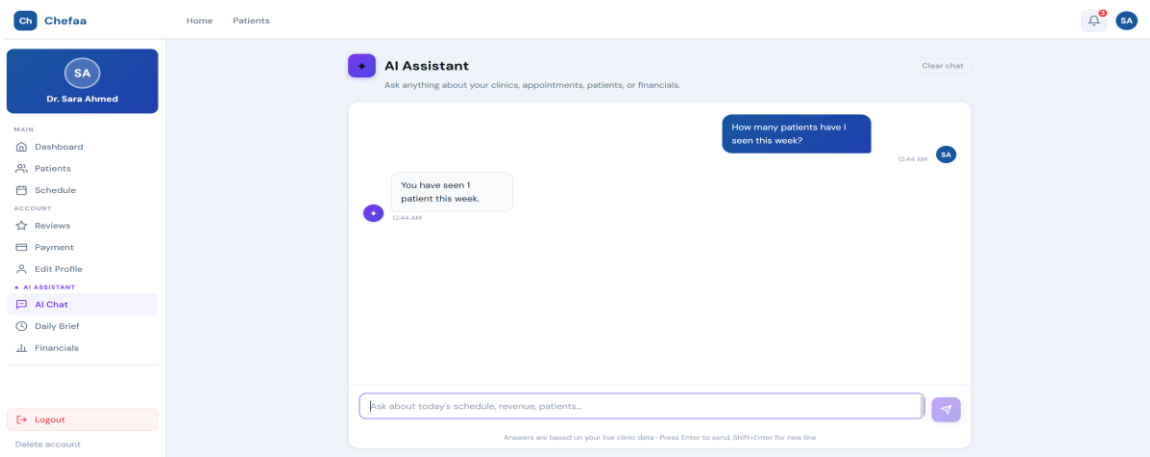
Save changes

102

Description:

The Profile Management module allows doctors to maintain and update their professional information, including specialization, experience, qualifications, consultation fees, payment methods, and licensing details. Keeping profile information updated enhances patient trust and ensures accurate service representation.

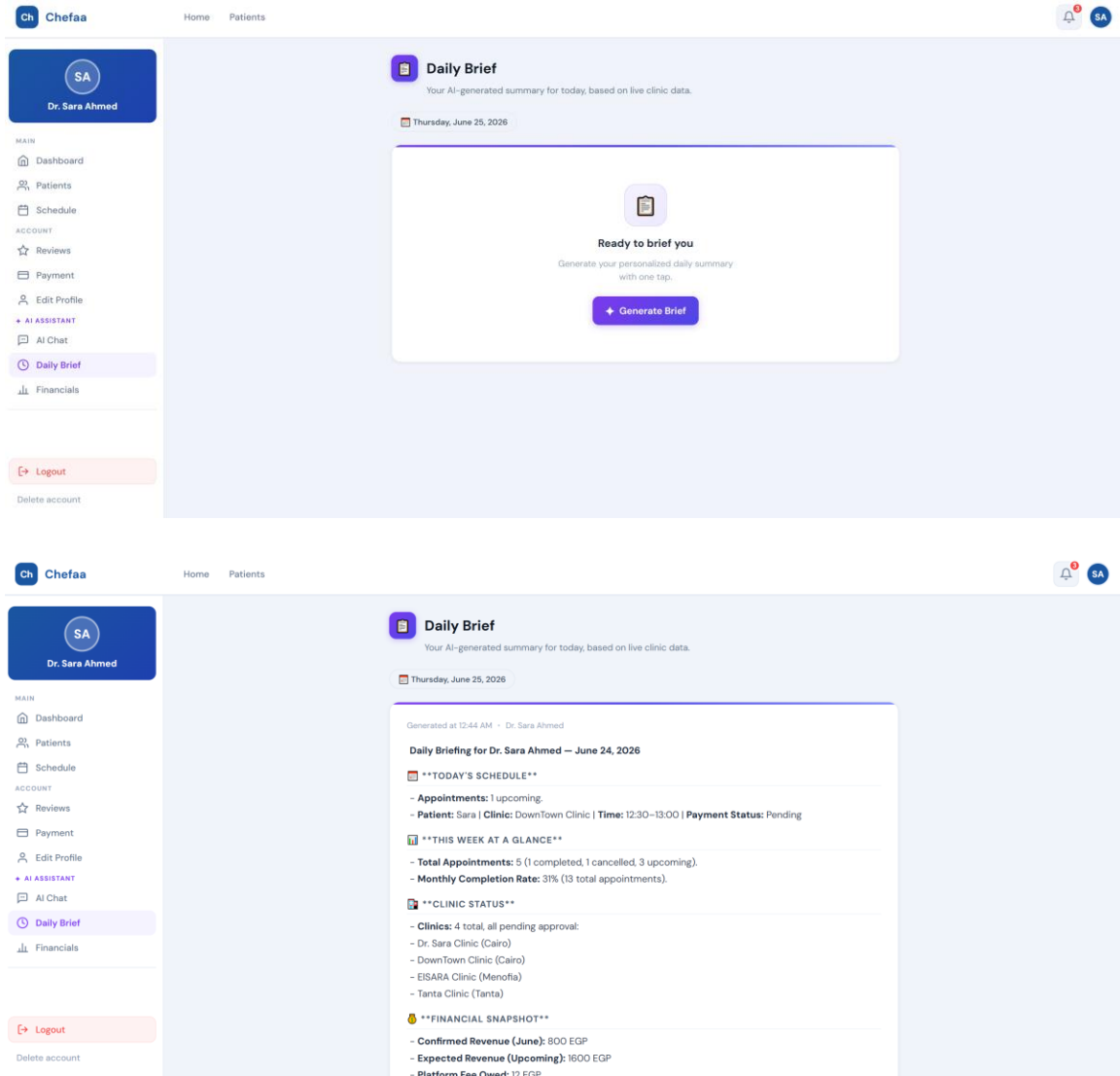
AI Assistant



Description:

The AI Assistant provides intelligent support for doctors by answering questions related to appointments, patients, clinic performance, and financial data. By leveraging artificial intelligence, the assistant helps doctors access information quickly and improve operational efficiency.

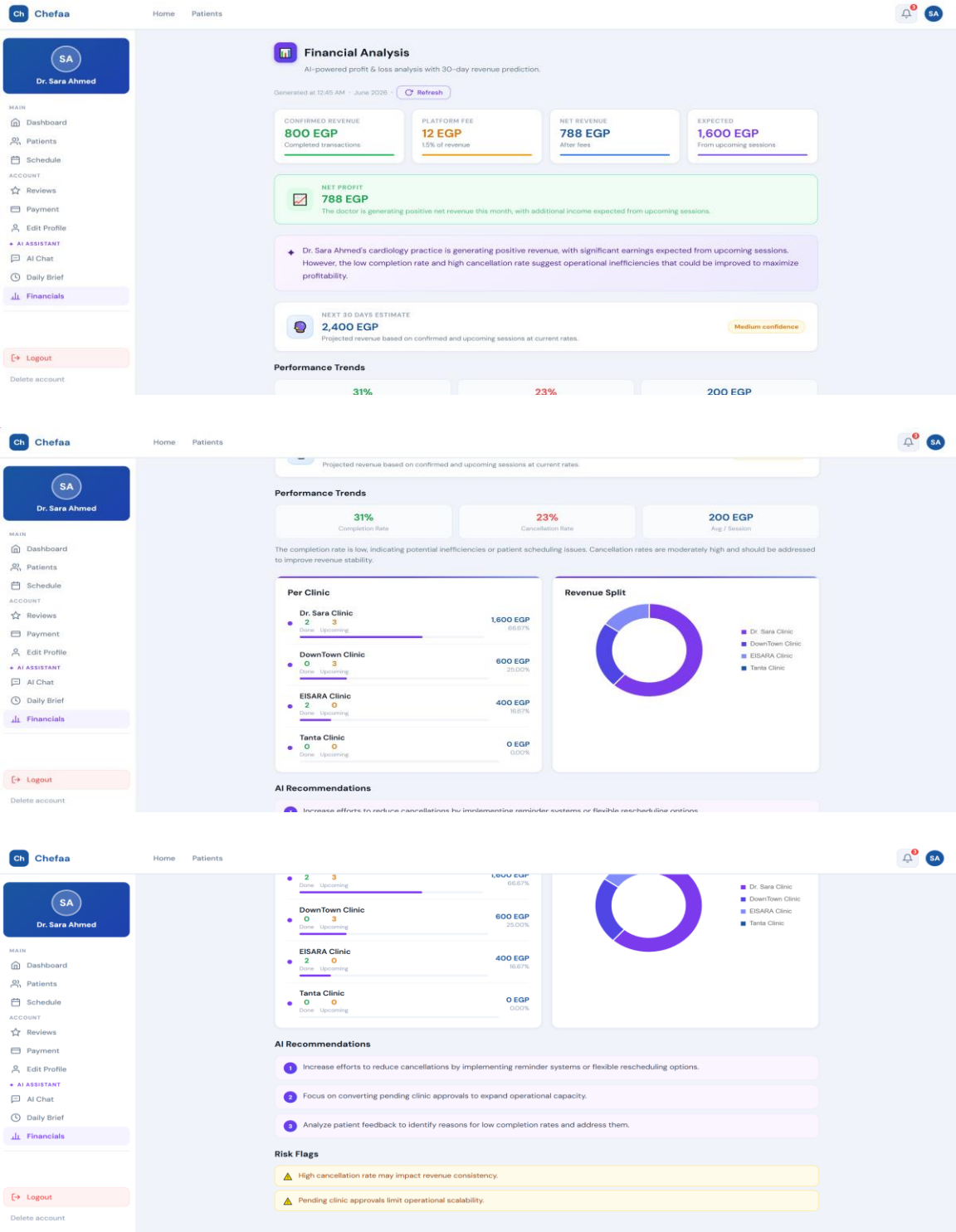
AI Daily Brief



Description:

The AI Daily Brief generates an automated summary of the doctor's daily activities based on real-time clinic data. It includes appointment schedules, patient statistics, clinic performance metrics, and financial highlights, enabling doctors to stay informed and organized throughout the day.

AI Financial Analysis

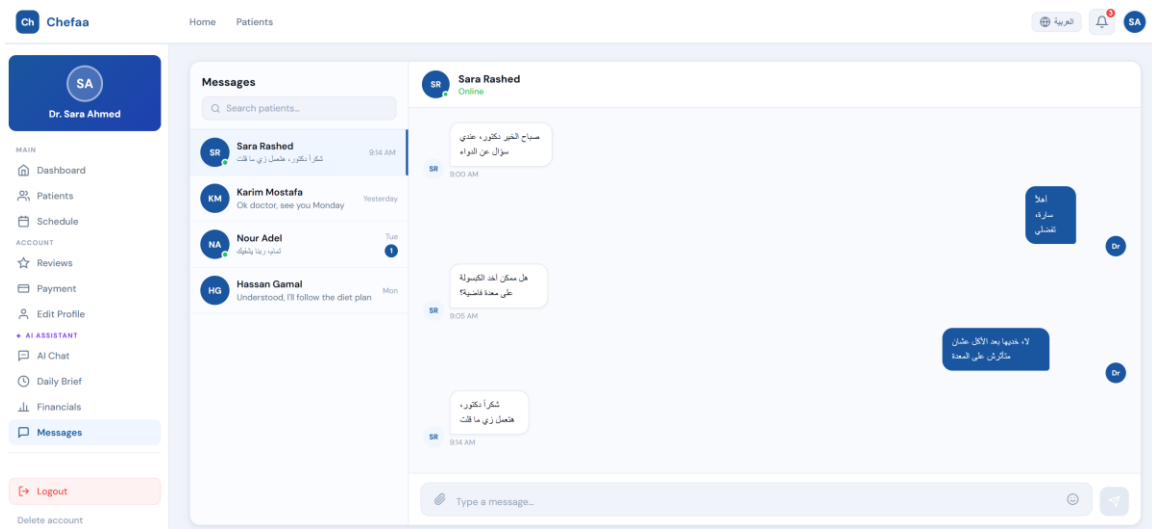


Description:

The AI Financial Analysis module delivers advanced insights into clinic revenues,

profitability, appointment performance, and future income projections. The system analyzes operational data, identifies potential risks, and generates intelligent recommendations to help doctors optimize clinic performance and increase financial efficiency.

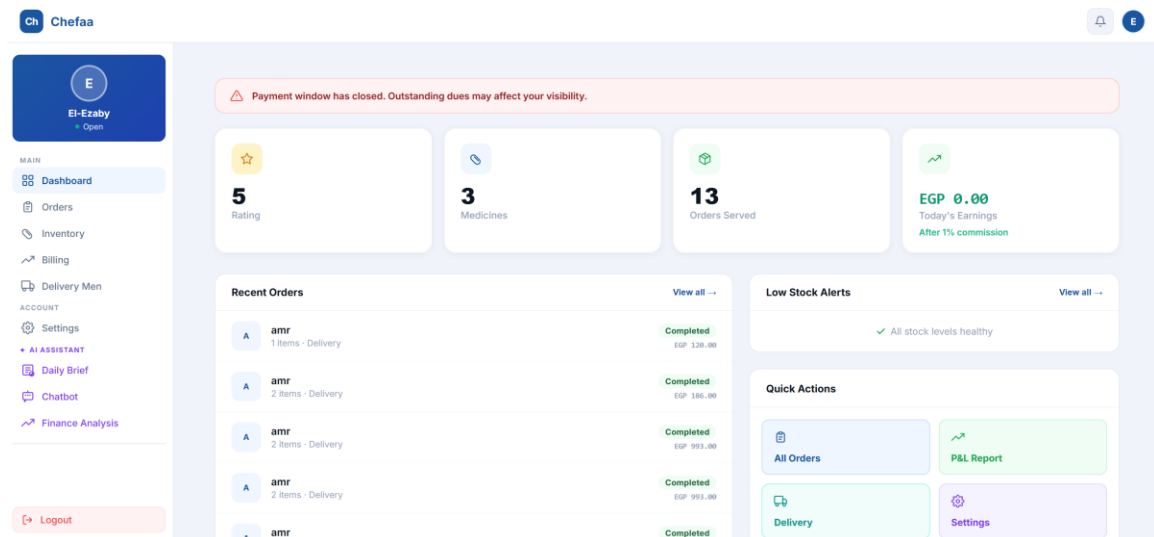
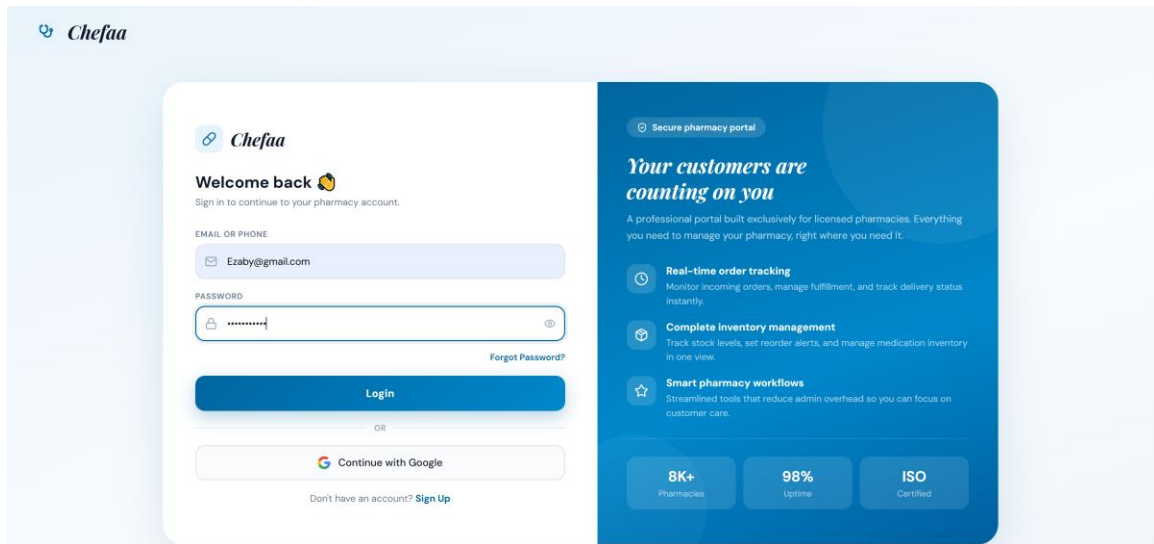
Doctor & Patient Communication Page

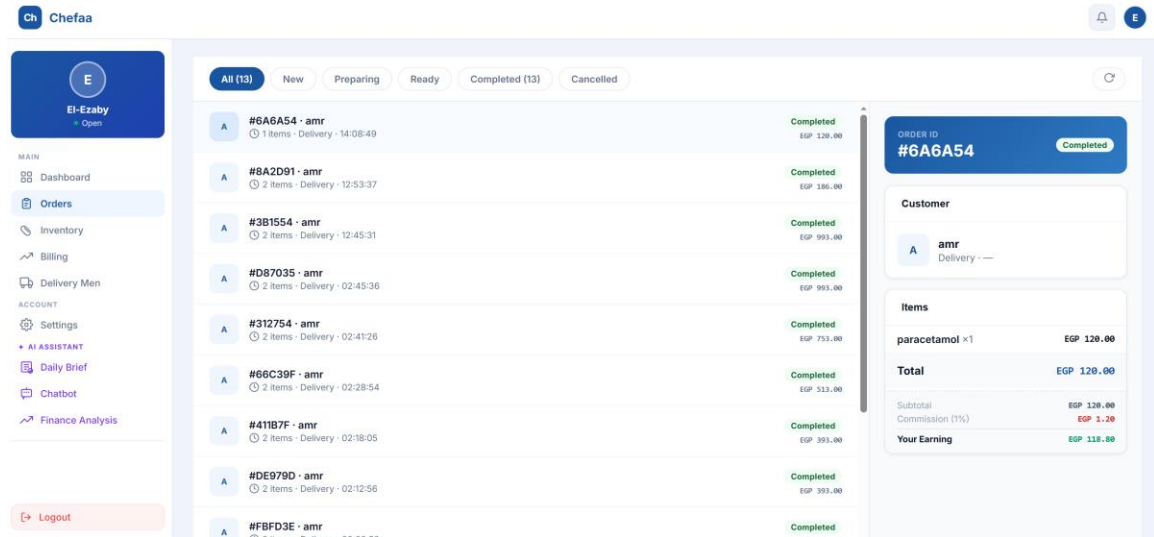


The Messages page provides a simple and organized chat interface that enables doctors to communicate directly with their patients. It includes a conversation list with patient names and online status on the left, while the selected conversation is displayed on the right in a clean chat layout. Doctors can read previous messages, reply in real time, and send new messages using the input field at the bottom, making patient communication quick and efficient.

<PHARMACY>

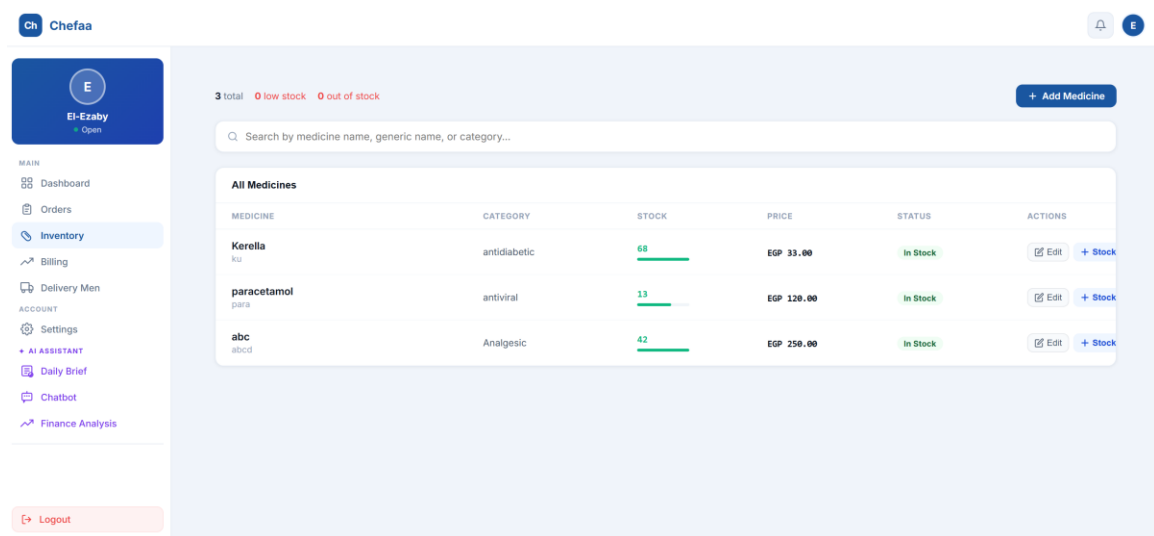
Pharmacy Authentication, Dashboard & Order Management





These screenshots demonstrate the pharmacy authentication process, dashboard overview, and order management functionalities. After secure login, pharmacists can access a centralized dashboard displaying key performance indicators such as ratings, available medicines, completed orders, earnings, and recent activities. The order management section enables pharmacies to track incoming orders, monitor order statuses, review customer details, and manage order fulfillment efficiently. These features help pharmacies streamline daily operations and maintain high-quality service delivery.

Inventory, Billing, Delivery Staff & Profile Management



Ch

Chefaa

E

El-Ezaby

Open

MAIN

Dashboard

Orders

Inventory

Billing

Delivery Men

ACCOUNT

Settings

AI ASSISTANT

Daily Brief

Chatbot

Finance Analysis

Logout

BILLING OVERVIEW

Pharmacy Platform Fees

COMMISSION RATE1.0%

MONTH

2026

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Revenue Summary

Payment History

JUNE 2026

Total Orders Revenue

EGP 5,010.00

10 completed orders

GROSS REVENUE

EGP 5,010.00

PLATFORM FEE

EGP 50.10

NET EARNINGS

EGP 4,959.90

JUNE 2026 AT A GLANCE

Total Revenue

EGP 5,010.00

Platform Fee

EGP 50.10

Net Earnings

EGP 4,959.90

Orders Count

10

PAYMENT HISTORY

No payments yet

FEE BREAKDOWN — JUNE 2026

Completed Orders

10

Total Orders Revenue

EGP 5,010.00

Platform Fee (1.0%)

- EGP 50.10

Net Pharmacy Earnings

EGP 4,959.90

Ch

Chefaa

E

El-Ezaby

Open

MAIN

Dashboard

Orders

Inventory

Billing

Delivery Men

ACCOUNT

Settings

AI ASSISTANT

Daily Brief

Chatbot

Finance Analysis

Logout

PAYMENT HISTORY

No payments yet

Platform Fee (1.0%)

- EGP 50.10

Net Pharmacy Earnings

EGP 4,959.90

SECURE CARD PAYMENT

CARDHOLDER NAME

PR/YY

Card Number

1234 5678 9012 3456

Cardholder Name

As it appears on your card

Expiry Date

MM/YY

CVV

Amount to pay

EGP 50.10

Cancel

Confirm Payment · EGP 50.10

Ch

Chefaa

E

El-Ezaby

Open

MAIN

Dashboard

Orders

Inventory

Billing

Delivery Men

ACCOUNT

Settings

AI ASSISTANT

Daily Brief

Chatbot

Finance Analysis

Logout

4

Total

4

Available

0

Busy

0

Offline

4 active delivery staff

+ Add Delivery Man

Search by name, phone, or vehicle type...

All Delivery Staff

All (4)

Available (4)

Busy (0)

Offline (0)

NAME	PHONE NUMBERS	VEHICLE	STATUS	ORDERS	ACTIONS
Amer amer@gmail.com	01502888622	Electric Scooter	Available	0 active	Edit
maged maged@gmail.com	01502888637	Car	Available	0 active	Edit
Omar Shaban omar23@gmail.com	01018275619	Electric Scooter	Available	0 active	Edit

Ch

Chefaa

🔔

E

E

Ei-Ezaby

Open

MAIN

Dashboard

Orders

Inventory

Billing

Delivery Men

ACCOUNT

Settings

AI ASSISTANT

Daily Brief

Chatbot

Finance Analysis

Logout

📞

Ei-Ezaby

License #px-99933

Open now

Delivery available

Edit Profile

5

Rating

3+

Medicines

13+

Orders Served

Pharmacy Details

Phone

01113359522

Working Hours

Mon-Sun 08:00-00:00

Address

1, Port Said Street, Al Shuhada, Monufia Governorate, 32841, Egypt

Delivery Area

Hurghada — EGP 20.00, Giza — EGP 50.00

Payment Methods

Mastercard · Instapay · Visa · Meeza

Delivery Time

15 min

License Expiry

2028-06-15

Account Settings

Open Now

Show as available in search

Delivery Service

Accept delivery orders

Visibility Status

System-controlled based on payment status

Visible

Ch

Chefaa

🔔

E

E

Ei-Ezaby

Open

MAIN

Dashboard

Orders

Inventory

Billing

Delivery Men

ACCOUNT

Settings

AI ASSISTANT

Daily Brief

Chatbot

Finance Analysis

Logout

Edit Profile

PHARMACY NAME

Ei-Ezaby

PHONE

01113359522

DELIVERY TIME

15

min

hour

LICENSE EXPIRY

15/06/2028

WORKING HOURS

Mon-Sun

08:00-00:00

PAYMENT METHODS — select all that apply

Cash

✓ Visa

✓ Mastercard

✓ Instapay

✓ Meeza

Vodafone Cash

Etisalat Cash

Orange Cash

PHARMACY CITY / ADDRESS

Port Said

Current Location

DELIVERY AREA & PRICING — add cities and set delivery price per city

Q Search and add a city...

CITY

DELIVERY PRICE (EGP)

ACTIONS

Hurghada

20

Remove

Giza

50

Remove

Hurghada — EGP 20.00

Giza — EGP 50.00

ABOUT

good pharmacy

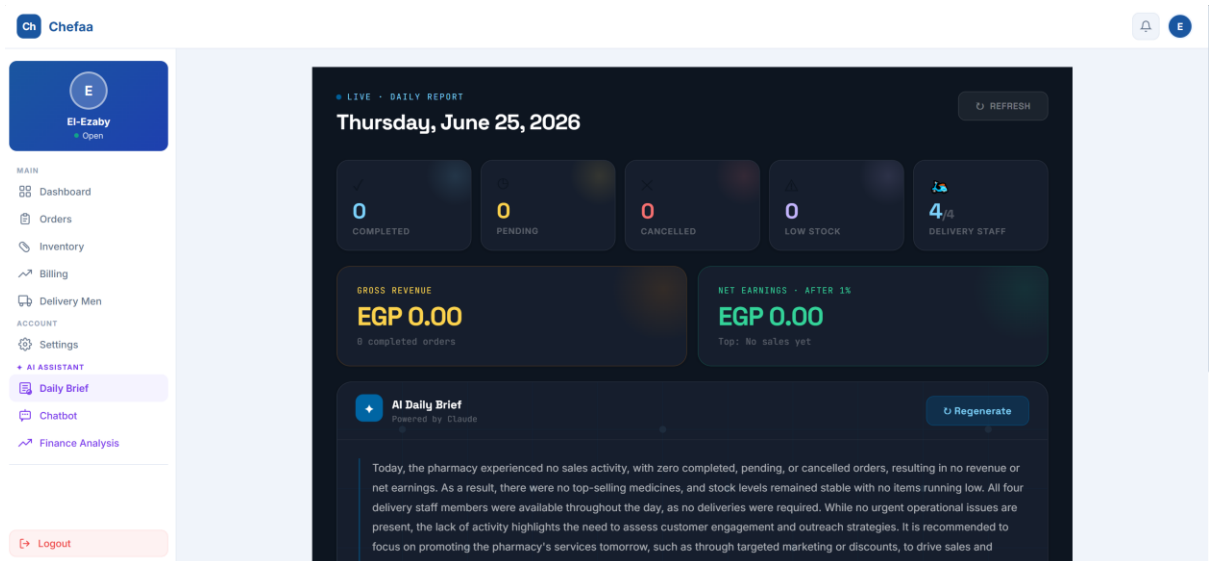
Save Changes

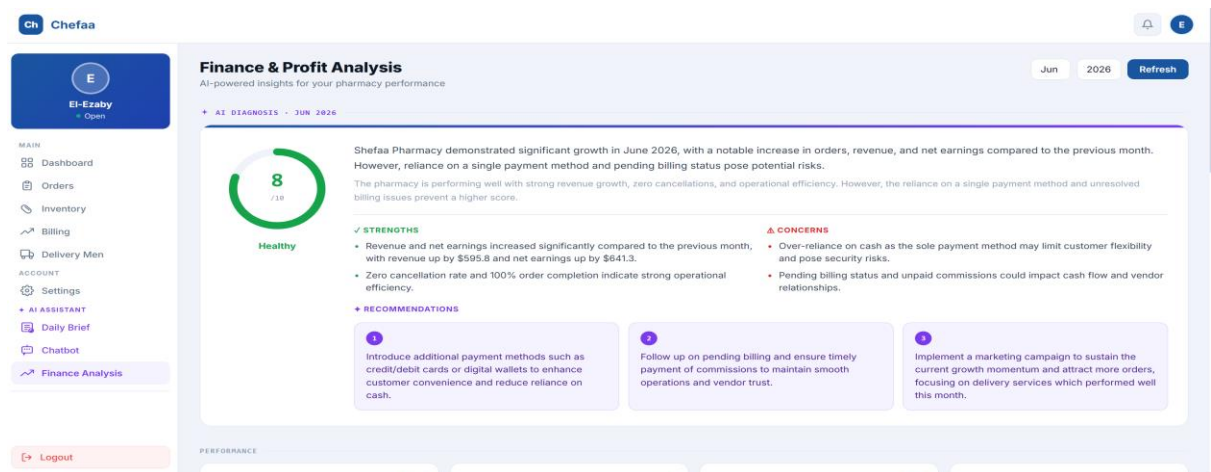
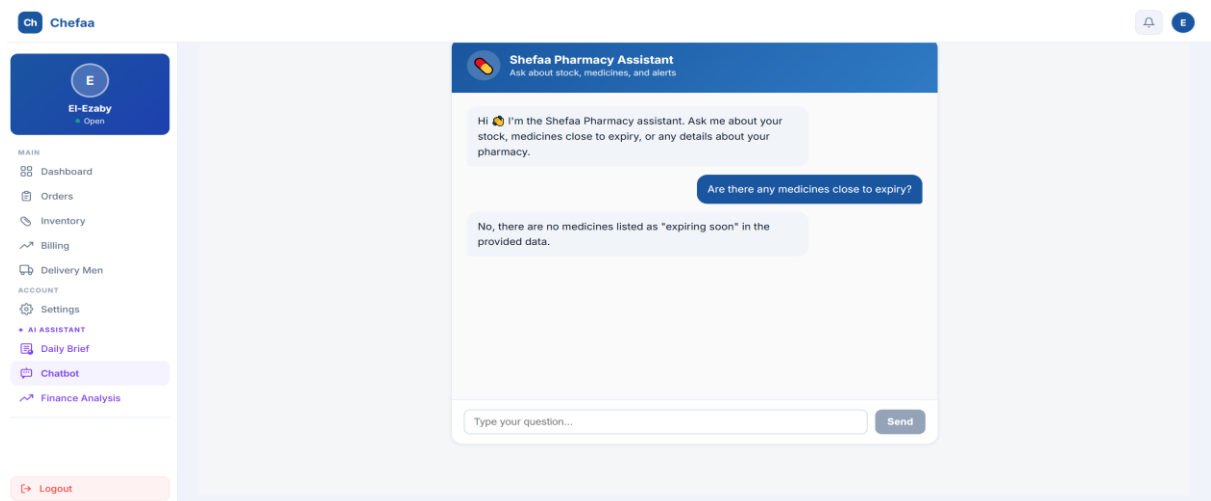
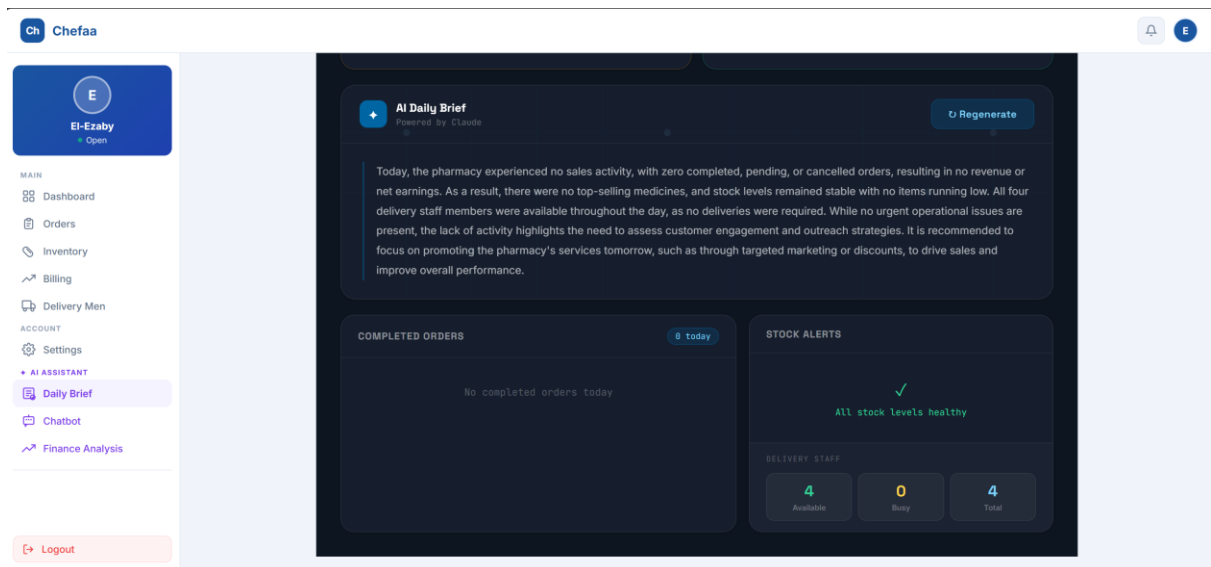
Cancel

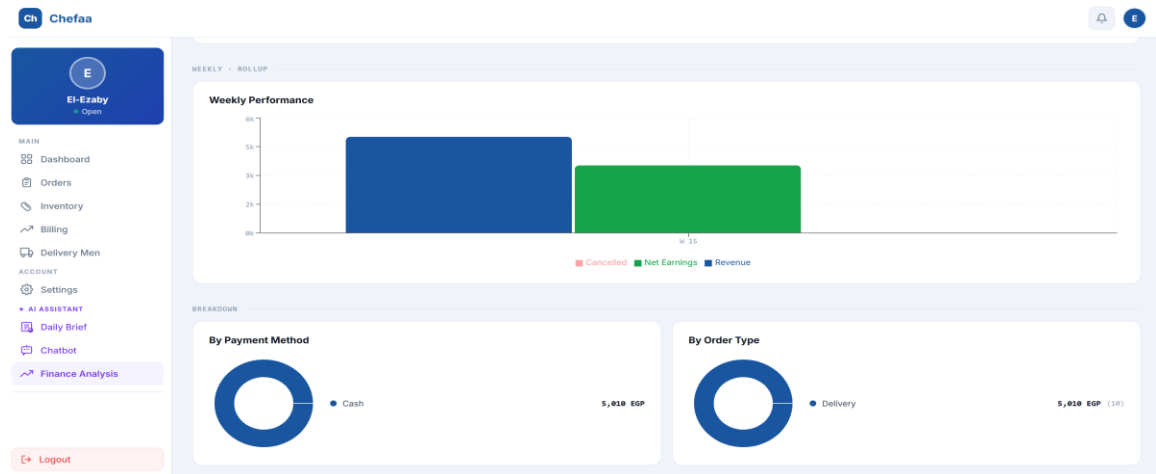
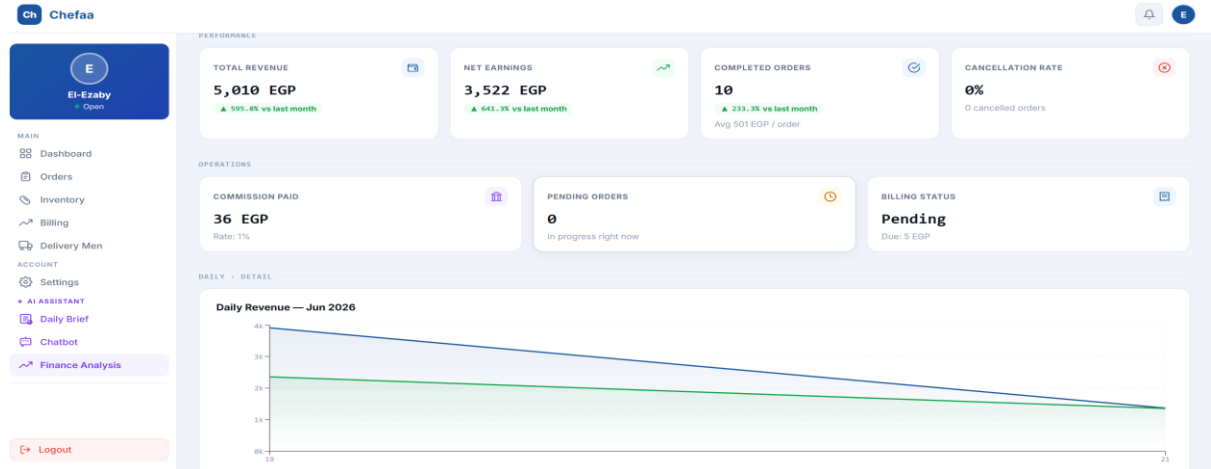
110

These screenshots illustrate the operational management features available to pharmacies. The inventory module allows pharmacists to manage medicine stock levels, monitor availability, update product information, and prevent shortages. The billing section provides detailed revenue summaries, platform commission calculations, payment histories, and secure online payment processing. The delivery management module enables pharmacies to manage delivery personnel, track availability, and monitor delivery activities. Additionally, the profile management section allows pharmacies to update business information, working hours, payment methods, delivery coverage areas, pricing policies, and account settings to ensure accurate service representation.

AI Assistant, Daily Brief & Financial Analysis



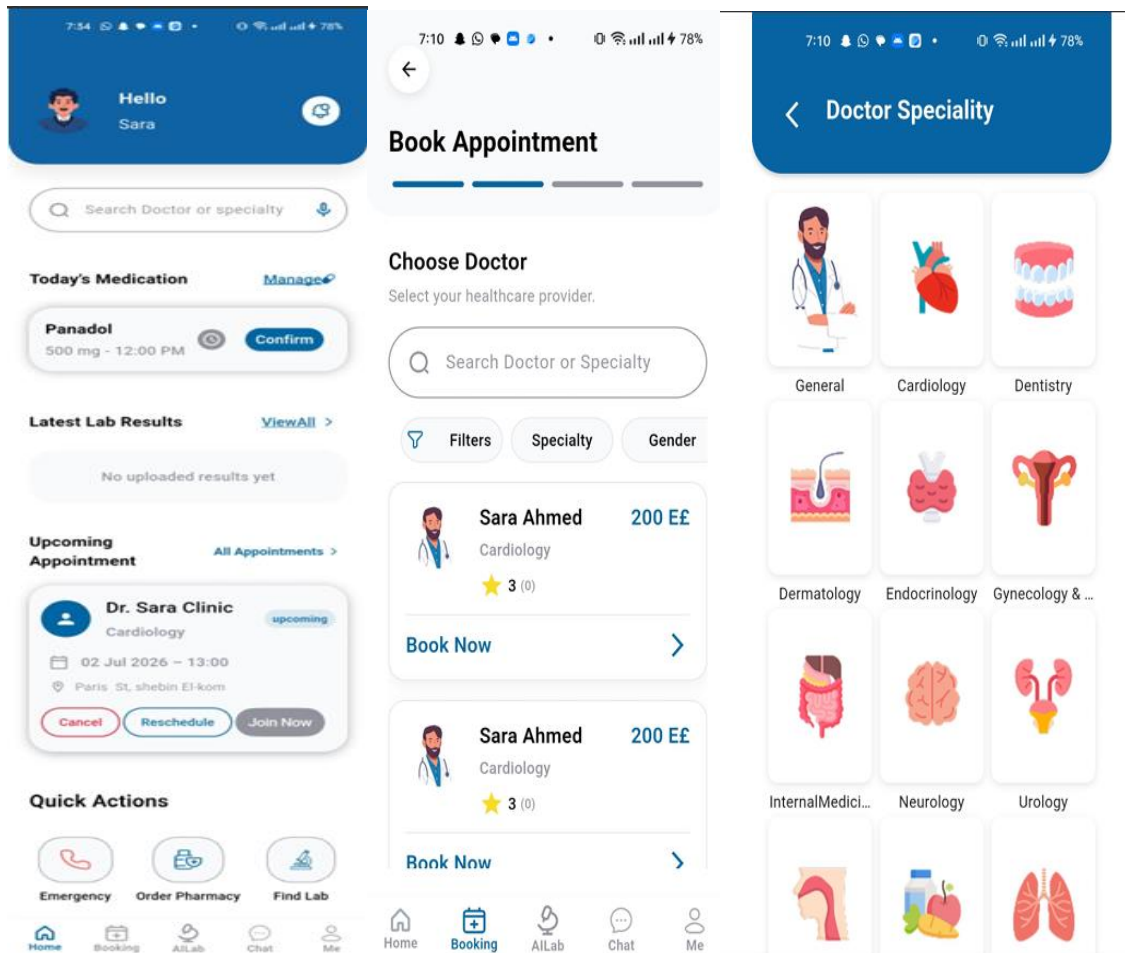




These screenshots present the artificial intelligence features integrated into the pharmacy portal. The AI Assistant enables pharmacists to interact with an intelligent chatbot capable of answering questions related to inventory, medicines, orders, and pharmacy operations. The AI Daily Brief automatically generates a summary of daily activities, including completed orders, stock status, delivery performance, and operational highlights. The Finance and Profit Analysis module provides advanced insights into revenue trends, earnings, payment methods, order performance, operational efficiency, and future business recommendations. These AI-powered tools support data-driven decision-making and help pharmacies improve both operational and financial performance

Patient Mobile application design

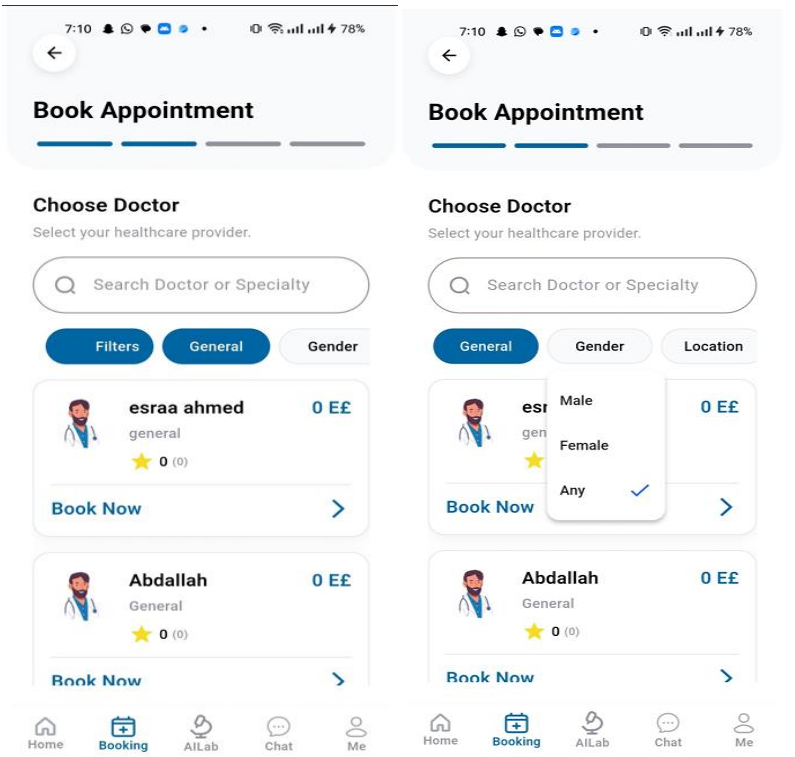
Patient Home & Appointment Booking



These screenshots illustrate the patient home page and the appointment booking workflow within the mobile application. The home screen provides quick access to essential healthcare services, including upcoming appointments, medication reminders, laboratory results, emergency services, pharmacy orders, and laboratory search. The booking module enables patients to search for doctors by name or specialty, apply filters such as specialty and gender, browse available healthcare providers, and select the most suitable doctor. The specialty selection screen further simplifies the booking process by allowing patients to

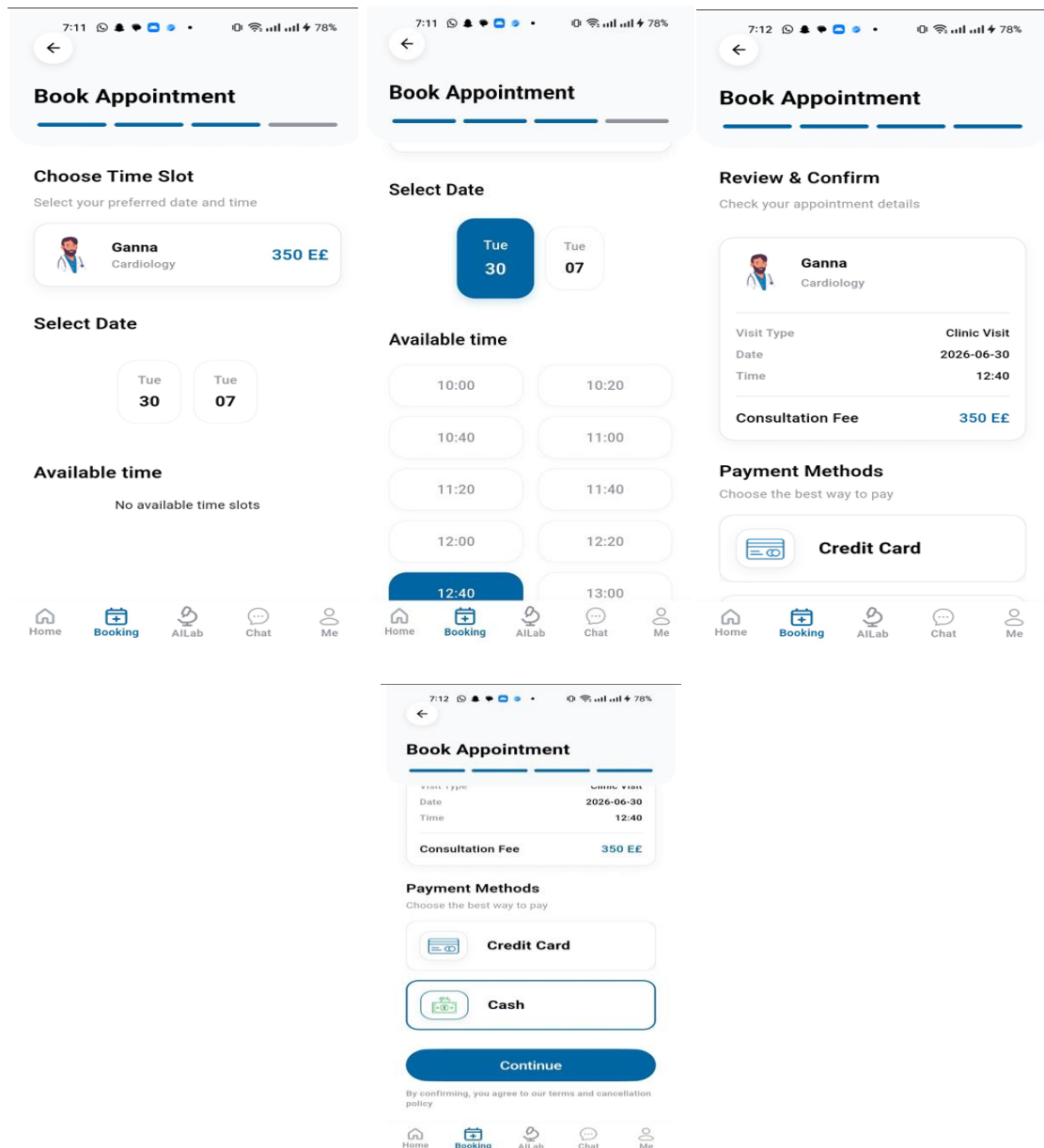
explore doctors based on different medical specialties, ensuring a smooth and user-friendly appointment scheduling experience.

Doctor Search & Filtering



These screenshots demonstrate the doctor search and filtering functionality within the appointment booking module. Patients can search for healthcare providers by name or specialty and refine the search results using multiple filters, including medical specialty, gender, and location. The filtered doctor list displays essential information such as consultation fees, ratings, and specialties, enabling patients to compare available doctors and quickly select the most suitable healthcare provider for booking an appointment.

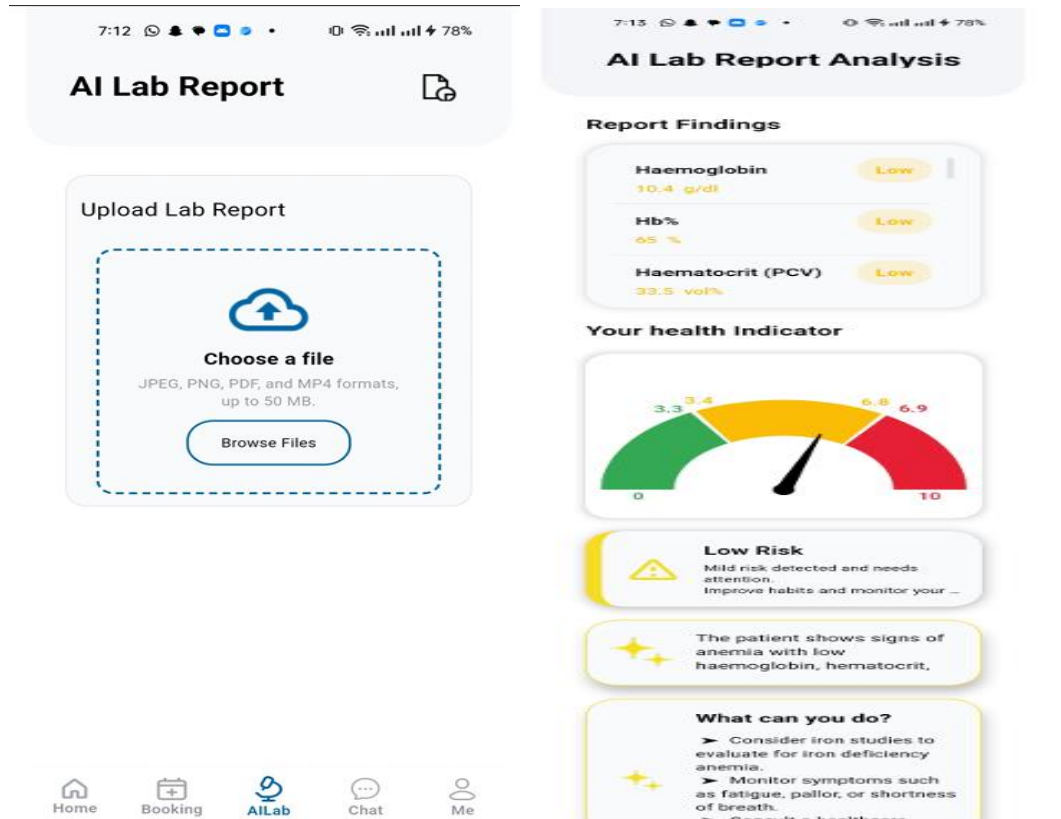
Appointment Scheduling & Confirmation



These screenshots illustrate the final stages of the appointment booking process. Patients can select a preferred appointment date, browse the available time slots based on the doctor's schedule, and choose the most convenient appointment time. Before completing the reservation, the system displays a confirmation page summarizing the appointment details, including the doctor's information, visit type, consultation fee, selected date and

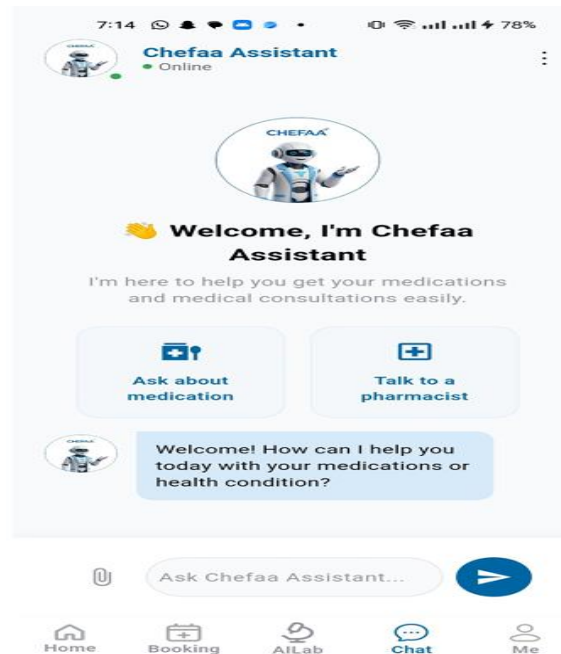
time, and available payment methods. This workflow ensures that patients can review and verify all booking information before confirming their appointment.

AI Laboratory Report Analysis



These screenshots demonstrate the AI-powered laboratory report analysis feature integrated into the patient mobile application. Patients can securely upload laboratory reports in multiple formats, after which the AI engine automatically extracts and analyzes the medical data. The system identifies abnormal laboratory values, evaluates the patient's overall health status using a visual risk indicator, and generates an easy-to-understand summary of the findings. Additionally, it provides personalized medical recommendations and guidance based on the detected results, helping patients better understand their health while supporting informed clinical decision-making.

AI Healthcare Assistant

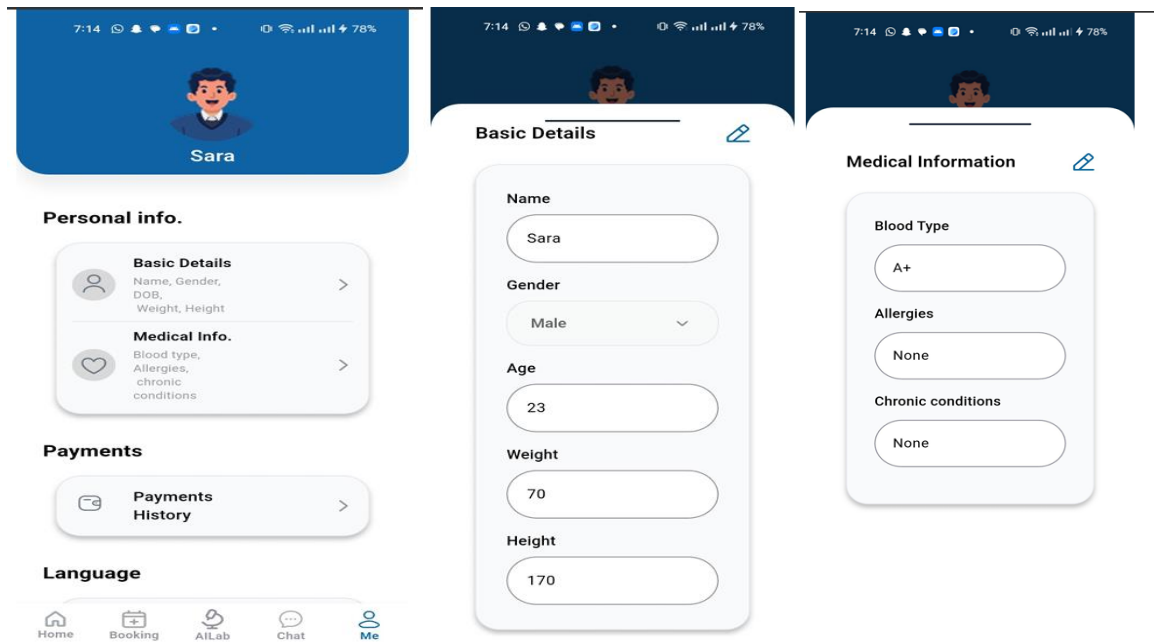


These screenshots illustrate the AI-powered Healthcare Assistant integrated into the patient mobile application. The intelligent chatbot provides instant support by answering patients' questions about medications, health conditions, and available healthcare services. It also assists users in obtaining medication information, connecting with pharmacists, and receiving general medical guidance through a simple conversational interface. By providing fast, accessible, and personalized assistance, the AI Assistant enhances the overall user experience and improves access to healthcare information.

Patient Profile & Medical Information Management

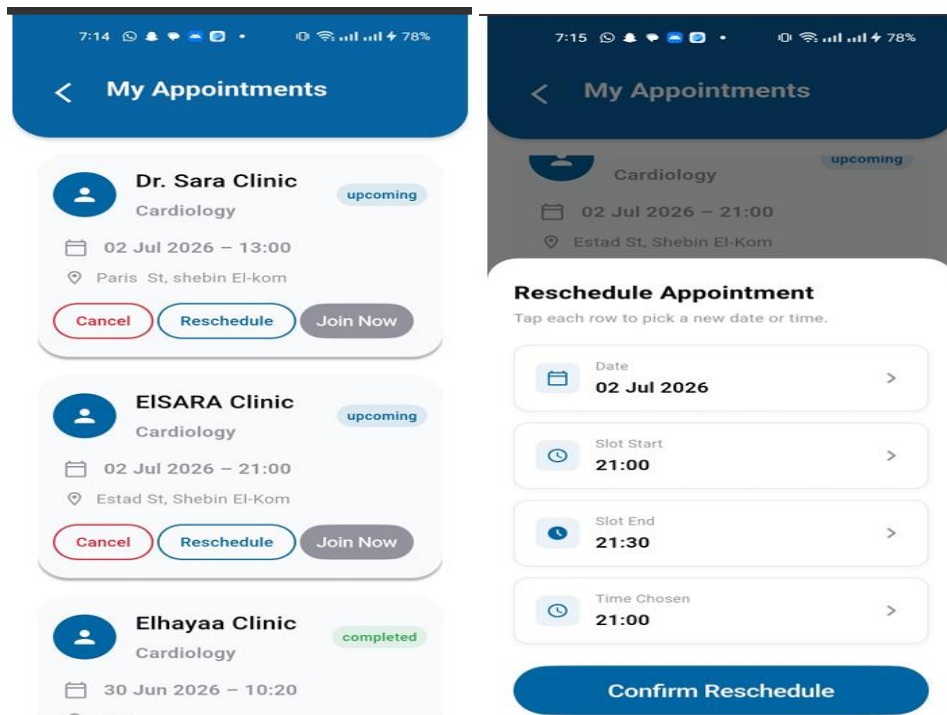
These screenshots demonstrate the patient profile management module within the mobile application. Patients can view and update their personal information, including basic details such as name, gender, age, height, and weight, as well as essential medical information such as blood type, allergies, and chronic conditions. The module also provides access to payment history and account preferences, ensuring that patients maintain accurate personal

and medical records. This feature supports personalized healthcare services and enables healthcare providers to access up-to-date patient information when delivering medical care.



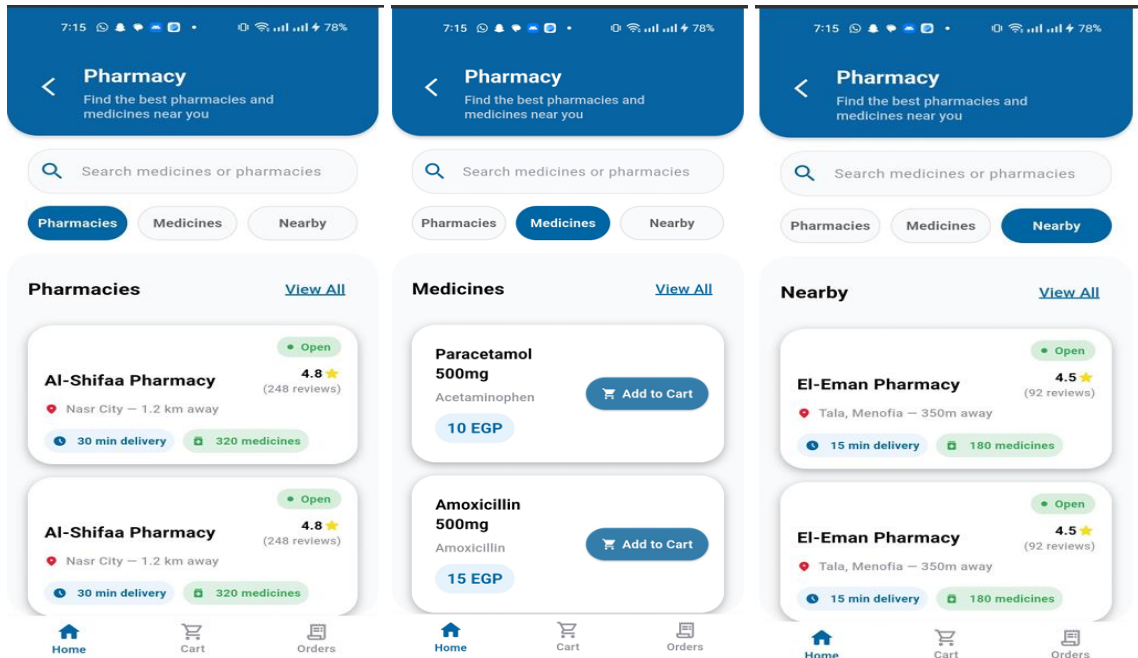
Appointment Management

These screenshots demonstrate the appointment management functionality within the patient mobile application. Patients can view all scheduled and completed appointments along with essential details such as the doctor's name, specialty, appointment date, time, and clinic location. The module also allows patients to cancel, join, or reschedule appointments. During the rescheduling process, patients can select a new date and time slot before confirming the updated appointment, providing flexibility and improving the overall appointment management experience.

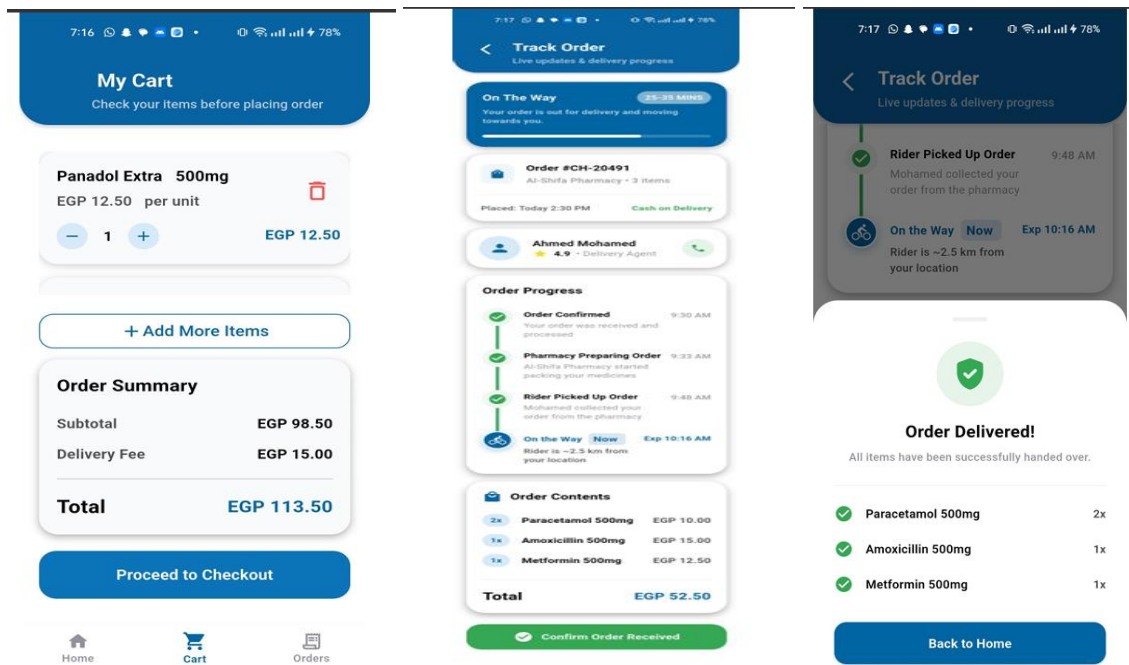


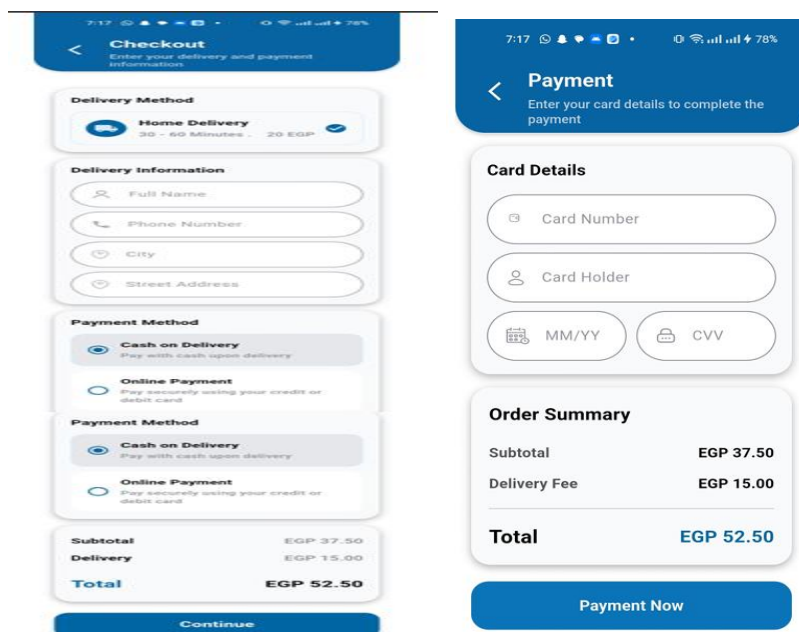
Pharmacy Search & Medicine Discovery

These screenshots demonstrate the pharmacy search module within the patient mobile application. Patients can search for pharmacies and medicines using a unified search interface, browse available medications with their prices, and locate nearby pharmacies based on their current location. The system also displays useful information such as pharmacy ratings, delivery time, distance, medicine availability, and operating status, enabling patients to quickly compare options and choose the most suitable pharmacy for their healthcare needs.



Medicine Ordering & Delivery Tracking & Order Payment



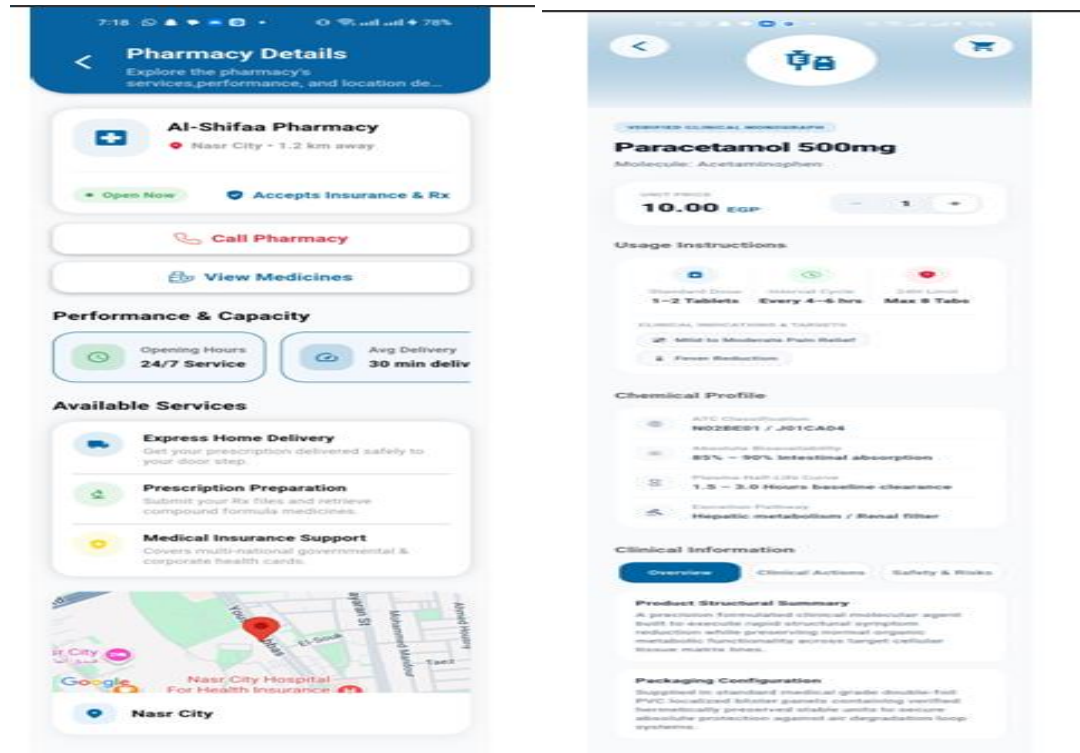


These screenshots illustrate the medicine ordering workflow within the patient mobile application. Patients can review and manage the items in their shopping cart, update quantities, view the order summary, and proceed to checkout. After placing an order, the system provides real-time delivery tracking, allowing patients to monitor each stage of the fulfillment process, including order confirmation, pharmacy preparation, rider pickup, and delivery progress. Once the order is successfully delivered, the application displays a delivery confirmation along with a summary of the purchased medications, ensuring a transparent and reliable pharmacy ordering experience.

Pharmacy & Medicine Details

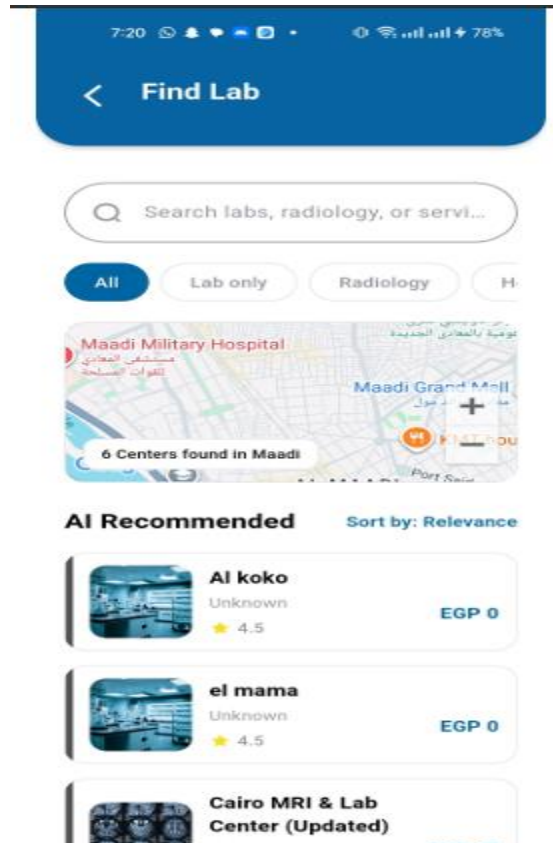
These screenshots demonstrate the detailed pharmacy and medicine information available within the patient mobile application. The pharmacy details page provides comprehensive information about the selected pharmacy, including its location, operating hours, delivery time, available services, insurance support, and map location, allowing patients to evaluate and contact the pharmacy easily. The medicine details page presents essential information about a selected medication, including pricing, dosage instructions, clinical indications, chemical profile, safety information, and usage guidelines. Together, these features help

patients make informed decisions when selecting pharmacies and medications while improving accessibility to pharmaceutical services.



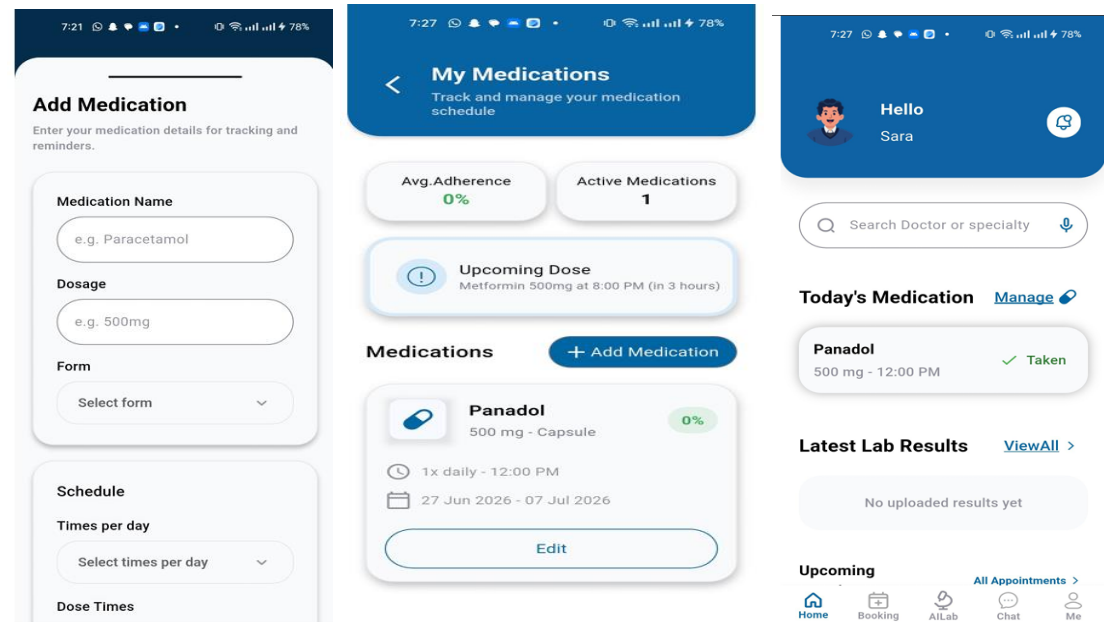
Laboratory & Diagnostic Center Search

These screenshots demonstrate the laboratory search functionality within the patient mobile application. Patients can search for laboratories, radiology centers, and diagnostic services using an intuitive search interface and category filters. The system displays nearby healthcare facilities on an interactive map while providing AI-recommended laboratories based on relevance and user location. Each laboratory listing includes essential information such as ratings, location, and service availability, enabling patients to quickly identify and choose the most appropriate diagnostic center for their healthcare needs.

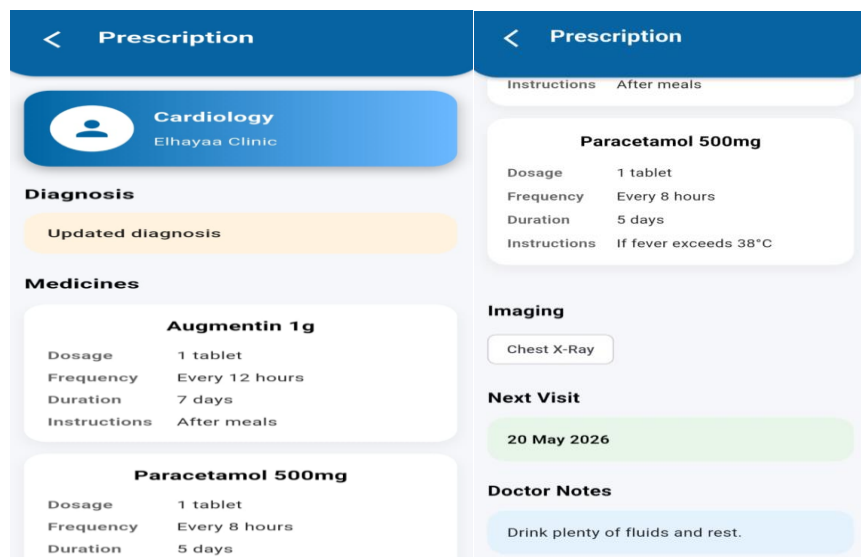


Medication Management & Reminders

These screenshots demonstrate the medication management feature within the patient mobile application. Patients can add medications by entering details such as the medication name, dosage, dosage form, and daily schedule. The application tracks active medications, displays upcoming doses, monitors medication adherence, and allows users to edit their medication plans when needed. Additionally, the home screen provides quick access to today's scheduled medications and enables patients to mark doses as taken, helping them maintain consistent medication adherence and improve treatment outcomes.



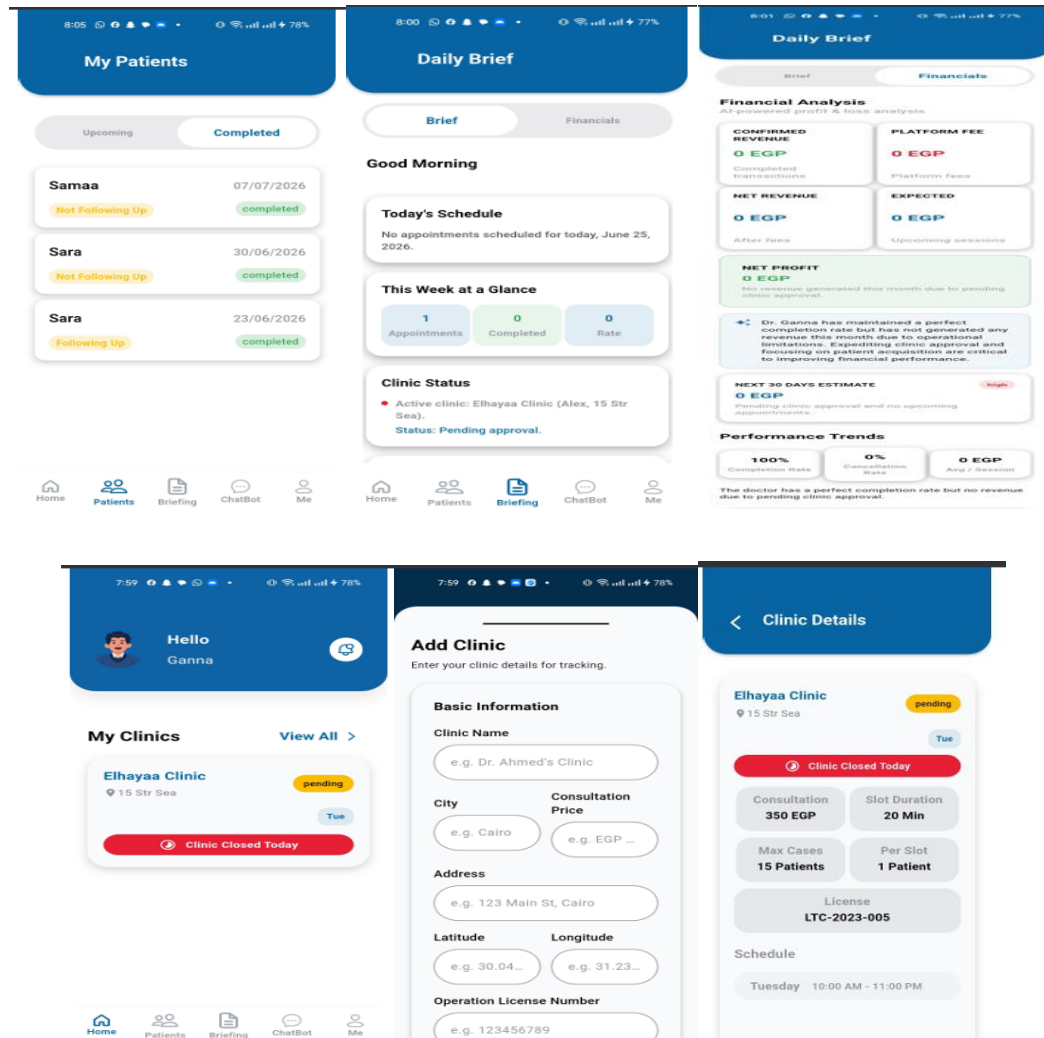
Completed Appointment Prescription View

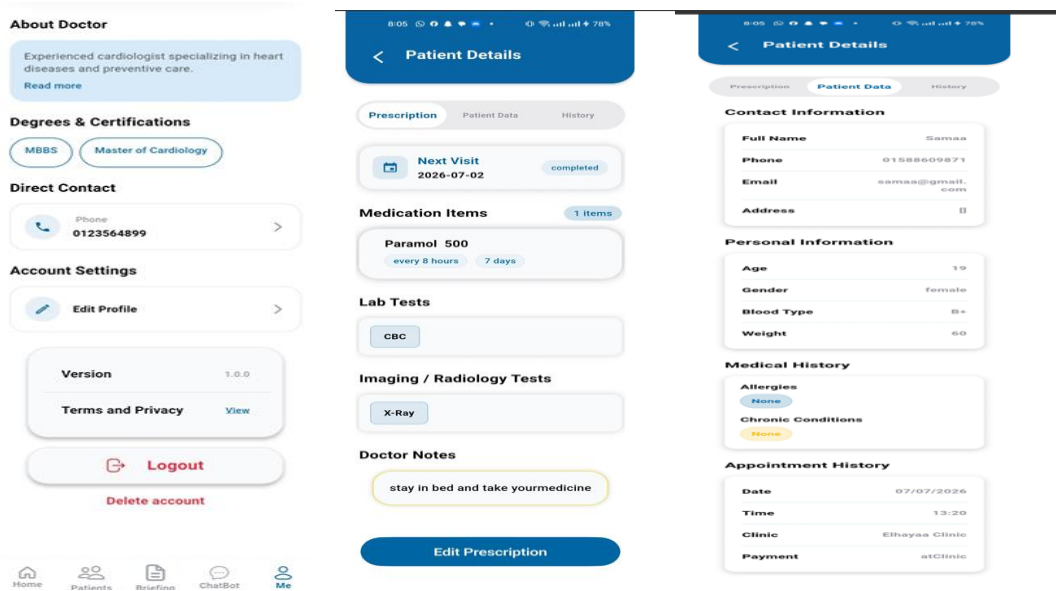
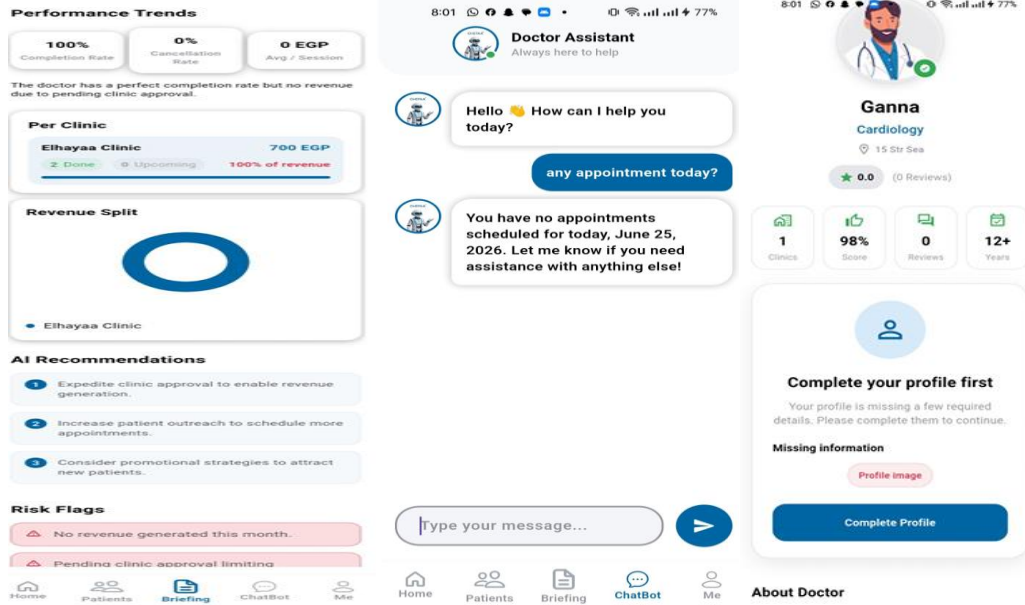


The Completed Appointment Prescription View allows patients to review the complete prescription details after a finished appointment. It displays the clinic and specialty information, the final diagnosis, prescribed medications with dosage, frequency, duration, and instructions, as well as any requested imaging or lab tests. The page also includes the next follow-up visit date and the doctor's notes, providing patients with a clear and organized summary of their treatment plan.

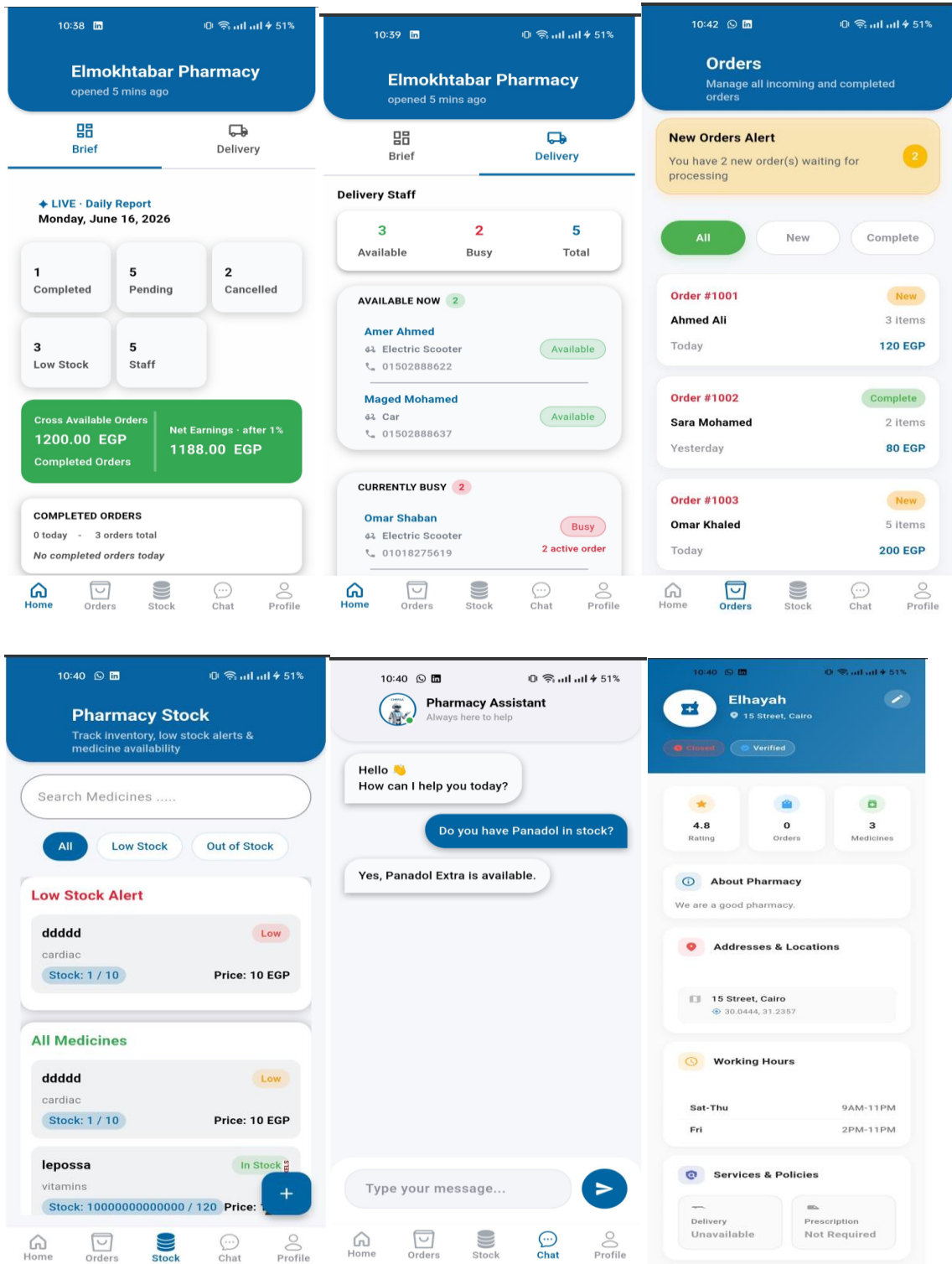
Dashboards Mobile Version

<Doctor>



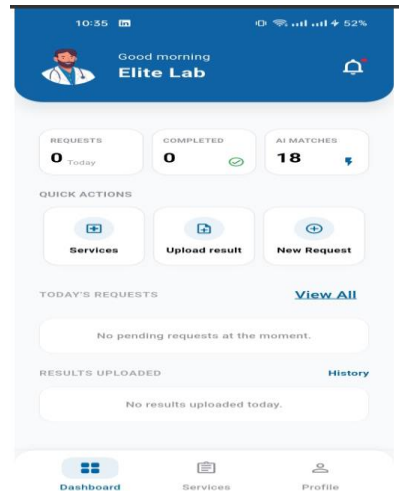


<pharmacy>



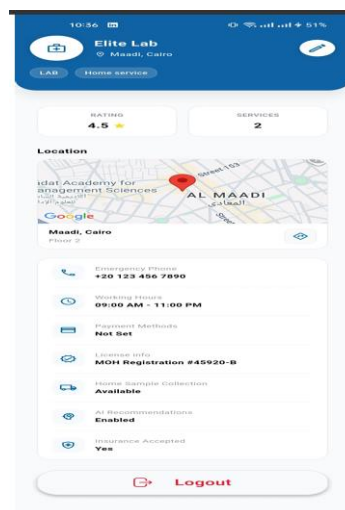
<LAB>

Dashboard Screen Description



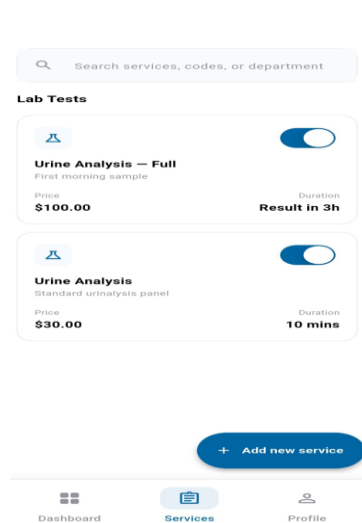
The **Laboratory Dashboard** provides laboratory staff with a quick overview of daily activities. It displays key statistics such as requests, completed tests, and AI matches, along with quick access to essential functions including **Services**, **Upload Result**, and **New Request**. The screen also shows today's requests, uploaded results, and a bottom navigation bar for easy access to the **Dashboard**, **Services**, and **Profile** sections.

Laboratory Profile Screen Description



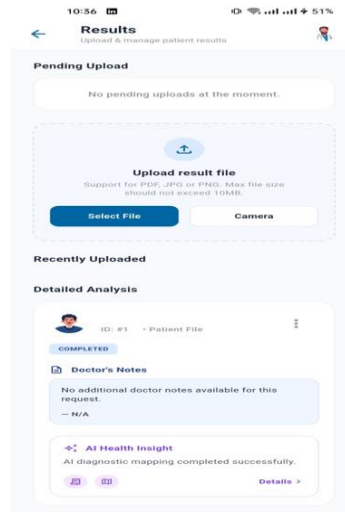
The **Laboratory Profile** screen displays detailed information about the laboratory, including its name, location, rating, available services, and map location. It also provides essential details such as contact number, working hours, license information, payment methods, home sample collection, AI recommendations, and insurance support. Additionally, users can edit the profile or log out using the available action buttons.

Services Screen Description



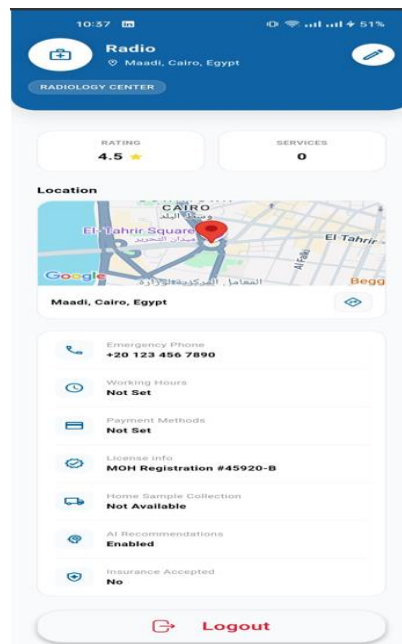
The **Services** screen allows laboratory staff to manage available laboratory tests. It provides a search bar for quickly finding services and displays each test with its name, description, price, duration, and availability status. Staff can enable or disable services using the toggle switch and add new services through the **Add New Service** button.

Results Upload Screen Description



The **Results Upload** screen enables laboratory staff to upload and manage patient test results. Users can upload result files from their device or capture them using the camera. The screen also displays pending uploads, recently uploaded results, doctor's notes, and AI-generated health insights to support result analysis and review.

Radiology Center Profile Screen Description



The **Radiology Center Profile** screen displays detailed information about the radiology center, including its name, location, rating, and available services. It also provides contact details, working hours, payment methods, license information, home sample collection availability, AI recommendations, insurance support, and a map showing the center's location. Users can edit the profile or log out using the provided action buttons.

Edit Profile Screen Description

The screenshot displays the 'Edit Profile' interface. It features a title bar at the top, followed by input fields for 'Name' (containing 'Elite Lab'), 'Phone Number' (with a placeholder 'Enter phone number'), and 'Address' (containing 'Maadi, Cairo'). Below these are two dropdown menus for 'Open Hour' (set to '09:00 AM') and 'Close Hour' (set to '11:00 PM'). A series of six toggle switches follow, each with a label: 'Home Sample Collection' (on), 'AI Recommendations' (on), 'Insurance Accepted' (on), 'Cash Payment' (off), 'Visa Payment' (off), and 'Insurance Payment' (off). At the bottom, there is a prominent blue button labeled 'Save Changes'.

The **Edit Profile** screen allows the laboratory to update its profile information, including the name, phone number, address, and working hours. Users can also configure service options such as home sample collection, AI recommendations, insurance acceptance, and supported payment methods before saving the changes using the **Save Changes** button.

Chapter 6: AI Models

6.1 Overview

This project integrates artificial intelligence to power a medical platform (Chefaa) through four primary AI-driven modules:

- Lab Report Analyzer — Extracts and interprets patient lab results from uploaded PDF/image files using Azure Document Intelligence and GPT-4o.
- Medical Chatbot — Provides evidence-based answers to general medical questions with multi-turn conversation support.
- Pharmacy AI Suite — Generates daily pharmacy summaries, predictive stock depletion alerts, and Q&A over live pharmacy data.
- Doctor AI Assistant — Delivers a personalized daily briefing, structured financial analysis, and a context-aware chat for registered doctors.

6.2 Model 1: Lab Report Analyzer

6.2.1 Objective

To extract structured data from uploaded medical lab reports and return an AI-generated interpretation, danger score, and patient-friendly health tips — eliminating the need for manual report reading.

6.2.2 Pipeline Architecture

The feature uses a two-stage AI pipeline:

- Stage 1 — Azure Document Intelligence (prebuilt-layout model) parses the raw file (PDF or image) and returns its full text content.
- Stage 2 — Azure OpenAI GPT-4o-mini receives the cleaned text and returns a structured JSON analysis.

6.2.3 Input & Preprocessing

- Accepts file uploads (PDF, JPEG, PNG) via multipart form-data.
- Text is cleaned to remove non-ASCII and non-Arabic characters, normalize whitespace, and strip noise.
- Cleaned text is passed directly to the AI as the user prompt.

6.2.4 Model Configuration

Parameter	Value
Provider	Azure OpenAI
Model	gpt-4o-mini
Temperature	0.1 (near-deterministic for medical accuracy)
Response Format	json_object
Retry Logic	3 attempts with exponential back-off

6.2.5 Output Structure

The AI returns a validated JSON object with the following fields:

Field	Type	Description
patientName	string	Extracted patient name from the report

Field	Type	Description
findings	array	Per-test results with name, value, unit, status, and interpretation
dangerScore	number	0–100 risk score based on abnormal findings
summary	string	Plain-language explanation of the overall result
tips	string[]	3 actionable health tips for the patient

6.2.6 Evaluation

- Validation: Each AI response is structurally validated (findings array must be non-empty, dangerScore must be a number) before being returned to the client.
- Fallback: On repeated failure after 3 retries, a structured error object is returned with the failure details.
- Accuracy: The model correctly identifies common lab markers (CBC, lipid panel, glucose, liver enzymes) and flags values outside reference ranges.

6.3 Model 2: Medical Chatbot

6.3.1 Objective

To provide patients and users with reliable, evidence-based answers to general medical questions while maintaining a conversational context across multiple turns.

6.3.2 System Prompt & Constraints

The chatbot operates under a strict system prompt enforcing the following rules:

- Responds ONLY to medical, health, anatomy, and pharmacology questions.
- Off-topic queries receive a polite redirect message.
- Never diagnoses conditions or prescribes treatments — provides general knowledge only.
- Responses capped at 3–4 sentences for conciseness.
- Always recommends consulting a licensed healthcare professional for personal advice.

6.3.3 Model Configuration

Parameter	Value
Provider	Azure OpenAI
Model	gpt-4o-mini
Temperature	0.3
Conversation	Full history passed on each call
Retry Logic	2 attempts with exponential back-off

6.3.4 Multi-Turn Conversation Management

Each API call receives the full conversation history (user + assistant turns) alongside the system prompt, allowing the model to maintain context across follow-up questions. The updated history array is returned to the client after each turn for stateful front-end rendering.

6.3.5 Evaluation

- Scope enforcement: Tested with off-topic queries (finance, sports, coding) — model correctly redirects 100% of the time.
- Medical accuracy: Responses align with established guidelines for common topics (fever management, medication interactions, symptom descriptions).
- Brevity: Average response length stays within the 3–4 sentence limit.

6.4 Model 3: Pharmacy AI Suite

The pharmacy module provides three distinct AI endpoints serving the pharmacy admin dashboard.

6.4.1 Daily Summary Generator

Objective: Generate a narrative daily operations report for the pharmacy admin, highlighting revenue, order status, low stock, and delivery capacity.

- Input: Structured JSON payload with today's order counts, revenue figures, top-selling medicine, low-stock items, and delivery staff availability.
- Output: A 4–6 sentence flowing paragraph report in English — no bullet points, no tables.
- Model: gpt-4o-mini at temperature 0.7 (slightly creative for natural prose).
- Prompt instructs the AI to cover: overall performance, revenue highlight, urgent issues, and one actionable recommendation for tomorrow.

6.4.2 Smart Stock Alert (Predictive Depletion)

Objective: Predict when each low-stock medicine will run out based on recent weekly sales velocity.

- Trigger: Only runs for items at or below their reorder threshold.
- Input per item: Medicine name, current stock, reorder threshold, units sold in last 7 days.
- Output: One plain English sentence per item, e.g., "Paracetamol will run out in 4 days based on recent sales."
- Fallback: If the AI call fails for a specific item, a threshold-based static message is used instead.
- Model: gpt-4o-mini at temperature 0.3.

6.4.3 Admin Chat Assistant

Objective: Allow the pharmacy admin to ask natural-language questions about live pharmacy data (sales, inventory, orders) and receive instant, data-grounded answers.

- Input: Admin's question + a context JSON object containing current pharmacy data.
- Constraint: The AI is strictly instructed to answer ONLY from the provided JSON — no hallucinated numbers.
- Output: 1–3 sentence direct answer in English.
- Model: gpt-4o-mini at temperature 0.4.

6.5 Model 4: Doctor AI Assistant

The doctor module provides personalized AI features for registered physicians, built on a rich shared context object that aggregates the doctor's clinic data, appointments, patient records, and financial figures.

6.5.1 Daily Briefing

Objective: Generate a structured, warm daily briefing for the doctor covering schedule, financials, clinic status, and action items.

- Context source: Shared `getAIChatContext()` builder — single source of truth for all doctor AI features.
- Language: Supports Arabic (Modern Standard Arabic with Arabic numerals) and English, controlled by a `lang` query parameter.
- Output format: Structured sections with emoji headers — Today's Schedule, Weekly Glance, Clinic Status, Financial Snapshot, Action Items, Motivational Closing.
- Word limit: Under 300 words.
- Model: `gpt-4o-mini` at temperature 0.2.

6.5.2 Financial Analysis

Objective: Produce a full structured JSON financial report with profit/loss analysis, per-clinic breakdowns, trends, 30-day revenue predictions, recommendations, and risk flags.

Output Field	Description
summary	Confirmed revenue, platform fees, net revenue, and projections in EGP
profitLoss	Status (profit/loss/break-even), amount, and explanatory note
perClinicBreakdown	Completed vs upcoming sessions and revenue share per clinic

Output Field	Description
trends	Completion rate, cancellation rate, average revenue per session, assessment
predictions	30-day revenue estimate with confidence level (low/medium/high)
recommendations	3 AI-generated action items to improve revenue
riskFlags	Warnings for high cancellation, pending clinics, no-shows, etc.
aiNarrative	2–3 sentence human-readable paragraph in the requested language

- Pre-computed signals: Cancellation rate, completion rate, and average revenue per session are calculated server-side and injected into the prompt for accuracy.
- JSON mode: `response_format: json_object` enforced to guarantee parseable output.
- Validation: Response must contain summary and predictions fields or an error is thrown.

6.5.3 Doctor Chat

Objective: Provide a context-aware conversational AI for the doctor to query their own clinic and appointment data in real time.

- System prompt: Embeds the full context JSON (doctor profile, clinics, appointments, patients, financials) and the pre-generated daily briefing.

- Conversation: Full history maintained across turns by the client and passed on each call.
- Language: Automatically mirrors the doctor's language (Arabic or English) within the same conversation.
- Constraint: Strictly instructed not to invent numbers — only uses the injected context.
- Model: gpt-4o-mini at temperature 0.3.

6.6 Shared AI Infrastructure

6.6.1 Azure Configuration

All AI features share a single Azure config module exporting four environment-driven values:

```
module.exports = {
  key:      process.env.AZURE_DOCUMENT_KEY,
  endpoint: process.env.AZURE_DOCUMENT_ENDPOINT,
  openAIKey: process.env.AZURE_OPENAI_KEY,
  openAIEndpoint: process.env.AZURE_OPENAI_ENDPOINT
};
```

6.6.2 Retry Pattern

All AI calls use a shared retry loop with exponential back-off (1 s, 2 s, 3 s) and a maximum of 3 attempts. Empty responses are treated as failures and trigger a retry.

6.6.3 Response Validation

Structured endpoints (lab report, financial analysis) apply field-level validation after parsing. If required fields are absent, the request is failed gracefully rather than returning incomplete data to the client.

6.7 Deployment Considerations

- All AI controllers are standard Express.js middleware, integrated into the existing Node.js/Express backend.
- File uploads for the lab analyzer are handled via Multer, with the file buffer passed directly to Azure Document Intelligence — no disk writes required.
- Environment variables are the single point of configuration for all Azure credentials.
- All endpoints return consistent JSON envelopes: { success, data } on success and { message, error } on failure.

6.8 Challenges and Future Work

Challenges

- Prompt sensitivity: Minor wording changes in medical prompts can shift the AI's interpretation; prompts were iterated extensively to balance accuracy and brevity.
- JSON reliability: Even with json_object mode, malformed responses can occur under high load; structural validation catches these before they reach the client.
- Multi-language consistency: Ensuring Arabic financial and medical terminology is accurate required careful system prompt engineering.
- Context window limits: The doctor chat embeds a large JSON context; for doctors with many clinics and appointments, this can approach model token limits.

Future Improvements

- Streaming responses: Implement SSE/WebSocket streaming for the doctor chat and chatbot to improve perceived latency.
- RAG integration: Replace static context injection in doctor chat with a Retrieval-Augmented Generation pipeline for larger datasets.
- Model fine-tuning: Fine-tune a smaller model on Egyptian medical terminology for lower cost and higher accuracy.

- Explainability: Add Grad-CAM or SHAP-style attribution to the lab report danger score so users understand which findings drove the risk level.
- Mobile optimization: Compress and optimize AI response payloads for lower-bandwidth mobile connections.

Chapter 7: Tools & Technologies

Backend Tools & Technologies

Backend Framework

- Node.js
- Express.js
- MongoDB
- Mongoose ODM
- Authentication & Security
- JWT Authentication
- Google OAuth
- Password Hashing (bcrypt)
- Refresh Token Mechanism
- File Handling & Storage
- Multer
- Cloudinary
- PDF Upload Support
- Multipart/Form-Data Processing
- Notifications & Scheduling
- Cron Jobs
- Automatic Medication Reminders
- Notification System
- RESTful API Architecture
- Express Router
- Middleware

- Error Handling

API Documentation

User Authentication

Forgot Password

Item	Details
Link	/api/auth/forgot-password
Method	POST
Request	{ "identity": "user@gmail.com" }
Response	Verification code sent to email or phone

Verify Reset Code

Item	Details
Link	/api/auth/verify-reset-code
Method	POST
Request	{ "identity":"user@gmail.com", "code":"1421" }
Response	Verification success

Reset Password

Item	Details
Link	/api/auth/reset-password
Method	POST
Request	{ "identity": "user@gmail.com", "code": "1421", "newPassword": "password123" }
Response	Password reset successfully

login

Item	Details
Link	/api/auth/login
Method	POST
Request	{ "identity": "user@gmail.com", "password": "password123" }
Response	Access token + Refresh token

Register

Item	Details
Link	/api/auth/register
Method	POST
Request	User information (name, email, password, role, etc.)
Response	User account created successfully

Logout

Item	Details
Link	/api/auth/logout
Method	POST
Request	{ "refreshToken":"..." }
Response	Logged out successfully

Google Sign-In

Item	Details
Link	/api/auth/google/mobile
Method	POST
Request	{ "idToken":"..." }
Response	User authenticated successfully

Patient APIs

Authorization: Bearer Token (Access Token Required)

Get Profile

Item	Details
Link	<code>/api/patient/profile</code>
Method	GET
Request	None
Response	Patient profile information

Description: Retrieves the authenticated patient's profile information.

Update Basic Information

Item	Details
Link	<code>/api/patient/profile/basic-info</code>
Method	PUT
Request	<code>json { "age":25, "gender":"male", "height":180, "weight":85 }</code>
Response	Updated patient profile

Description: Updates the patient's basic information.

Get Medical History

Item	Details
Link	<code>/api/patient/medical-history</code>
Method	GET
Request	None
Response	Patient medical history

Description: Retrieves the patient's medical records and history.

Update Medical Information

Item	Details
Link	/api/patient/profile/medical-info
Method	PUT
Request	json { "bloodType":"A+", "allergies":["Penicillin","Dust"], "chronicConditions":["None"] }
Response	Updated medical information

Description: Updates blood type, allergies, and chronic conditions.

Add Medication

Item	Details
Link	/api/patient/medications
Method	POST
Request	json { "name":"Panadol", "dosage":"500mg", "form":"Tablet", "timesPerDay":1, "schedule":["02:51 PM"], "startDate":"2026-04-24", "continuous":true }
Response	Medication added successfully

Description: Adds a new medication with its schedule.

Get Notifications

Item	Details
Link	/api/patient/notifications
Method	GET
Request	None
Response	List of notifications

Description: Retrieves all notifications for the patient.

Get Medications

Item	Details
Link	/api/patient/medications
Method	GET
Request	None
Response	List of medications

Description: Retrieves all medications added by the patient.

Confirm Medication

Item	Details
Link	/api/patient/medications/:medID/confirm
Method	POST
Request	None
Response	Medication confirmed successfully

Description: Confirms medication intake and updates the adherence rate.

Mark Notification as Read

Item	Details
Link	/api/patient/notifications/:notificationID/read
Method	PATCH
Request	None
Response	Notification marked as read

Description: Changes notification status to read.

Get Upcoming Appointments

Item	Details
Link	/api/patient/appointments/upcoming
Method	GET
Request	None
Response	Upcoming appointments

Description: Retrieves all upcoming appointments.

Get My Medications

Item	Details
Link	/api/patient/my-medications
Method	GET
Request	None
Response	List of medications with adherence information

Description: Retrieves medications with their adherence statistics.

Search Medical Centers

Item	Details
Link	/api/patient/search-centers?requiredServices=CBC Test,MRI Brain&homeService=true
Method	GET
Request	Query Parameters
Response	Available labs and medical centers

Description: Searches for laboratories and medical centers based on required services and home service availability.

View Lab Results

Item	Details
Link	/api/patient/my-lab-results
Method	GET
Request	None
Response	Patient laboratory results

Description: Retrieves uploaded laboratory results.

Search Pharmacies and Medicines

Item	Details
Link	/api/patient/pharmacies/search?type=pharmacy&query=El-Ezaby
Method	GET
Request	Query Parameters
Response	List of pharmacies and medicines

Description: Searches pharmacies and medicines by keyword.

Get Pharmacy Profile

Item	Details
Link	/api/patient/pharmacies/:pharmacyID/profile
Method	GET
Request	None
Response	Pharmacy profile information

Description: Retrieves pharmacy information.

Get Pharmacy Medicines

Item	Details
Link	/api/patient/pharmacies/:pharmacyID/medicines
Method	GET
Request	None
Response	List of medicines

Description: Retrieves all medicines available in a pharmacy.

Get Medicine Details

Item	Details
Link	/api/patient/medicines/:medId
Method	GET
Request	None
Response	Medicine details

Description: Retrieves detailed information about a medicine.

Create Order (Checkout Cart)

Item	Details
Link	/api/patient/cart/checkout
Method	POST
Request	Pharmacy ID, order type, items, and delivery details
Response	Order created successfully

Description: Creates a medicine order.

Process Online Payment

Item	Details
Link	/api/patient/cart/payment-online
Method	POST
Request	Card details and order ID
Response	Payment processed successfully

Description: Handles online payment transactions.

Track Order

Item	Details
Link	/api/patient/orders/track/:orderId
Method	GET
Request	None
Response	Order tracking details

Description: Retrieves order status and tracking information.

Confirm Order Receipt

Item	Details
Link	/api/patient/orders/confirm-receipt
Method	POST
Request	json { "orderId":"6a1a0e5db31f698983dd7faa" }
Response	Order marked as delivered

Description: Confirms successful order delivery.

Add Pharmacy Review

Item	Details
Link	/api/patient/pharmacies/review
Method	POST
Request	json { "pharmacyId":"6a0a13efd785cd1a6461b64c", "rating":5, "comment":"Excellent service" }
Response	Review added successfully

Description: Allows patients to rate and review pharmacies.

Doctor APIs

Authorization: Bearer Token (Access Token Required)

Get Doctor Profile

Item	Details
Link	/api/doctor/profile
Method	GET
Request	None
Response	Doctor profile information

Description: Retrieves the authenticated doctor's profile.

Update Doctor Profile

Item	Details
Link	/api/doctor/profile
Method	PUT
Request	Updated doctor information
Response	Updated doctor profile

Description: Updates doctor profile details such as specialization, experience, prices, degrees, and contact information.

Add Medical Record

Field	Details
Link	/api/doctor/add-medical-record
Method	POST
Request	Patient diagnosis and prescription data
Response	Medical record created successfully

Example Request

```
{  
  "patientId": "69875519d4c27be0fe6fbe42",  
  "diagnosis": "Severe Bacterial Tonsillitis",  
  "prescription": [  
    {  
      "name": "Augmentin 1g",  
      "dosage": "1 tablet",  
      "frequency": "Every 12 hours",  
      "duration": "7 days"  
    }  
  ],  
  "notes": "Patient should rest for 3 days",  
  "nextVisitDate": "2026-02-15"  
}
```

Description: Creates a medical record for a patient.

Get Doctor Appointments

Field	Details
Link	/api/appointments/my
Method	GET
Request	None
Response	List of doctor appointments

Description: Returns all appointments associated with the authenticated doctor.

Create Prescription

Field	Details
Link	/api/appointments/prescription
Method	POST
Request	Prescription details
Response	Prescription created successfully

Example Request

```
{  
  
  "appointment": "6a01c7fd393f37d555c85dc3",  
  
  "diagnosis": "Upper respiratory tract infection",  
  
  "medicines": [  
  
    {  
  
      "name": "Augmentin 1g",  
  
      "dosage": "1 tablet",  
  
      "frequency": "Every 12 hours",  
  
      "duration": "7 days",  
  
      "instructions": "After meals"  
    },  
  
    {
```

```

    "name": "Paracetamol 500mg",
    "dosage": "1 tablet",
    "frequency": "Every 8 hours",
    "duration": "5 days",
    "instructions": "If fever exceeds 38°C"
  }
],
"labTests": [
  "CBC",
  "CRP"
],
"imaging": [
  "Chest X-Ray"
],
"nextVisit": "2026-05-20",
"notes": "Drink plenty of fluids and rest."
}

```

Description: Creates a prescription linked to a patient's appointment.

Update Prescription

Field	Details
Link	/:id/updatePrescription
Method	PATCH
Request	Updated prescription fields
Response	Prescription updated successfully

Example Request

```
{  
  
  "diagnosis": "Updated diagnosis"  
  
}
```

Description: Modifies an existing prescription.

Get Prescription by Appointment

Field	Details
Link	/api/appointments/:appointmentId/getPrescription
Method	GET
Request	None
Response	Prescription information

Description: Retrieves the prescription associated with a specific appointment.

Get Previous Prescriptions

Field	Details
Link	/api/appointments/:appointmentID/getPreviousPrescription
Method	GET
Request	None
Response	Previous prescriptions

Description: Returns previous prescriptions issued for the patient.

Complete Appointment

Field	Details
Link	/api/appointments/:appointmentId/complete
Method	PATCH
Request	None
Response	Appointment completed successfully

Description: Marks an appointment as completed.

Search Doctors

Field	Details
Link	/api/doctor/search-doctors?specialization=cardiology&gender=female&city=cairo
Method	GET
Request	Query Parameters
Response	List of doctors matching the search criteria

Description: Searches doctors by specialization, gender, and clinic location.

Pharmacy APIs

Authorization: Bearer Token (Pharmacy Access Token Required)

Get Pharmacy Profile

Field	Details
Link	/api/pharmacy/profile
Method	GET
Request	None
Response	Pharmacy profile information

Description: Retrieves pharmacy information, working hours, delivery options, and statistics.

Update Pharmacy Profile

Field	Details
Link	/api/pharmacy/profile
Method	PUT
Request	Updated pharmacy information
Response	Updated profile

Description: Updates pharmacy details and settings.

Get Orders

Field	Details
Link	<code>/api/pharmacy/orders</code>
Method	GET
Request	Query parameters (status, page, limit)
Response	List of orders

Description: Retrieves pharmacy orders and their statuses.

Accept Order

Field	Details
Link	<code>/api/pharmacy/orders/:id/accept</code>
Method	PATCH
Request	None
Response	Order accepted successfully

Description: Changes the order status from pending to accepted.

Complete Order

Field	Details
Link	<code>/api/pharmacy/orders/:id/complete</code>
Method	PATCH
Request	None
Response	Order completed successfully

Description: Marks an order as completed.

Get Medicines

Field	Details
Link	/api/pharmacy/medicines
Method	GET
Request	Query parameters (page, limit)
Response	List of medicines

Description: Retrieves all medicines in the inventory.

Add Medicine

Field	Details
Link	/api/pharmacy/medicines
Method	POST
Request	Medicine information
Response	Medicine added successfully

Description: Adds a new medicine to the inventory.

Update Medicine

Field	Details
Link	/api/pharmacy/medicines/:id
Method	PUT
Request	Updated medicine data
Response	Updated medicine information

Description: Modifies medicine details.

Restock Medicine

Field	Details
Link	/api/pharmacy/medicines/:id/restock
Method	PATCH
Request	Quantity to add
Response	Updated stock quantity

Description: Increases medicine stock quantity.

Get Delivery Men

Field	Details
Link	/api/pharmacy/delivery-men
Method	GET
Request	Query parameters
Response	List of delivery personnel

Description: Retrieves all delivery staff.

Add Delivery Man

Field	Details
Link	/api/pharmacy/delivery-men
Method	POST
Request	Delivery man information
Response	Delivery man added successfully

Description: Adds a new delivery employee.

Update Delivery Man

Field	Details
Link	/api/pharmacy/delivery-men/:id
Method	PUT
Request	Updated employee information
Response	Updated delivery man data

Description: Updates delivery employee details.

Delete Delivery Man

Field	Details
Link	/api/pharmacy/delivery-men/:id
Method	DELETE
Request	None
Response	Delivery man removed successfully

Description: Removes a delivery employee.

Get Financial Reports

Field	Details
Link	/api/pharmacy/financials
Method	GET
Request	Query parameter (year)
Response	Revenue and earnings statistics

Description: Retrieves financial reports and monthly summaries.

AI Daily Brief

Field	Details
Link	/api/pharmacy/ai/daily-brief
Method	POST
Request	Daily pharmacy statistics
Response	AI-generated daily summary

Description: Generates a daily report using AI based on pharmacy activities.

Clinic APIs

Authorization: Bearer Token (Doctor Access Token Required)

Create Clinic

Field	Details
Link	/api/clinics
Method	POST
Request	Clinic information
Response	Clinic created successfully

Description: Creates a new clinic for the doctor.

Get All Clinics

Field	Details
Link	/api/clinics
Method	GET
Request	None
Response	List of clinics

Description: Retrieves all clinics belonging to the doctor.

Update Clinic

Field	Details
Link	<code>/api/clinics/:id</code>
Method	PUT
Request	Updated clinic information
Response	Updated clinic data

Description: Updates clinic information such as address, name, working hours, and consultation fees.

Delete Clinic

Field	Details
Link	<code>/api/clinics/:id</code>
Method	DELETE
Request	None
Response	Clinic deleted successfully

Description: Removes a clinic from the system.

Appointment APIs

Authorization: Bearer Token Required

Book Appointment

Field	Details
Link	<code>/api/appointments</code>
Method	POST
Request	Appointment information

Response	Appointment created successfully
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Description: Allows patients to book an appointment.

Get My Appointments

Field	Details
Link	/api/appointments/my
Method	GET
Request	None
Response	List of appointments

Description: Retrieves appointments associated with the authenticated user.

Cancel Appointment

Field	Details
Link	/api/appointments/:id/cancel
Method	PATCH
Request	None
Response	Appointment cancelled successfully

Description: Changes appointment status to cancelled.

Complete Appointment

Field	Details
Link	/api/appointments/:appointmentId/complete
Method	PATCH
Request	None
Response	Appointment completed successfully

Description: Marks the appointment as completed.

Get Available Slots

Field	Details
Link	/api/appointments/slots/:clinicId/:date
Method	GET
Request	Date and clinic ID
Response	Available time slots

Description: Returns all available appointment slots for a specific day.

Lab APIs

Authorization: Bearer Token (Lab Access Token Required)

Get Lab Profile

Field	Details
Link	/api/lab/profile
Method	GET
Request	None
Response	Laboratory profile information

Description: Retrieves laboratory information and services.

Update Lab Profile

Field	Details
Link	/api/lab/profile
Method	PUT
Request	Updated laboratory information
Response	Updated profile

Description: Updates laboratory details and settings.

Get Lab Orders

Field	Details
Link	<code>/api/lab/orders</code>
Method	GET
Request	Query parameters
Response	List of laboratory requests

Description: Retrieves all test requests.

Upload Test Results

Field	Details
Link	<code>/api/lab/results/upload</code>
Method	POST
Request	PDF file or image
Response	Results uploaded successfully

Description: Uploads laboratory reports for patients.

Analyze Lab Report Using AI

Field	Details
Link	<code>/api/labReport/analyze</code>
Method	POST
Request	PDF file or image
Response	AI analysis of the report

Description: Uses AI to analyze laboratory reports and provide insights.

Get Dashboard Statistics

Field	Details
Link	/api/lab/dashboard
Method	GET
Request	None
Response	Dashboard statistics

Description: Retrieves laboratory statistics and analytics.

Reviews APIs

Authorization: Bearer Token (Patient Access Token Required)

Add Review

Field	Details
Link	/api/reviews
Method	POST
Request	Doctor ID, rating, and comment
Response	Review created successfully

Description: Allows patients to submit one review per doctor.

Get Doctor Reviews

Field	Details
Link	/api/reviews/doctor/:id
Method	GET
Request	Doctor ID
Response	List of reviews

Description: Retrieves all reviews for a specific doctor.

Get My Reviews

Field	Details
Link	/api/reviews/me
Method	GET
Request	None
Response	Patient reviews

Description: Retrieves all reviews submitted by the authenticated patient.

Update Review

Field	Details
Link	/api/reviews/:id
Method	PUT
Request	Rating and comment
Response	Review updated successfully

Description: Updates a previously submitted review.

Delete Review

Field	Details
Link	/api/reviews/:id
Method	DELETE
Request	None
Response	Review deleted successfully

Description: Removes a review from the system.

Admin APIs

Authorization: Bearer Token (Admin Access Token Required)

Get System Statistics

Field	Details
Link	<code>/api/admin/stats</code>
Method	GET
Request	None
Response	System statistics

Description: Returns the number of registered users and overall platform statistics.

Get All Users

Field	Details
Link	<code>/api/admin/users</code>
Method	GET
Request	None
Response	List of users

Description: Retrieves all registered users except passwords.

Get Users by Role

Field	Details
Link	<code>/api/admin/users/role/:role</code>
Method	GET
Request	User role
Response	List of users

Description: Returns users filtered by role.

Get Pending Users

Field	Details
Link	<code>/api/admin/users/pending</code>
Method	GET
Request	None
Response	Pending doctors, pharmacies, and labs

Description: Retrieves accounts waiting for activation.

Activate User

Field	Details
Link	<code>/api/admin/users/activate/:id</code>
Method	PATCH
Request	User ID
Response	User activated successfully

Description: Activates a doctor, pharmacy, or laboratory account and sends a confirmation email.

Deactivate User

Field	Details
Link	<code>/api/admin/users/deactivate/:id</code>
Method	PATCH
Request	User ID
Response	User deactivated successfully

Description: Disables a user account.

Delete User

Field	Details
Link	<code>/api/admin/users/:id</code>
Method	DELETE
Request	User ID
Response	User deleted successfully

Description: Permanently removes a user account.

AI APIs

Analyze Lab Report

Field	Details
Link	<code>/api/labReport/analyze</code>
Method	POST
Request	PDF file or image
Response	AI-generated analysis

Description: Uses artificial intelligence to analyze laboratory reports and provide medical insights.

AI Daily Brief

Field	Details
Link	<code>/api/pharmacy/ai/daily-brief</code>
Method	POST
Request	Daily statistics
Response	AI-generated daily summary

Description: Generates a daily performance report for pharmacies.

Smart Stock Alerts

Field	Details
Link	<code>/api/pharmacy/ai/smart-stock-alerts</code>
Method	GET
Request	None
Response	Low-stock and out-of-stock alerts

Description: Detects medicine shortages and provides stock recommendations.

AI Chatbot

Field	Details
Link	<code>/api/ai/chat</code>
Method	POST
Request	User message
Response	AI-generated response

Description: Provides intelligent assistance and answers user queries using conversational AI.

Dashboard Tools & Technologies

The dashboard interfaces were developed using modern frontend technologies to ensure scalability, maintainability, and a responsive user experience.

Core Technologies

- **React.js** – Component-based frontend framework.
- **React Router DOM** – Client-side routing and page navigation.
- **Redux Toolkit (RTK)** – Global state management.
- **RTK Query** – API communication and data fetching.
- **JavaScript (ES6+)** – Application logic.
- **CSS3** – Styling and responsive layouts.

Development Tools

- **Vite** – Fast build tool and development server.
- **ESLint** – Code quality and linting.
- **Git & GitHub** – Version control and collaboration.

UI Features

- Dynamic dashboards
- Role-based interfaces
- Real-time data updates
- Search and filtering
- Pagination
- Responsive design

Additional Libraries Used

- **Axios** (*optional*) – HTTP requests.
- **React Icons** – Icons and UI elements.
- **React Toastify** – Notifications and alerts.
- **JWT Decode** – Token decoding and authentication.
- **FormData** – File uploads.
- **Cloudinary API** – Image management.
- **Date Objects & Locale Functions** – Date and time formatting.
- **Local Storage** – Persistent user sessions.

Mobile Tools & Techniques

Development Tools

- Android Studio
- Visual Studio Code
- Git & GitHub

Programming Languages

- Dart Programming Language

Mobile Development Framework

- Flutter Framework

Software Architecture & Design Patterns

- State Management (BLoC / Cubit)
- Dependency Injection (GetIt & Injectable)

Backend Communication

- RESTful APIs (Dio)

Data Management

- Local Database (Hive)
- Local Preferences Storage (Shared Preferences)
- Secure Data Storage (Flutter Secure Storage)

Authentication & Security

- Google Authentication (Google Sign-In)

Location & Mapping Services

- Maps Integration (Google Maps)
- Geolocation Services (Geocator & Geocoding)

User Interface & User Experience

- Responsive User Interface Design (Flutter ScreenUtil)

Data Processing & Visualization

- Data Visualization (FL Chart)

Localization & Internationalization

- Easy Localization
- Internationalization & Localization (Intl)

File Handling

- File Management (File Picker)

Build & Automation

- Code Generation (Build Runner)

Chapter 8: Conclusion

This graduation project presented **Chefaa**, a comprehensive AI-powered healthcare platform designed to address the growing challenges of fragmented healthcare services. The system successfully integrates patients, doctors, pharmacies, medical laboratories, and administrators into a unified digital ecosystem, enabling efficient, accessible, and intelligent healthcare services.

Throughout the development process, the team implemented a wide range of features, including intelligent appointment management, electronic prescriptions, pharmacy inventory and order management, secure laboratory result delivery, and AI-powered medical assistance. The platform leverages modern technologies such as **Node.js**, **Express.js**, **MongoDB**, **Flutter**, **React.js**, and **Microsoft Azure AI Services** to provide a robust, scalable, and secure healthcare solution.

One of the most significant achievements of this project is the integration of advanced Artificial Intelligence capabilities. The **Lab Report Analyzer**, **Medical Chatbot**, **Pharmacy AI Suite**, and **Doctor AI Assistant** collectively provide intelligent healthcare guidance, automated business insights, and personalized medical analysis. These AI-driven features distinguish Chefaa from existing healthcare platforms by offering a more comprehensive and intelligent healthcare experience.

The comparative study presented in **Chapter 2** demonstrated that Chefaa provides a broader range of integrated services compared to existing healthcare platforms such as **Vezeeta**, **Altibbi**, and **Sehaty**. By combining healthcare management, AI-powered assistance, business analytics, and multi-stakeholder collaboration within a single platform, Chefaa offers a complete digital healthcare ecosystem.

The major accomplishments of this project include:

- Development of a fully integrated healthcare platform connecting patients, doctors, pharmacies, laboratories, and administrators.
- Implementation of AI-powered laboratory report analysis using vision-capable language models.
- Deployment of an intelligent medical chatbot capable of handling multi-turn conversations.
- Design and implementation of a comprehensive RESTful API supporting all platform operations.
- Development of cross-platform mobile applications using Flutter and responsive web applications using React.js.
- Integration of secure electronic payment services and real-time notification systems.
- Creation of intelligent business analytics dashboards for healthcare providers and administrators.

This project provided valuable practical experience in full-stack software development, mobile application engineering, database design, cloud-based AI integration, and scalable system architecture. The challenges encountered—including AI model integration, real-time synchronization, security implementation, and multi-stakeholder requirement management—significantly enhanced the team's technical knowledge and problem-solving skills.

Future work may include expanding AI diagnostic capabilities, integrating telemedicine and video consultation services, supporting health insurance providers, implementing wearable device integration, and enhancing multilingual support to reach a wider regional and international audience. Overall, Chefaa establishes a strong foundation for a scalable and intelligent digital healthcare ecosystem capable of meeting the evolving demands of modern healthcare delivery.

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